

Finite Elasticity

Arrangements

- Twelve lectures on seismology and Earth structure
- Two linked problem sets – solutions online immediately
- Supervisions in the weeks of **17th November** and **1st December** – sign up information to come
- Change in lecture style – **fewer topics, more detail**
- Lecture content builds directly on each other

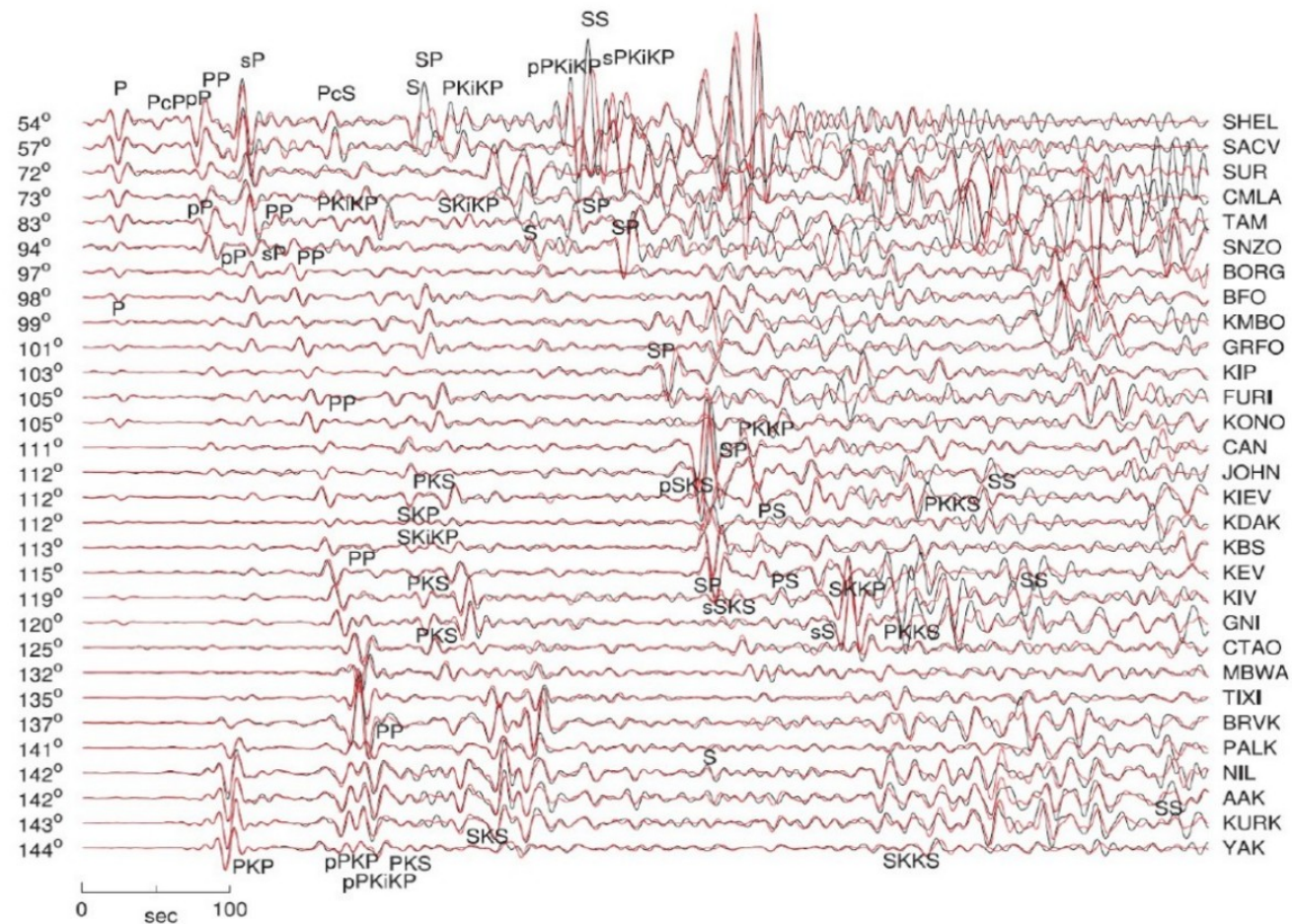
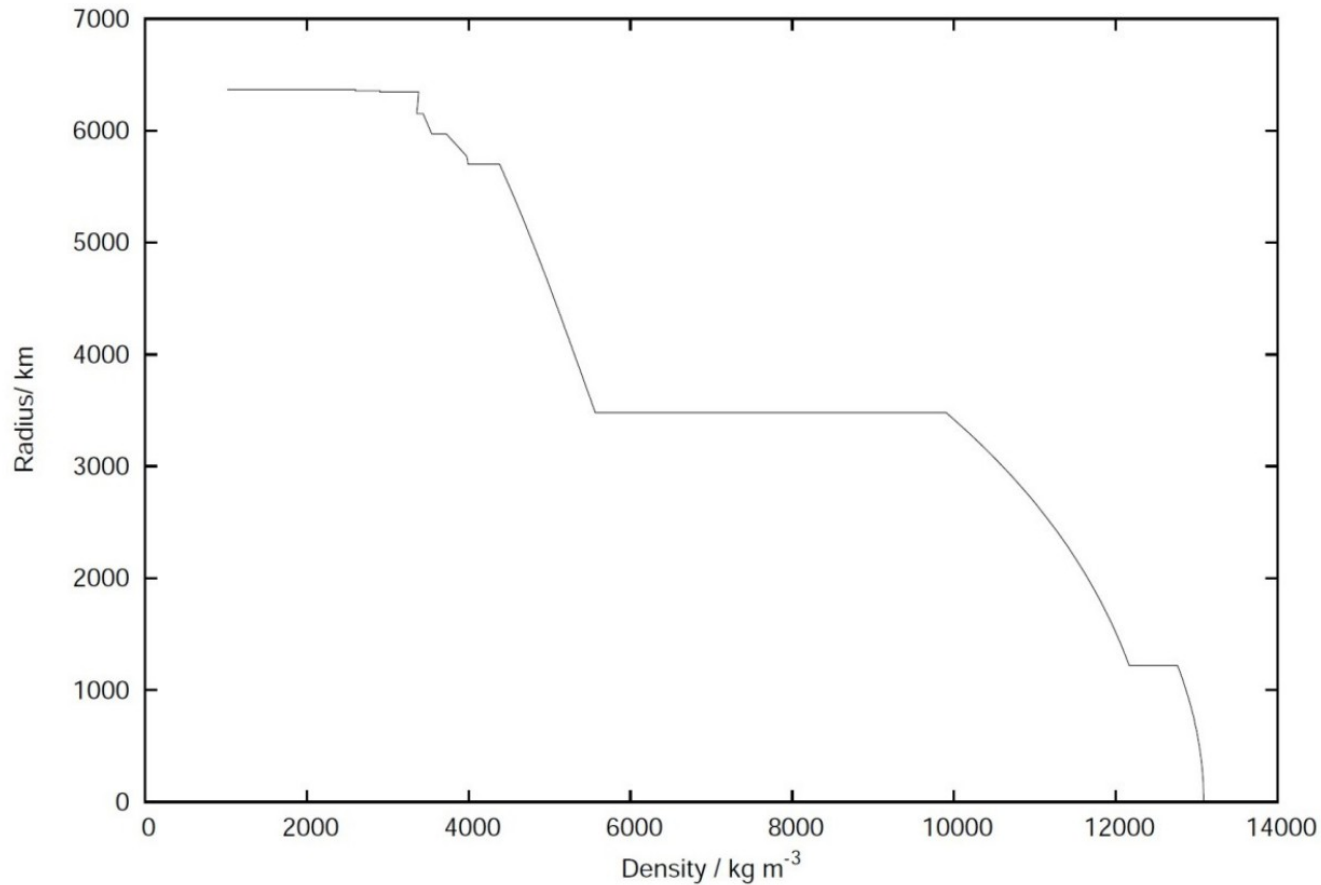


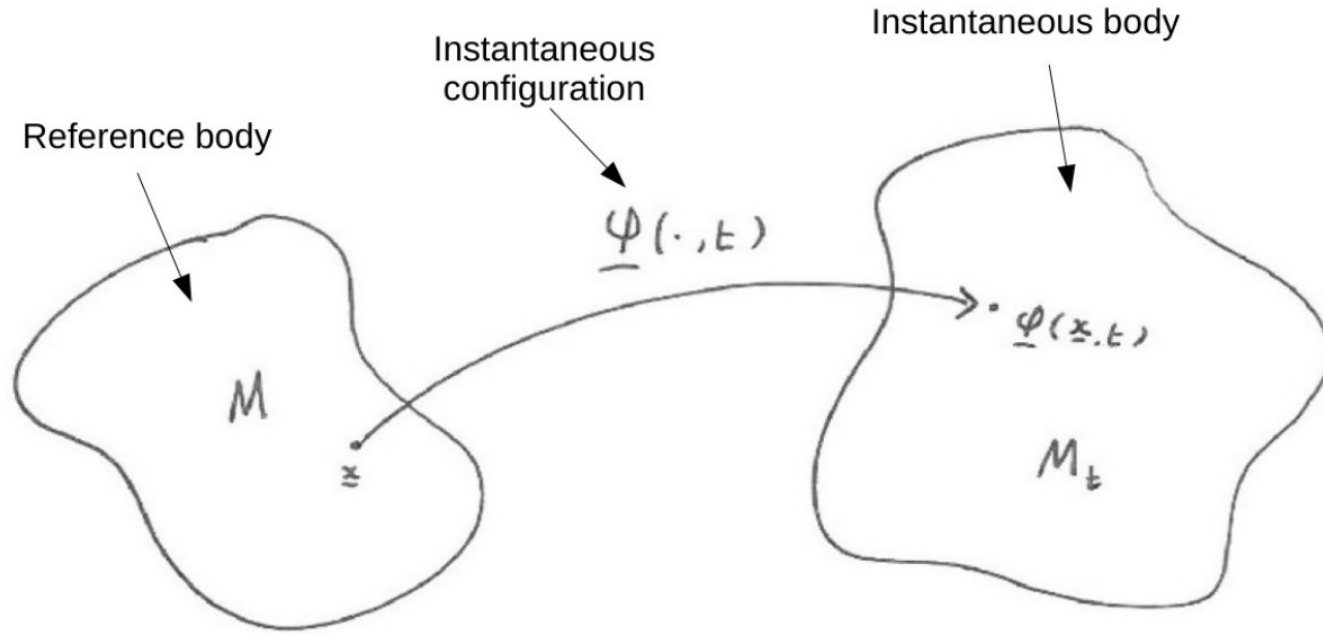
Figure 7. Vertical component record section comparing data (black) and SEM synthetics (red) for the 2008 September 3, $M_w = 6.3$ Santiago del Estero, Argentina earthquake, which occurred at a depth of 571 km. The records are aligned on the P wave, plotted as a function of epicentral distance, and bandpass filtered between 17 and 60 s. Major seismicological body wave arrivals are labelled. Epicentral distance is plotted to the left of each set of traces, and FDSN station identification codes are plotted to the right.



Model of Earth's interior density structure. Believed to be accurate enough to conclude that the outer core is less dense than would be expected for an iron-nickel alloy. Suggests there is a “light element” in the core. Tells us something about planetary formation and differentiation.

Kinematics

Motion of a body



$$\varphi(\mathbf{x}, t_0) = \mathbf{x}$$

Referential vs spatial descriptions

- **Referential quantities** are functions of position within the **reference body**
- **Spatial quantities** are functions of position within **physical space**
- In either case, functions might also depend on time
- In fluid mechanics, spatial (or **Eulerian**) descriptions most common
- In solid mechanics, referential (or **Lagrangian**) descriptions most common
- These conventions follow ultimately from the **constitutive relations** involved

Referential velocity

Referential velocity is the velocity of fixed particles:

$$v_i(\mathbf{x}, t) = \frac{\partial \varphi_i}{\partial t}(\mathbf{x}, t)$$

Can be related to the **spatial velocity** using the motion

We will always use the referential velocity, and typically just call it *the* velocity

Deformation gradient

First-order Taylor expansion of the motion about a fixed point:

$$\varphi_i(\mathbf{x} + \delta\mathbf{x}, t) = \varphi_i(\mathbf{x}, t) + F_{ij}(\mathbf{x}, t)\delta x_j + \dots$$

Deformation gradient has components:

$$F_{ij} = \frac{\partial \varphi_i}{\partial x_j}$$

Related to the **strain tensor** – more details in the next lecture

Jacobian

The **Jacobian** of the motion is defined by:

$$J(\mathbf{x}, t) = \det \mathbf{F}(\mathbf{x}, t).$$

Must be non-zero if the instantaneous configurations are invertible. We can assume it is **always positive** so that the motion is **orientation preserving**

Can be used to transform spatial integrations to the reference body:

$$\int_{M_t} f(\mathbf{y}) \, d^3\mathbf{y} = \int_M f[\varphi(\mathbf{x}, t)] J(\mathbf{x}, t) \, d^3\mathbf{x}$$

Conservation of mass

Integral conservation laws

For an arbitrary sub-body, we must have:

$$\frac{d}{dt} \int_{U_t} \varrho(\mathbf{y}, t) d^3\mathbf{y} = 0.$$

Transforming the integral to the reference body, we find:

$$\frac{d}{dt} \int_U J(\mathbf{x}, t) \varrho[\boldsymbol{\varphi}(\mathbf{x}, t), t] d^3\mathbf{x} = 0.$$

Local conservation laws

Because the sub-body was arbitrary, we must have:

$$\frac{\partial}{\partial t} \{J(\mathbf{x}, t) \varrho[\varphi(\mathbf{x}, t), t]\} = 0.$$

This leads us to define the **referential density**:

$$\rho(\mathbf{x}) = J(\mathbf{x}, t) \varrho[\varphi(\mathbf{x}, t), t]$$

Note slight notation change between printed handouts

Referential density is just the density within the reference configuration

Kinetic and potential energies

Kinetic energy

Kinetic energy density is defined by analogy with particle mechanics:

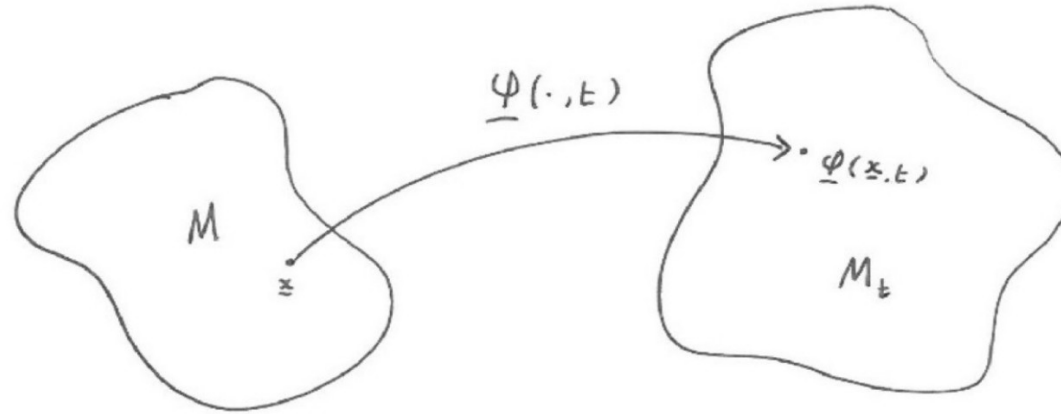
$$\frac{1}{2}\rho(\mathbf{x})||\mathbf{v}(\mathbf{x}, t)||^2,$$

Note this definition is per unit volume in the reference body

Kinetic energy of the body at an instance in time is given by:

$$\mathcal{T}(t) = \frac{1}{2} \int_M \rho(\mathbf{x}) ||\mathbf{v}(\mathbf{x}, t)||^2 \mathrm{d}^3\mathbf{x}.$$

Hyperelastic bodies



Elastic bodies have a **preferred shape and/or volume**, with work being necessary to deform them away from this preferred state

If the work required depends only on the **final configuration**, there is a well-defined **potential energy**. Such bodies are said to be **hyperelastic**

Elastic potential energy

We assume that the potential energy takes the form:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, t, \varphi(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x},$$

This choice *defines* a **simple hyperelastic** body

The term **simple** means that the potential energy density at a point depends only on the motion and its first spatial derivative at the same point

It's possible to consider dependence on higher-order spatial derivatives at a point, or a non-local form for the potential energy density, but this is not obviously needed within real materials

The form of the potential energy can be simplified using symmetry arguments – this is an application of the **principle of material frame indifference**

Time-translation invariance

If two motions differ only by a time-translation, we require that:

$$\mathcal{V}'(t) = \mathcal{V}(t - \tau).$$

This condition is met if and only if the potential energy density simplifies:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \varphi(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x}.$$

No explicit time-dependence of the potential energy density

Super-imposed rigid body motions

Two motions differing by a **rigid body motion** can be related by:

$$\varphi'(\mathbf{x}, t) = \mathbf{a}(t) + \mathbf{Q}(t)\varphi(\mathbf{x}, t)$$

We require the motions to have the same potential energy

A sufficient condition is that:

$$W[\mathbf{x}, \varphi(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{a}(t) + \mathbf{Q}(t)\varphi(\mathbf{x}, t), \mathbf{Q}(t)\mathbf{F}(\mathbf{x}, t)]$$

This can also be shown to be a necessary condition

Super-imposed rigid body motions

Focusing first on spatial translations, we conclude that

$$W[\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{a}(t) + \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)].$$

No explicit dependence on the **particle's position in space**

Potential energy now reduced to:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x}.$$

$$W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{Q}(t)\mathbf{F}(\mathbf{x}, t)],$$

The potential energy density is conventionally called the **strain energy function**

Hamilton's principle

Action and Lagrangian

Variational arguments covered from **first-principles** and in full detail in the handout – please ask if you need any additional help

Action for the for a motion is defined by:

$$\mathcal{S}(\varphi) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \varphi(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x} dt,$$

Where the **Lagrangian** takes the conventional form:

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}||^2 - W(\mathbf{x}, \mathbf{F})$$

To each possible motion of the body, the action returns a single real number – it defines a **functional** of the body's motion

Hamilton's principle

Functional derivative of the action must vanish at the true motion:

$$\delta S = \lim_{h \rightarrow 0} \frac{1}{h} [\mathcal{S}(\varphi + h \delta\varphi) - \mathcal{S}(\varphi)]$$

so long as the variations to the motion have **fixed-endpoints**

$$\delta\varphi(\mathbf{x}, 0) = \delta\varphi(\mathbf{x}, T) = 0.$$

This is **Hamilton's principle**, and it is **equivalent to Newton's laws of motion** for systems in which a potential energy is defined

Variation of the action

To first-order in the perturbed motion, we have:

$$\delta \mathcal{S} = \int_0^T \int_M \left\{ \frac{\partial \mathcal{L}}{\partial \varphi_i} \delta \varphi_i + \frac{\partial \mathcal{L}}{\partial v_i} \delta v_i + \frac{\partial \mathcal{L}}{\partial F_{ij}} \delta F_{ij} \right\} d^3 \mathbf{x} dt,$$

This can be simplified by relating variations in velocity and deformation gradient to the underlying motion:

$$\begin{aligned} \delta v_i &= \frac{\partial \delta \varphi_i}{\partial t} \\ \delta F_{ij} &= \frac{\partial \delta \varphi_i}{\partial x_j} \end{aligned}$$

Variation of the action

Using the product rule, we see that:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial v_i} \delta v_i &= \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \delta \varphi_i \right) - \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \right) \delta \varphi_i \\ \frac{\partial \mathcal{L}}{\partial F_{ij}} \delta F_{ij} &= \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial F_{ij}} \delta \varphi_i \right) - \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial F_{ij}} \right) \delta \varphi_i,\end{aligned}$$

We can then integrate by parts in time, and use the divergence theorem to simplify the expression:

$$\delta \mathcal{S} = \int_0^T \int_M \left\{ \frac{\partial \mathcal{L}}{\partial \varphi_i} \delta \varphi_i + \frac{\partial \mathcal{L}}{\partial v_i} \delta v_i + \frac{\partial \mathcal{L}}{\partial F_{ij}} \delta F_{ij} \right\} d^3 \mathbf{x} dt,$$

Variation of the action

Following this effort, the first order change in the action is:

$$\delta \mathcal{S} = \int_0^T \int_M \left\{ \frac{\partial \mathcal{L}}{\partial \varphi_i} - \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial F_{ij}} \right) \right\} \delta \varphi_i d^3 \mathbf{x} dt$$
$$+ \int_0^T \int_{\partial M} \frac{\partial \mathcal{L}}{\partial F_{ij}} \hat{n}_j \delta \varphi_i dS dt$$

Fixed-endpoint conditions have been used

We need the above expression to vanish for all such variations of the motion

Euler-Lagrange equations

From Hamilton's principle, we have obtained the **Euler-Lagrange equation**:

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} - \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial F_{ij}} \right) = 0,$$

along with the **natural boundary condition**

$$\frac{\partial \mathcal{L}}{\partial F_{ij}} \hat{n}_j = 0,$$

These equations govern the motion of a simple hyperelastic body

First Piola-Kirchhoff stress

For our Lagrangian, we see that

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} = 0, \quad \frac{\partial \mathcal{L}}{\partial v_i} = \rho v_i \quad \frac{\partial \mathcal{L}}{\partial F_{ij}} = -\frac{\partial W}{\partial F_{ij}}.$$

We define the **first Piola-Kirchhoff stress tensor** to be

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)],$$

Conservation of linear momentum

Euler-Lagrange equations can now be written:

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0,$$

subject to the boundary conditions

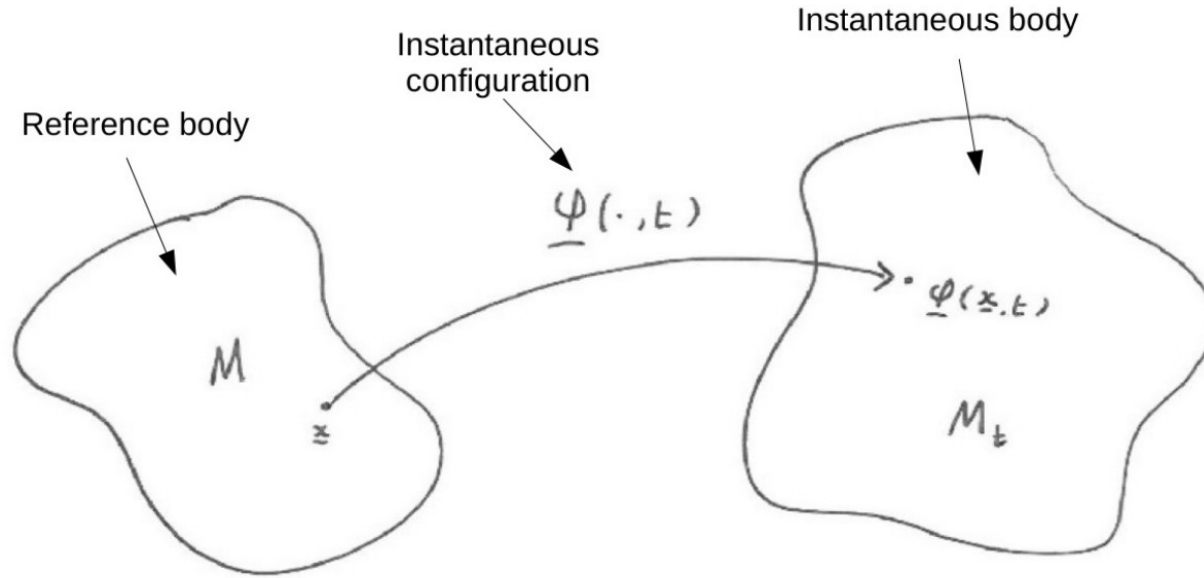
$$T_{ij} \hat{n}_j = 0.$$

In the third problem set, you will show these equations are equivalent to the **conservation of linear momentum**

You will also learn about the physical meaning of this new stress tensor

Constitutive theory

From last time



$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2} \rho(\mathbf{x}) \|\mathbf{v}\|^2 - W(\mathbf{x}, \mathbf{F})$$

$$W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{Q}(t) \mathbf{F}(\mathbf{x}, t)],$$

Material frame indifference

Polar decomposition theorem

For any **invertible matrix** with a **positive determinant**, we can write:

Rotation matrix

$$\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}$$

Symmetric positive-definite matrices

The diagram shows the equation $\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}$. Above the equation, the text "Rotation matrix" has two arrows pointing down to the $\mathbf{R}$ matrices in $\mathbf{R}\mathbf{U}$ and $\mathbf{V}\mathbf{R}$. Below the equation, the text "Symmetric positive-definite matrices" has two arrows pointing up to the $\mathbf{U}$ and $\mathbf{V}$ matrices.

U and **V** have the **same eigenvalues** – same magnitude of stretching about rotated principal axes

Proof is non-examinable

Rotational invariance of strain energy

We require the invariance property:

$$W(\mathbf{x}, \mathbf{QF}) = W(\mathbf{x}, \mathbf{F})$$

Writing $\mathbf{F} = \mathbf{RU}$, it follows that:

$$W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \mathbf{RU}) = W(\mathbf{x}, \mathbf{U}),$$

This shows that strain energy is only affected by the **stretching part** of the deformation gradient

Reduced strain energy function

We have shown that the strain energy must take the form:

$$W(\mathbf{x}, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}),$$

where we have introduced the **right Cauchy-Green deformation tensor**:

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}$$

Note that $\mathbf{U}$ is the square-root of $\mathbf{C}$. We use $\mathbf{C}$ in practice because its simpler

Equivalently, we could use the **finite strain tensor**:

$$\mathbf{E} = \frac{1}{2}[\mathbf{C} - \mathbf{I}]$$

Consequences of the result

Recall the first Piola-Kirchhoff stress tensor defined by:

$$T_{ij} = \frac{\partial W}{\partial F_{ij}}$$

Using the chain rule, we can then write:

$$T_{ij} = \frac{\partial U}{\partial C_{kl}} \frac{\partial C_{kl}}{\partial F_{ij}}.$$

From a careful application of index-notation

$$\frac{\partial C_{kl}}{\partial F_{ij}} = \frac{\partial}{\partial F_{ij}} (F_{mk} F_{ml}) = \delta_{im} \delta_{jk} F_{ml} + F_{mk} \delta_{im} \delta_{jl} = \delta_{jk} F_{il} + \delta_{jl} F_{ik},$$

Consequences of the result

We have now shown that:

$$T_{ij} = F_{ik} S_{kj}$$

Where we have defined the **second Piola-Kirchhoff stress tensor**:

$$S_{ij} = 2 \frac{\partial U}{\partial C_{ij}}$$

Note that **S** is a **symmetric tensor** – in the third problem set you will see that this symmetry is equivalent to the **conservation of angular momentum**

Equilibria and linearisation

Equilibrium configurations

Equations of motion and boundary conditions:

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0, \quad T_{ij} \hat{n}_j = 0.$$

Equilibrium conditions:

$$\varphi^0(\mathbf{x}) = \mathbf{x} \quad - \frac{\partial T_{ij}^0}{\partial x_j} = 0 \quad T_{ij}^0 \hat{n}_j = 0.$$

Stress-free equilibrium configurations are a possible solution!

Linearisation about equilibrium

We consider an initial disturbance about a stress-free equilibrium:

$$\varphi_i(\mathbf{x}, 0) = x_i + s u_i^0(\mathbf{x}) \quad v_i(\mathbf{x}, 0) = s v_i^0(\mathbf{x}).$$

Here “s” is a perturbation parameter that is used for book-keeping. The case $s=0$ corresponds to equilibrium

We expand the motion in powers of s:

$$\varphi_i(\mathbf{x}, t) = x_i + s u_i(\mathbf{x}, t) + O(s^2)$$

The first-order coefficient, $\mathbf{u}$, is known as the **displacement vector**

First order expansions

First-order expansions of the velocity and deformation gradient:

$$v_i = s \frac{\partial u_i}{\partial t} + O(s^2)$$

$$F_{ij} = \delta_{ij} + s \frac{\partial u_i}{\partial x_j} + O(s^2).$$

First order expansions

From the definition of the first Piola-Kirchhoff stress, we find:

$$T_{ij} = s A_{ijkl} \frac{\partial u_k}{\partial x_l} + O(s^2).$$

where we have defined the **elastic tensor**:

$$A_{ijkl}(\mathbf{x}) = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{x}, \mathbf{I}),$$

This is **Hooke's law**, but its nothing more than the result of linearisation

Note the **hyperelastic symmetry** which will be important:

$$A_{ijkl} = A_{klij}.$$

Linearised equations of motion

Linearised equation of motion:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

Linearised boundary condition:

$$n_j A_{ijkl} \frac{\partial u_k}{\partial x_l} = 0.$$

These linearised equations are sufficient for describing most aspects of seismic wave propagation within the Earth

Additional symmetries

About a stress-free equilibrium, the elastic tensor possess additional symmetries:

$$A_{ijkl} = A_{jikl} = A_{ijlk}$$

You will show this in the third problem set

This allows us to write the **linearised first Piola-Kirchhoff** stress tensor:

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} = A_{ijkl} \frac{1}{2} \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right)$$

where on the right we see the **linearised strain tensor**, which is often called *the* strain tensor in elementary (but inaccurate) discussions of elasticity

Material symmetries

The material symmetry group

Recall the invariance property under super-imposed rotations:

$$W(\mathbf{x}, \mathbf{QF}) = W(\mathbf{x}, \mathbf{F})$$

What happens if we do the rotation first?

$$W(\mathbf{x}, \mathbf{FQ}) = W(\mathbf{x}, \mathbf{F}),$$

This condition does not hold in general for all rotation matrices, $\mathbf{Q}$

When it does, we say that $\mathbf{Q}$ is in the **material symmetry group** at this point

This reflects an invariance of the strain energy with respect to **orientation of stretching**. Physically, it reflects **lattices symmetries** in crystalline materials

Isotropic materials

An elastic body is **isotropic** at a point if the corresponding material symmetry group comprises **all rotation matrices**

Otherwise, the body is **anisotropic** at this point

Making use of material frame indifference, isotropy requires:

$$U(\mathbf{C}) = U(\mathbf{Q}^T \mathbf{C} \mathbf{Q}).$$

for all rotation matrices

It follows that the strain energy depends only on the **eigenvalues** of the right Cauchy-Green deformation tensor

Isotropic materials

It is only a matter of tedious algebra to show that:

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{kj}),$$

Specification of the full elastic tensor requires only the two **Lame parameters**

If the material is also homogeneous, the linearised equations reduce to:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} - (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) - \mu \nabla^2 \mathbf{u} = 0.$$

This is the form presented in **1B dynamics**, but there are a **lot of assumptions** necessary to get here, most of which **don't hold in the Earth**

Elastic fluids

In an **elastic fluid**, there is no preferred shape, only a **preferred volume**

The strain energy depends only on the Jacobian of the motion:

$$W(\mathbf{x}, \mathbf{F}) = V(\mathbf{x}, J),$$

Equivalently, the material symmetry group comprises all matrices with **determinant equal to one** (the special linear group)

In this case, the elastic tensor takes the simple form:

$$A_{ijkl} = \kappa \delta_{ij} \delta_{kl}$$

Seismic sources

From last time

Linearised equations of motion:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

subject to boundary conditions:

$$n_j A_{ijkl} \frac{\partial u_k}{\partial x_l} = 0.$$

Along with initial conditions on the displacement vector and its time-derivative

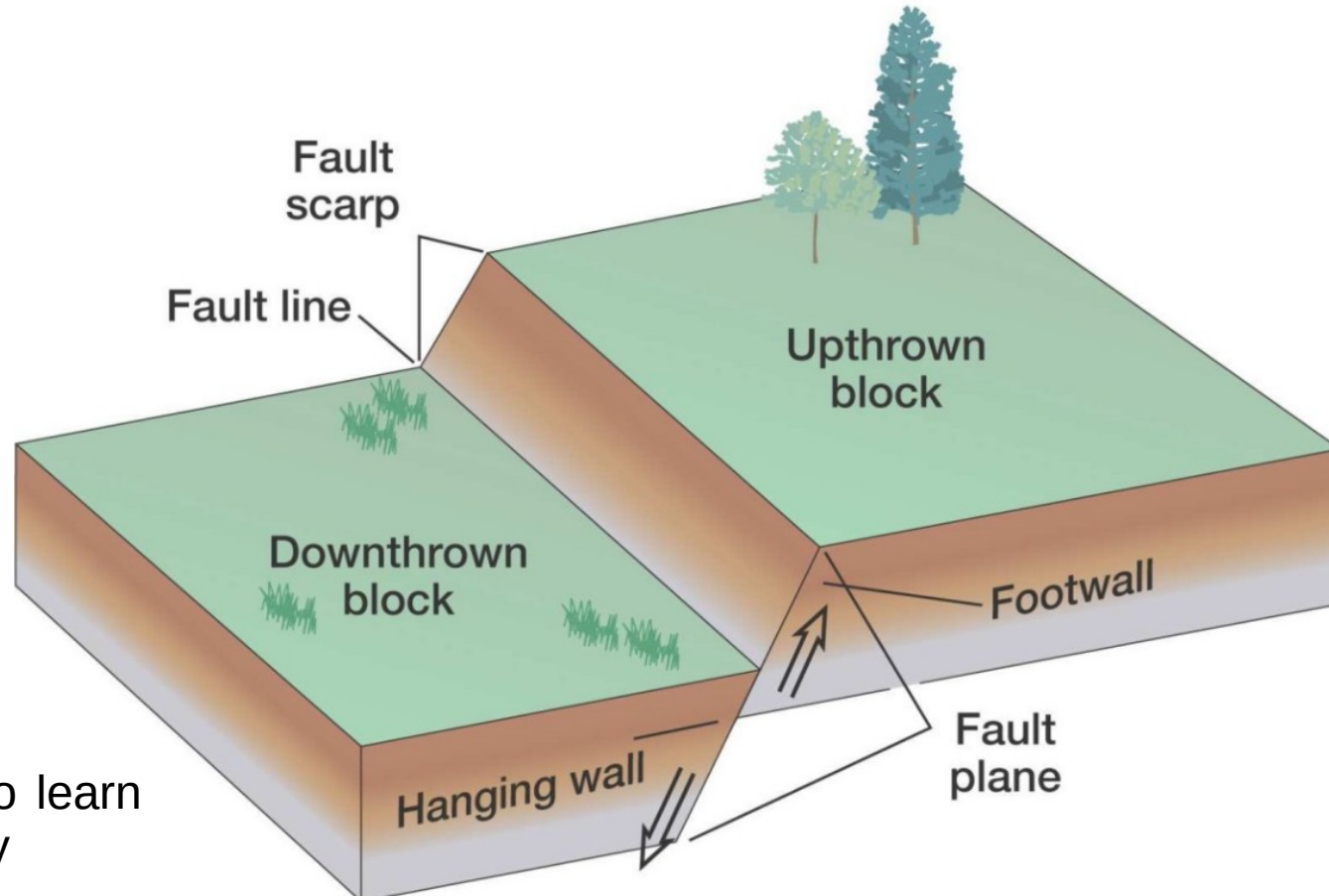
But where do earthquakes enter into the theory?

Earthquakes and tectonics



The San Andreas Fault in the Carrizo Plain of California

Idealised fault planes

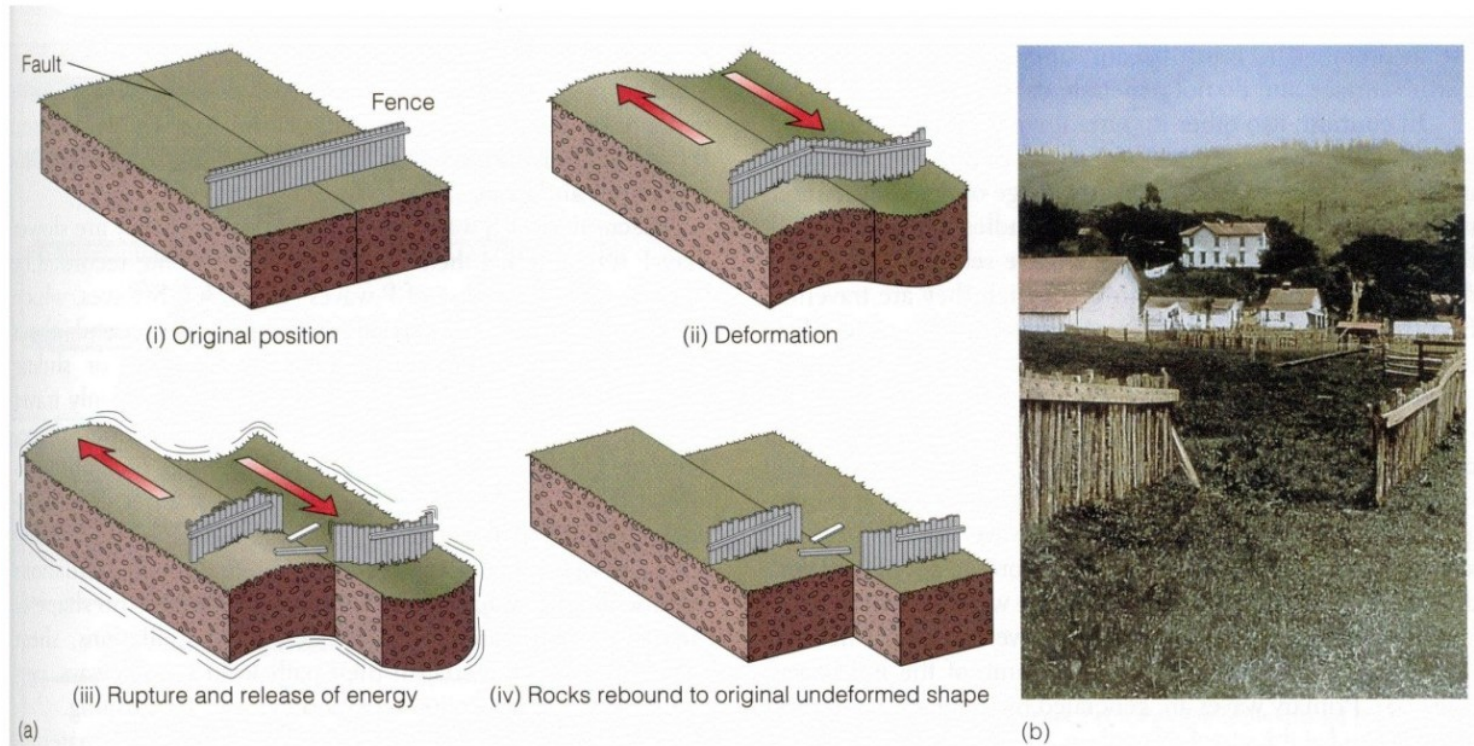


You **don't** need to learn
all this terminology

San Francisco 1906



Elastic rebound theory



Harry Fielding Reid (1859—1944)

Kinematic source approximation

Conservation of energy

Lagrangian density for finite elasticity:

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}||^2 - W(\mathbf{x}, \mathbf{F}),$$

Motion of the body satisfies conservation of energy:

$$\frac{d}{dt} \int_M E(\mathbf{x}, t) d^3\mathbf{x} = 0,$$

$$E(\mathbf{x}, t) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}(\mathbf{x}, t)||^2 + W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)].$$

A body in a **stable** equilibrium configuration will remain in it – no earthquakes!

Breaking the time-translation symmetry

To model an earthquake phenomenologically, we just add an explicit time-dependence to the strain energy function:

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}||^2 - W(\mathbf{x}, t, \mathbf{F}),$$

Equations of motion are unchanged

Constitutive equation now has an explicit time-dependence:

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, t, \mathbf{F}(\mathbf{x}, t)].$$

Stress glut

We assume that the time-dependent strain energy function can be written:

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij}$$

The corresponding first Piola-Kirchhoff stress tensor is

$$T_{ij}(\mathbf{x}, t) = F_{ik}(\mathbf{x}, t) \left\{ 2 \frac{\partial U}{\partial C_{kj}} [\mathbf{x}, \mathbf{C}(\mathbf{x}, t)] - \bar{S}_{kj}(\mathbf{x}, t) \right\}$$

First term on right hand side is the stress predicted by elasticity, but this differs from the true stress due to the the second term

Stress glut

The **stress glut** is defined by:

$$\begin{aligned}\bar{T}_{ij}(\mathbf{x}, t) &= F_{ik}(\mathbf{x}, t) \bar{S}_{kj}(\mathbf{x}, t) \\ &= 2F_{ik}(\mathbf{x}, t) \frac{\partial U}{\partial C_{kj}}[\mathbf{x}, \mathbf{C}(\mathbf{x}, t)] - T_{ij}(\mathbf{x}, t).\end{aligned}$$

This term quantifies the failure of elasticity to correctly predict stresses in the body

By definition, the stress glut is **non-zero** within the **source region**

In the full problem, the stress glut would be determined dynamically

Under the kinematic source approximation, the stress glut is reduced to a parameter

Linearised equations of motion

We add a stress glut as a perturbation to a stress-free equilibrium configuration:

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - s \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij},$$

Linearisation proceeds almost identically to yield:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

Seismic source associated with an **effective body force** and **effective traction** (force per unit area) on the boundary

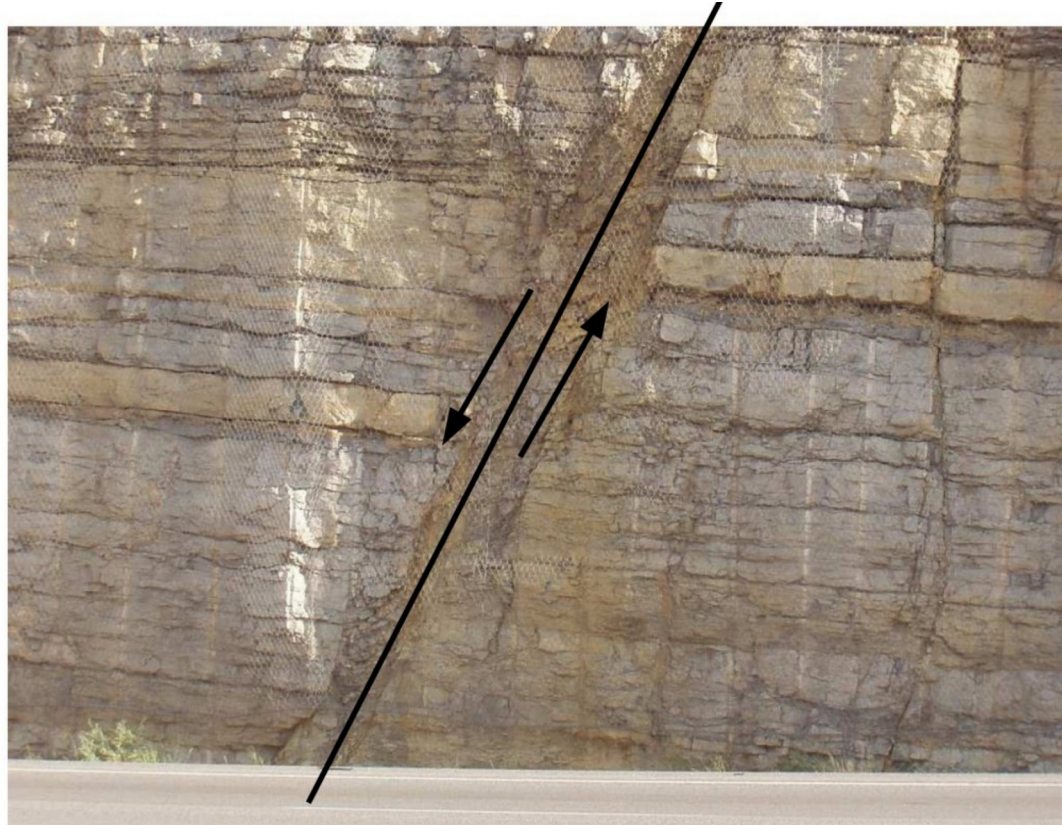
Model for an ideal fault

Real and ideal faults



Real faults involve strong non-elastic deformation across a **finite fault zone**

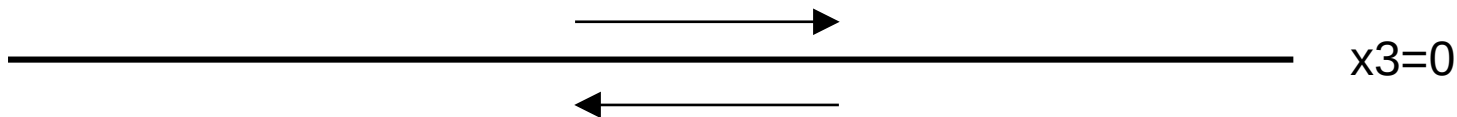
Real and ideal faults



For an **ideal fault** we collapse the fault zone to a surface across which **displacement is discontinuous**

Stress glut for an ideal fault

We consider a **planar fault** and adapt our coordinates suitably



Displacement takes the form:

$$u_i(x_1, x_2, x_3, t) = u_i^0(x_1, x_2, x_3, t) + w_i(x_1, x_2, t)H(x_3),$$

The **displacement discontinuity vector**, w , is tangential to the fault plane

Stress glut for an ideal fault

Spatial derivative of the displacement vector:

$$\frac{\partial u_i}{\partial x_j} = \frac{\partial u_i^0}{\partial x_j} + \frac{\partial w_i}{\partial x_j} H(x_3) + w_i \delta_{j3} \delta(x_3),$$

Resulting linearised stress predicted by elasticity:

$$A_{ijkl} \frac{\partial u_k^0}{\partial x_l} + A_{ijkl} \frac{\partial w_k}{\partial x_l} H(x_3) + A_{ijk3} w_k \delta(x_3).$$

Motivates the **definition** of the stress glut:

$$\bar{S}_{ij} = A_{ijk3} w_k \delta(x_3).$$

Simple generalisation to non-planar faults

Equations of motion

Equations of motion with an idealised fault:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j},$$

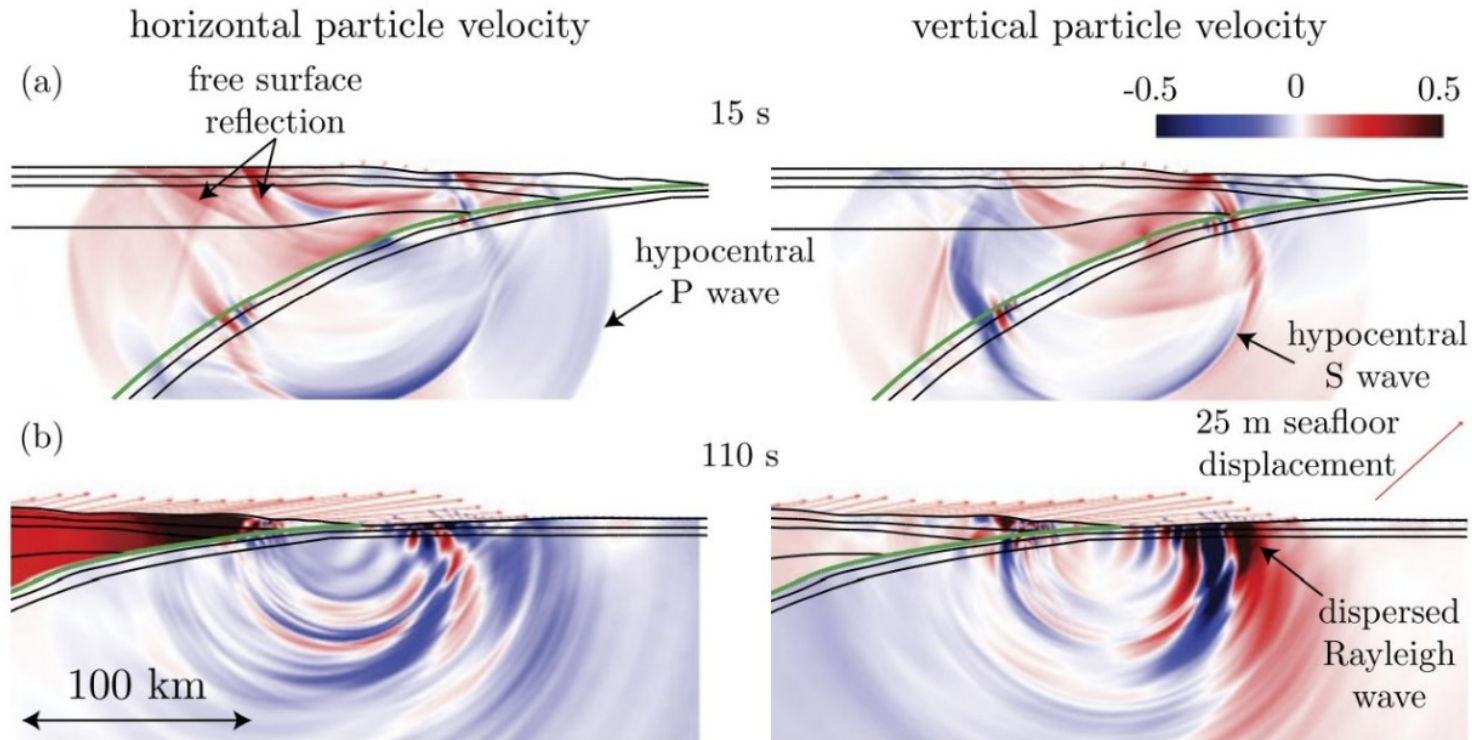
$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

$$\bar{S}_{ij} = A_{ijkl} w_k n_l \delta_{\Sigma}.$$

Parameters in these equations:

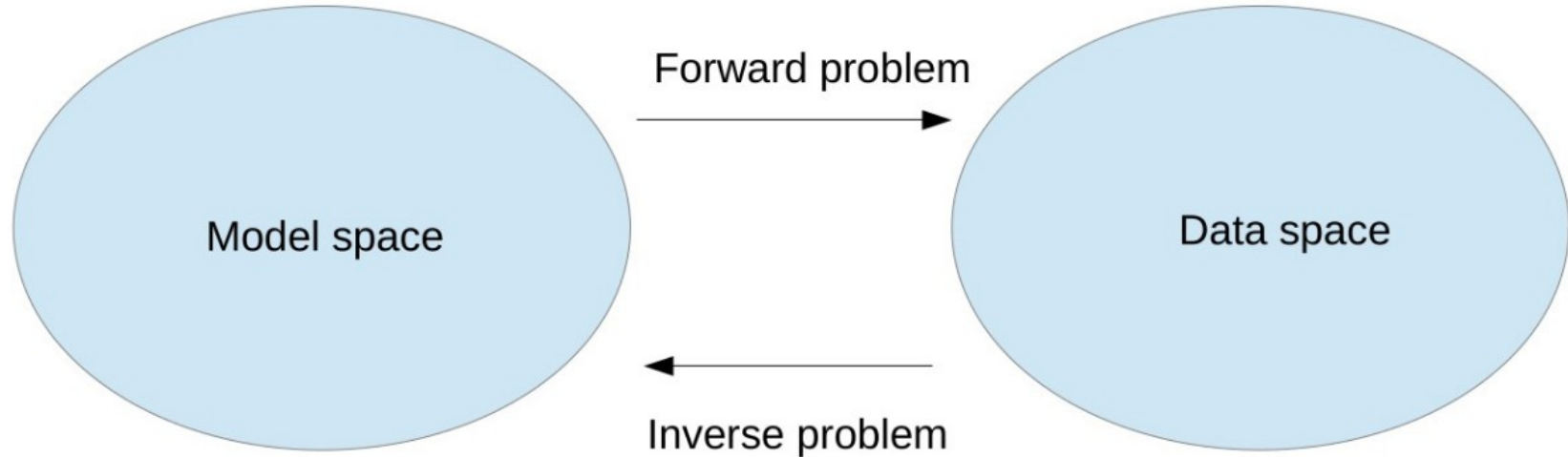
- Geometry of the Earth model
- Density and the elastic tensor
- Geometry of the fault plane and the displacement discontinuity vector

Numerical simulations



Inverse problems

Common structure



This graphical representation suggests a **symmetry** between forward and inverse problems – **this is not the case!**

Well-posed and ill-posed problems

A mathematical problem can be viewed as mapping between inputs and outputs

Such a problem is said to be **well-posed** if:

- 1) For each input there **exists** an output
- 2) For each input there exists a **unique** output
- 3) The output depends **continuously** on the input

A problem that is not well-posed is said to be **ill-posed**

The equations of mathematical physics are well-posed (or thought/hoped to be)

Inverse problems are almost always ill-posed

Plane wave propagation

Previously...

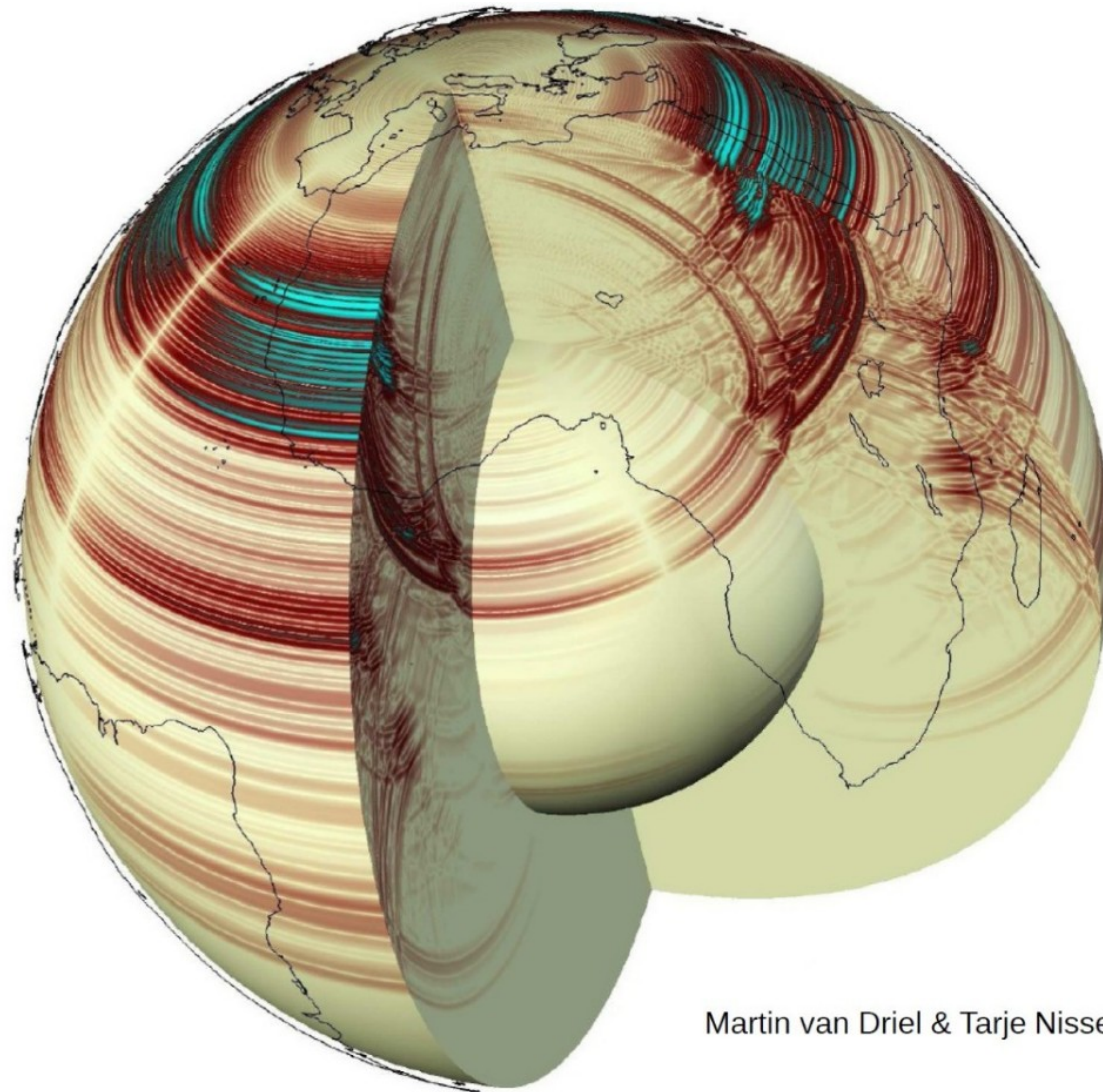
Linearised equations of motion including a stress glut:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

Definition of the elastic tensor:

$$A_{ijkl}(\mathbf{x}) = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{x}, \mathbf{I}),$$



Martin van Driel & Tarje Nissen Meyer

The Christoffel equation

Plane wave ansatz

We consider an infinite and homogeneous body

Look for a plane wave solution of the form:

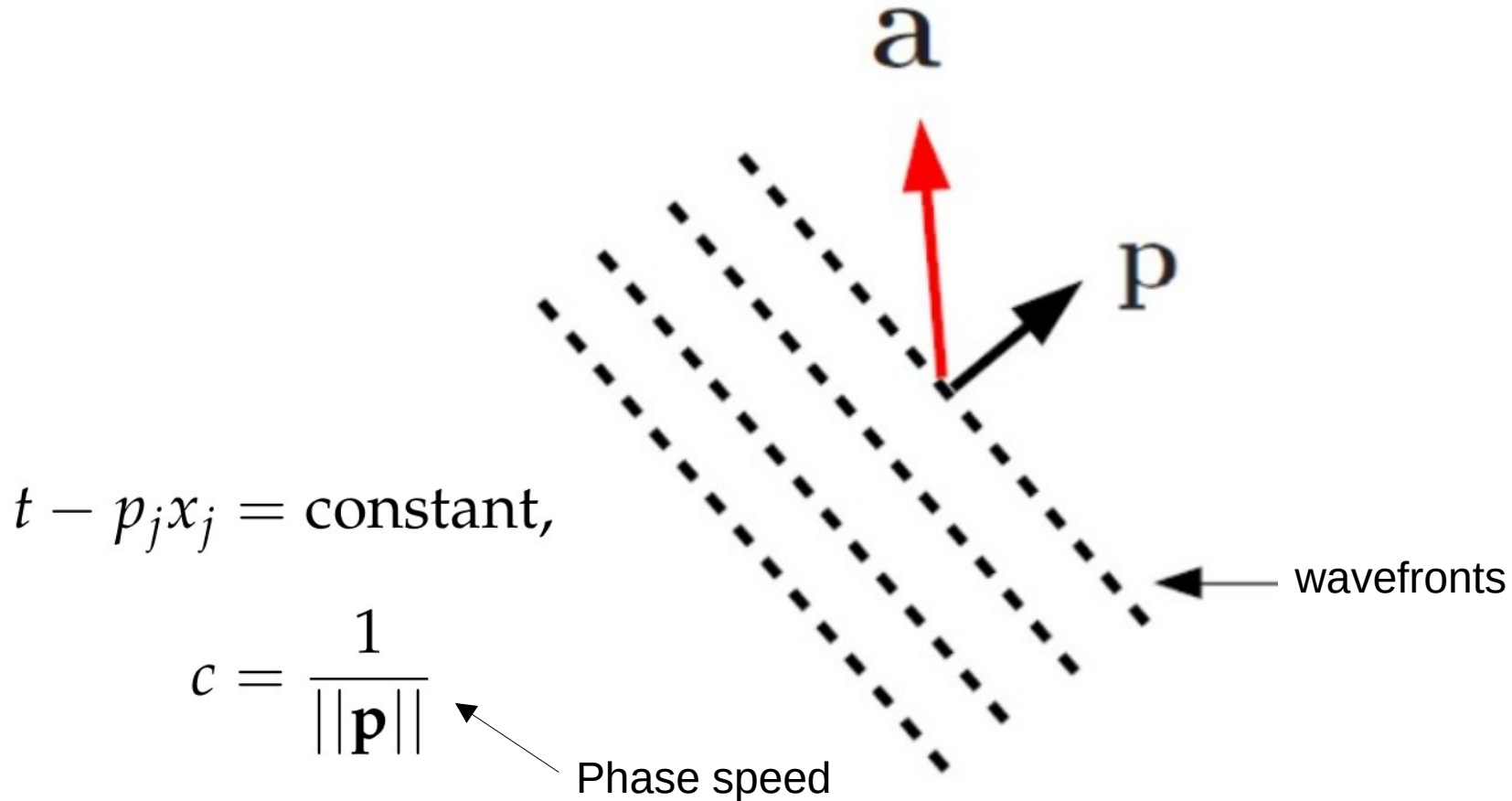
$$u_i(x, t) = f(t - p_j x_j) a_i$$

The **slowness vector**, $\mathbf{p}$, sets the orientation and speed of the wavefronts

The **polarisation vector**, $\mathbf{a}$, sets the direction of particle motion

The **pulse shape** set by f , and does not vary as the wave propagates – in this sense the waves are **non-dispersive** (but...)

Plane wave ansatz



The Christoffel equation

For the plane wave ansatz:

$$\frac{\partial^2 u_i}{\partial t^2} = f''(t - p_j x_j) a_i,$$

$$\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = f''(t - p_m x_m) A_{ijkl} p_j p_l a_k$$

A non-trivial plane wave solution exists if and only if:

$$(\rho \delta_{ik} - A_{ijkl} p_j p_l) a_k = 0.$$

This is known as the **Christoffel equation** – it places a constraint on both the slowness and polarisation vectors

The Christoffel matrix

It will be useful to define the **Christoffel matrix**, which is a function of $\mathbf{p}$:

$$\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l$$

The Christoffel equation can then be written:

$$[\Gamma(\mathbf{p}) - \mathbf{I}] \mathbf{a} = 0$$

This is a **homogeneous linear equation** for $\mathbf{a}$, that can have non-trivial solutions if and only if:

$$\det[\Gamma(\mathbf{p}) - \mathbf{I}] = 0$$

The roots of this equation define the **slowness surface**

Reduction to an eigenvalue problem

Two useful properties

From the definition of the Christoffel matrix, we have:

$$\Gamma(\lambda \mathbf{p}) = \lambda^2 \Gamma(\mathbf{p}),$$

It is a **homogeneous function of order two**

The hyperelastic symmetry implies that the Christoffel matrix is **symmetric**:

$$\Gamma_{ki}(\mathbf{p}) = \frac{1}{\rho} A_{kjil} p_j p_l = \frac{1}{\rho} A_{ilkj} p_j p_l = \frac{1}{\rho} A_{ijkl} p_j p_l = \Gamma_{ik}(\mathbf{p}),$$

Plane waves in a fixed direction

Fixing the orientation of the slowness vector, we can write:

$$\mathbf{p} = \frac{1}{c} \hat{\mathbf{p}},$$

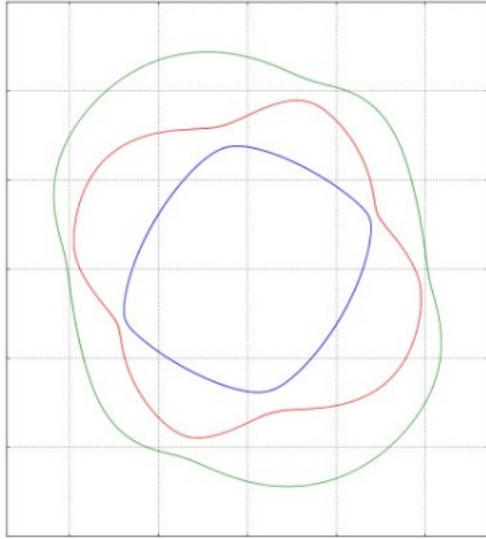
and hence the Christoffel equation reduces to:

$$\Gamma(\hat{\mathbf{p}}) \mathbf{a} = c^2 \mathbf{a}.$$

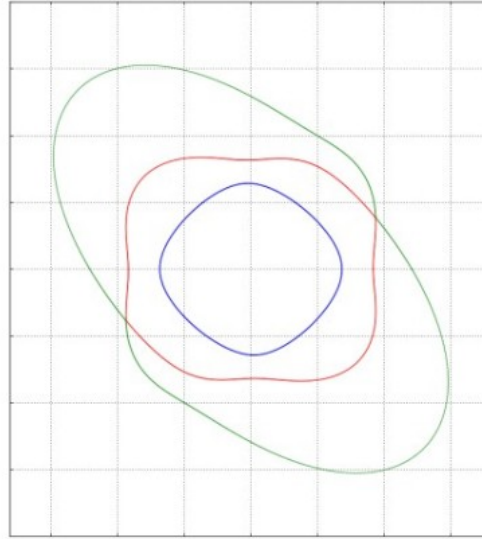
This is a **symmetric eigenvalue problem**:

- 1) Squared phase speeds are **real**, generally **three distinct values**
- 2) Polarisation vectors for distinct squares phase speeds are **orthogonal**
- 3) Degenerate squared phase speeds are possible

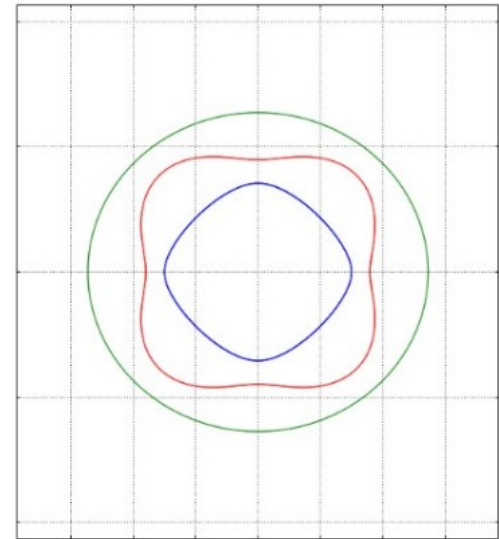
The slowness surface



Three distinct sheets



Two sheets touch



Cubic symmetry

$$[||\mathbf{p}||^2 c_1(\hat{\mathbf{p}})^2 - 1][||\mathbf{p}||^2 c_2(\hat{\mathbf{p}})^2 - 1][||\mathbf{p}||^2 c_3(\hat{\mathbf{p}})^2 - 1] = 0,$$

$$c_1(\hat{\mathbf{p}}) \leq c_2(\hat{\mathbf{p}}) \leq c_3(\hat{\mathbf{p}}),$$

Isotropic media

Isotropic Christoffel matrix

We have seen that the elastic tensor in an isotropic solid takes the form:

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}),$$

Putting into the definition of the Christoffel matrix, we arrive at:

$$\begin{aligned} \Gamma_{ik}(\hat{\mathbf{p}}) &= \frac{\lambda + \mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} \delta_{ik}. \\ &= \frac{\lambda + 2\mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} (\delta_{ik} - \hat{p}_i \hat{p}_k). \end{aligned}$$

In the final expression, note **orthogonal projections** parallel and perpendicular to the direction of the slowness vector

P-waves

Isotropic Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + 2\mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} (\delta_{ik} - \hat{p}_i \hat{p}_k).$$

By inspection we identify an eigenvector and eigenvalue:

$$\mathbf{a} \propto \hat{\mathbf{p}} \quad \alpha = \sqrt{\frac{\lambda + 2\mu}{\rho}}.$$

These plane waves are known as **P-waves**:

- Particle motion along the propagation direction
- Phase speed the same in all directions
- Associated with a non-degenerate eigenspace

S-waves

Isotropic Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + 2\mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} (\delta_{ik} - \hat{p}_i \hat{p}_k).$$

By inspection we identify an eigenvector and eigenvalue:

$$\mathbf{a} \perp \hat{\mathbf{p}} \qquad \beta = \sqrt{\frac{\mu}{\rho}}.$$

These plane waves are known as **S-waves**:

- Particle motion perpendicular to the propagation direction
- Phase speed the same in all directions
- Associated with a two-fold degenerate eigenspace

Slightly anisotropic media

Anisotropy in the Earth



Perturbed eigenvalue problem

We consider a slightly anisotropic elastic tensor:

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + s \delta A_{ijkl}$$

Resulting Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + \mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} \delta_{ik} + s \delta \Gamma_{ik}(\hat{\mathbf{p}}),$$

Eigenvalue problem to be solved:

$$[\mathbf{\Gamma}(\hat{\mathbf{p}}) + s \delta \mathbf{\Gamma}(\hat{\mathbf{p}})] \mathbf{a} = c^2 \mathbf{a}$$

Quasi P-waves

Apply **non-degenerate** perturbation theory:

$$c = \alpha + s \delta\alpha + \dots,$$

$$\mathbf{a} = \hat{\mathbf{p}} + s \delta\mathbf{a} + \dots.$$

First-order term in eigenvalue problem:

$$\Gamma(\hat{\mathbf{p}})\delta\mathbf{a} + \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{p}} = \alpha^2\delta\mathbf{a} + 2\alpha\delta\alpha\hat{\mathbf{p}}.$$

Through a standard argument, we arrive at:

$$\delta\alpha = \frac{1}{2\alpha} \hat{\mathbf{p}} \cdot \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{p}},$$

Polarisation **almost aligned** with propagation direction, phase speed now **varies with the orientation**

Quasi S-waves

Apply **degenerate** perturbation theory:

$$c = \beta + s \delta\beta + \dots ,$$

$$\mathbf{a} = a_1 \hat{\mathbf{t}}_1 + a_2 \hat{\mathbf{t}}_2 + s \delta\mathbf{a} + \dots ,$$

Leads to 2D eigenvalue problem:

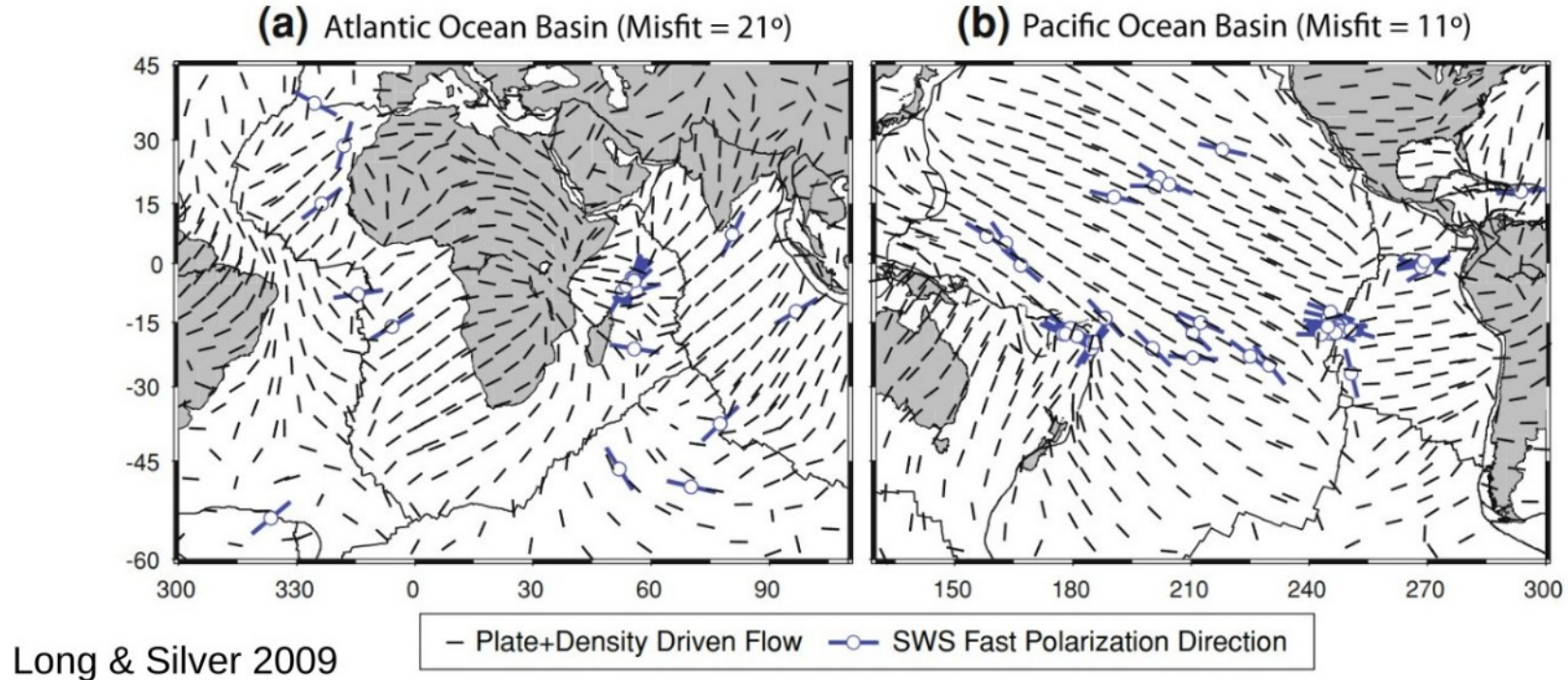
$$\begin{pmatrix} \hat{\mathbf{t}}_1 \cdot \delta\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_1 & \hat{\mathbf{t}}_1 \cdot \delta\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_2 \\ \hat{\mathbf{t}}_2 \cdot \delta\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_1 & \hat{\mathbf{t}}_2 \cdot \delta\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 2\beta\delta\beta \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} .$$

Solution gives **two perturbed phase speeds** which are usually distinct – the initial **degeneracy is split**

Zeroth-order eigenvectors orthogonal to each other and propagation direction, first order correction will have a component along propagation direction

This behaviour is known as **shear wave splitting**

Shear wave splitting and mantle flow



Convection in the mantle creates large-scale **alignment** of anisotropic mineral grains with the **flow direction** – produces an effective seismic anisotropy. Example here shows the relation between shallow anisotropy and plate motion

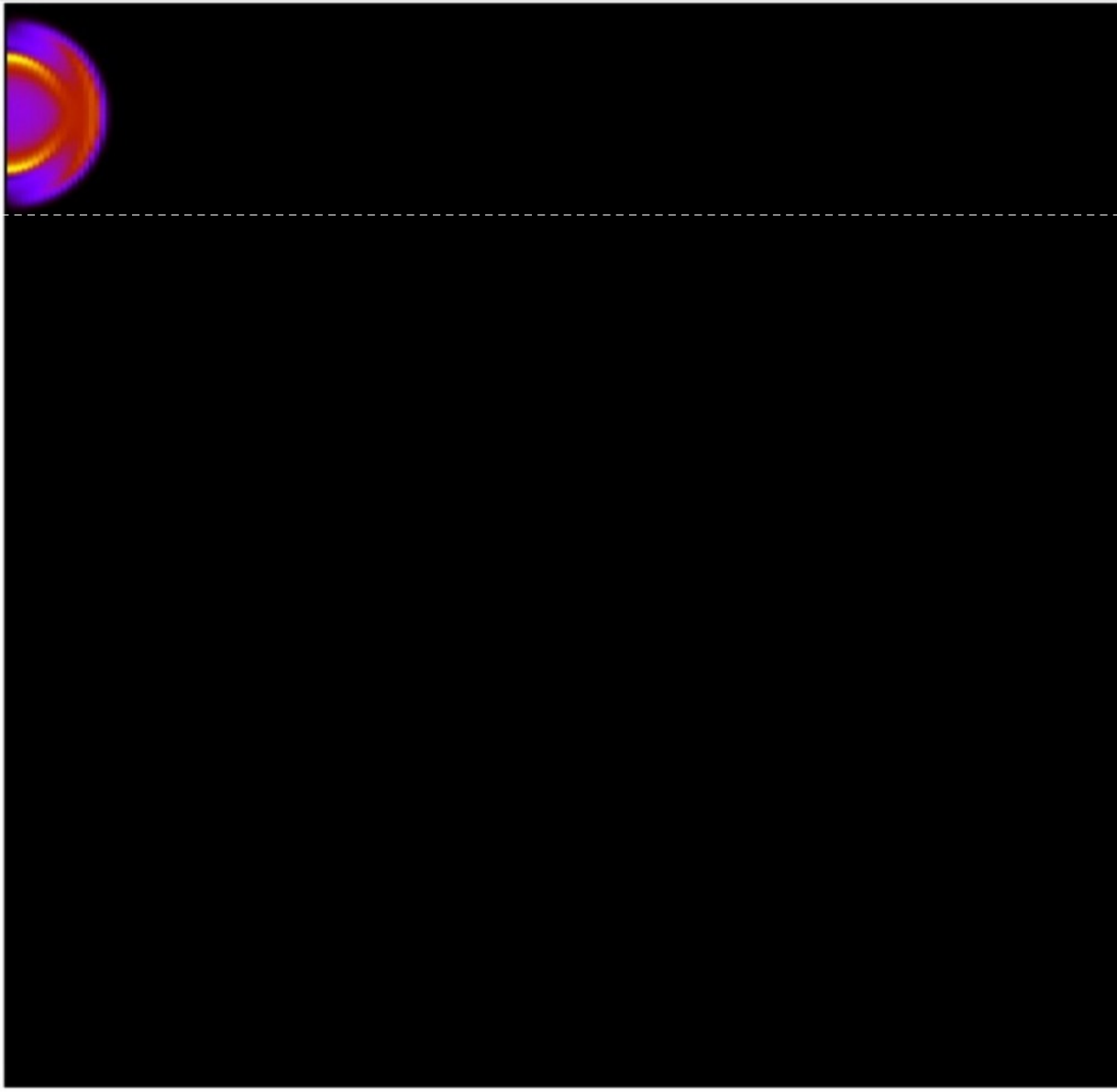
Interaction of plane waves with boundaries

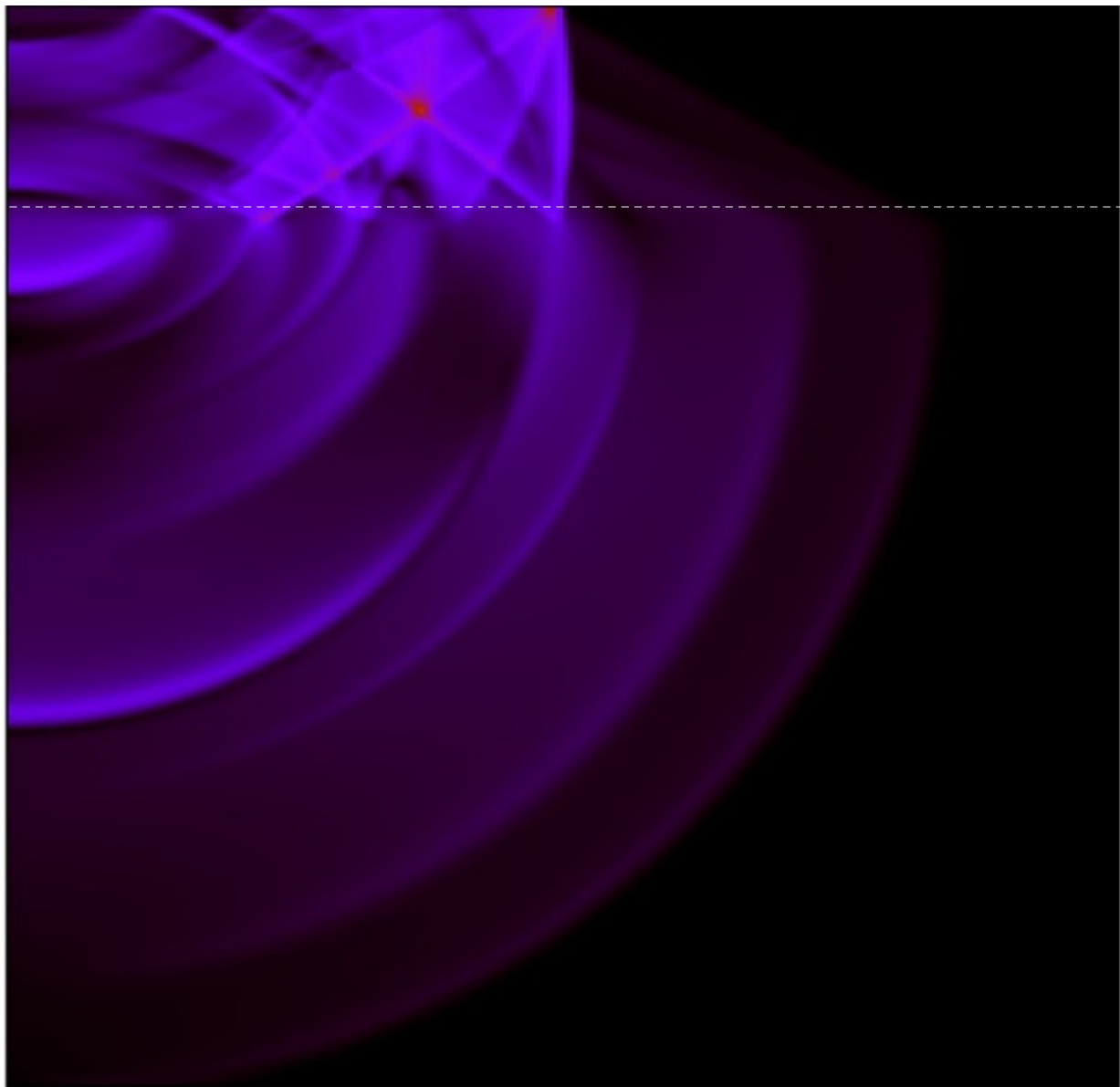
Topic covered quantitatively in previous years but never examined

Quantitative details now removed from the course for reasons of time

The usual **reflections** and **transmissions** of waves occur when a plane elastic wave interacts with a boundary.

The only interesting feature is the **conversion** between different wave types. For example, in an isotropic medium, an incident P-wave can produce reflected and transmitted P- and S- waves





Ray theory

From last lecture

Plane wave ansatz

$$u_i(x, t) = f(t - p_j x_j) a_i$$

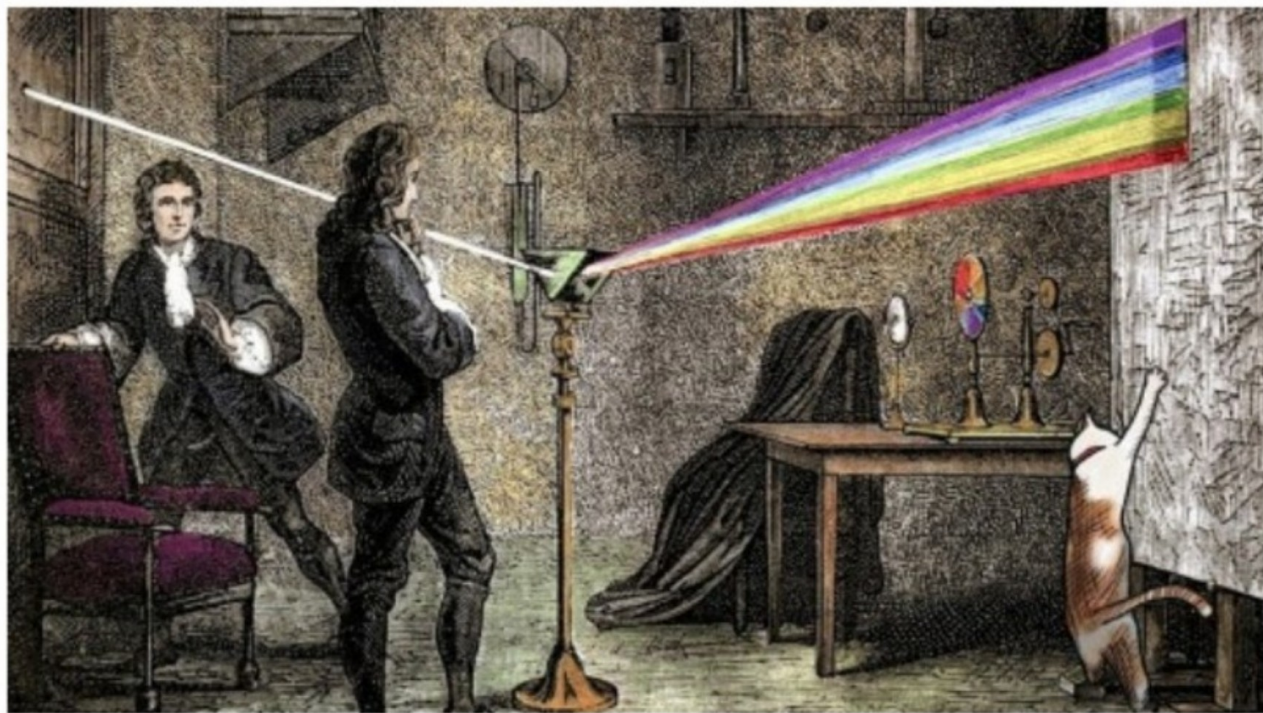
Christoffel equation for plane waves:

$$\det[\Gamma(\mathbf{p}) - \mathbf{I}] = 0$$

$$\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l$$

Definition of the slowness surface:

$$\det[\Gamma(\mathbf{p}) - \mathbf{I}] = 0$$



$$\begin{aligned}\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{H} + \frac{\partial \mathbf{D}}{\partial t} &= \mathbf{J}, & \nabla \cdot \mathbf{D} &= \rho,\end{aligned}$$

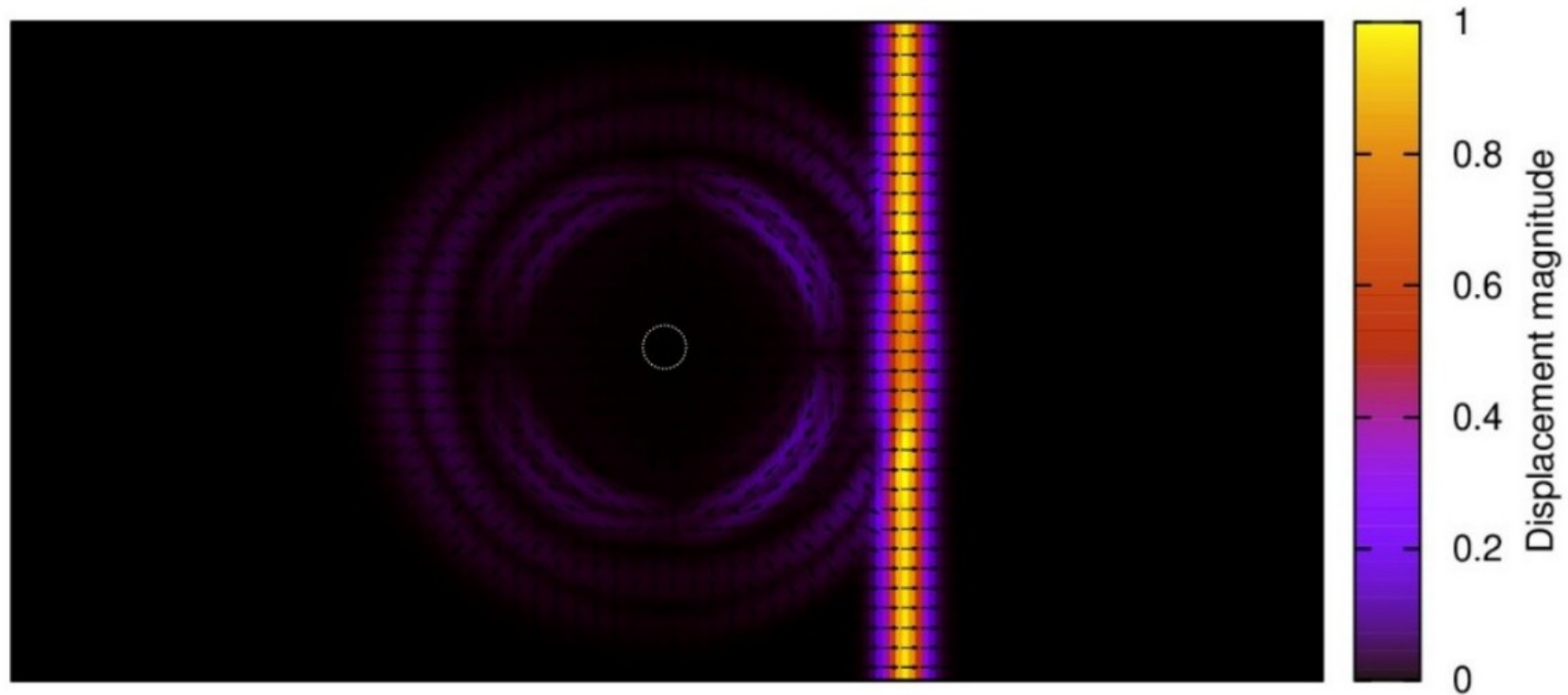
Quantum mechanics

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi$$

Classical mechanics

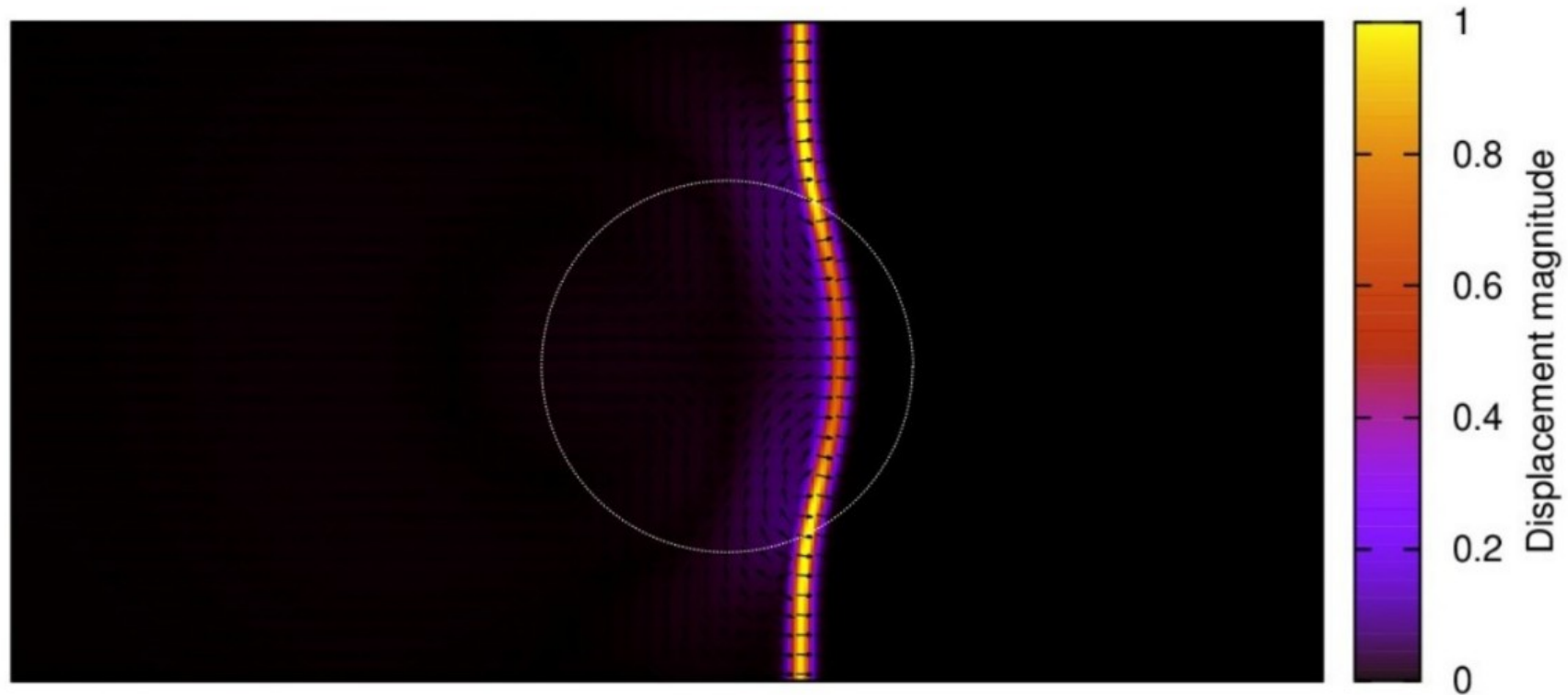
$$\begin{aligned} \frac{d\mathbf{q}}{dt} &= \frac{\partial H}{\partial \mathbf{p}} \\ \frac{d\mathbf{p}}{dt} &= -\frac{\partial H}{\partial \mathbf{q}} \end{aligned}$$

Length scales of heterogeneity



Scattering regime:

$$\frac{L}{\lambda} \approx 1$$

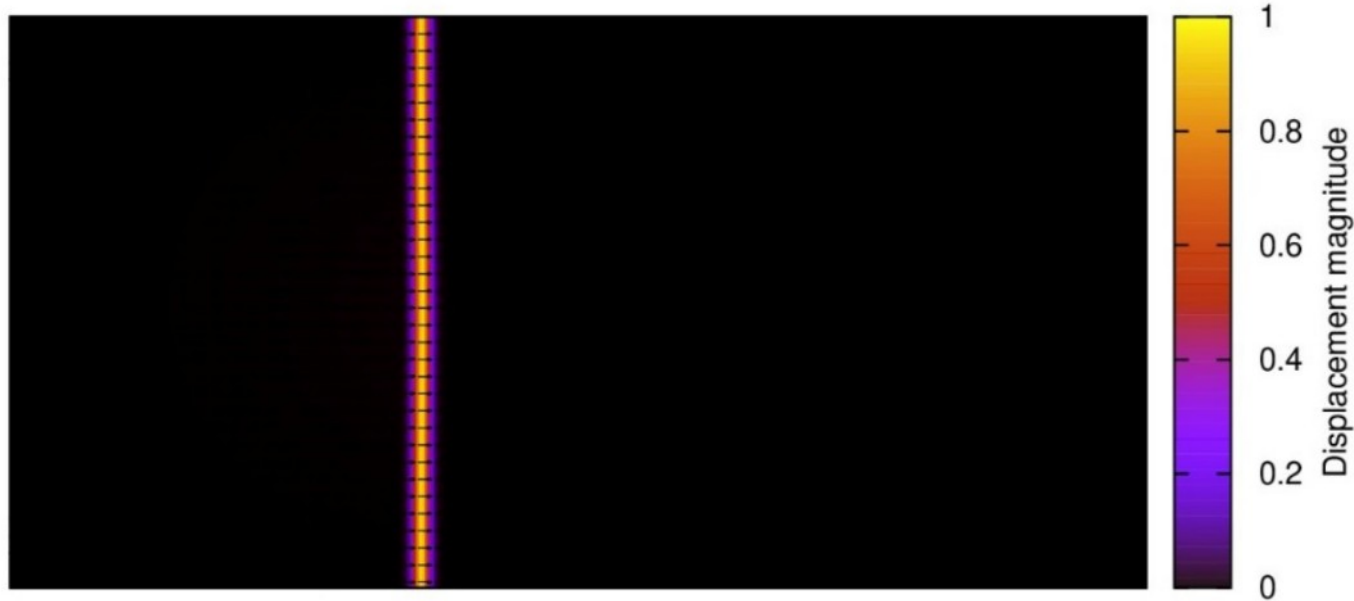


Geometric optics regime:

$$\frac{L}{\lambda} \gg 1$$

Ray series ansatz

Canonical problem

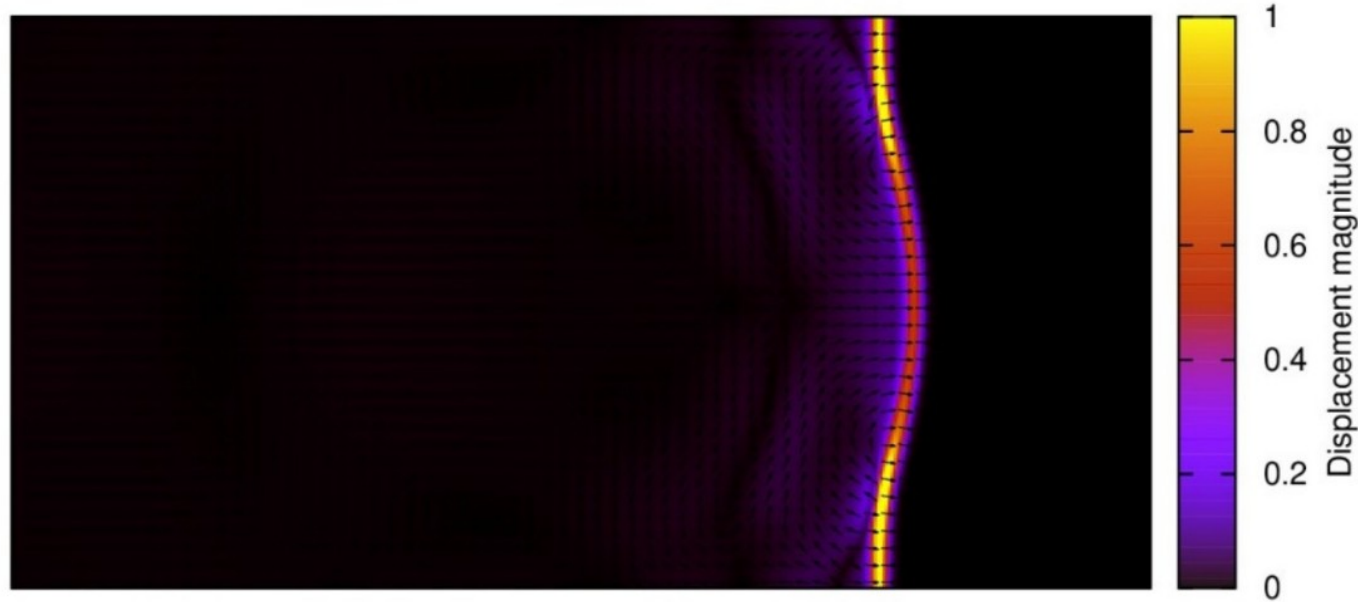


Start with a plane wave within the homogeneous left half plane:

$$u_i(x, t) = f(t - p_i^0 x_i) a_i^0$$

Can we extend this solution into the heterogeneous right half plane?

Initial ansatz



As an initial guess, we look for a solution of the form:

$$u_i(x, t) = f[t - T(x)]a_i(x)$$

Travel time and local slowness

Assumed solution takes the form:

$$u_i(x, t) = f[t - T(x)]a_i(x)$$

Introduction of the **travel time** function allows for curved wavefronts:

$$T(x) = \text{constant.}$$

The **local slowness vector** is orthogonal to the wavefronts:

$$p_i = \frac{\partial T}{\partial x_i}$$

Polarisation vector allowed to vary with position – accounts for **focusing effects** and for variations in anisotropic properties with position

The ray series

Harmonic decomposition solution:

$$u_i(x, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \tilde{f}(\omega) e^{i\omega[t-T(x)]} a_i(x) d\omega,$$

This is a convolution of f with the singular solution:

$$u_i(x, t) = \delta[t - T(x)] a_i(x)$$

Experience has shown that we should generalise the singular part to:

$$u_i(x, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{n=0}^{\infty} \frac{a_i^n(x)}{(i\omega)^n} e^{i\omega[t-T(x)]} d\omega,$$

First term ($n=0$) in the **ray series** is the singular one, while successive ones become smoother and smoother

The eikonal equation

Frequency-domain equations

Linearised equations with no force term:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0,$$

Transforming to the frequency-domain:

$$-\omega^2 \rho u_i - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

Ray series in the frequency-domain:

$$u_i(x, \omega) = \sum_{n=0}^{\infty} \frac{a_i^n(x)}{(i\omega)^n} e^{-i\omega T(x)}.$$

We can now substitute in our ansatz and see what happens!

Lots of differentiation...

Example step from the calculation:

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left(\frac{\partial a_k^n}{\partial x_l} - i\omega p_l a_k^n \right) e^{-i\omega T},$$

Factor of frequency is added each time we differentiate the phase term

Result of the calculation

Equations of motion imply:

$$\sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left[\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial a_k^n}{\partial x_l} \right) - i\omega \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^n) \right. \\ \left. - i\omega A_{ijkl} p_j \frac{\partial a_k^n}{\partial x_l} + (i\omega)^2 A_{ijkl} p_j p_l a_k^n - (i\omega)^2 \rho a_i^n \right] e^{-i\omega T} = 0.$$

Common phase term cancels

These equations must hold for **all frequencies** – this implies that the **coefficients of each power of frequency must separately vanish**

Compare with power series solution of ODEs

Results of the calculation

From the highest order coefficient, we have:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^0 = 0,$$

At the next order:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^1 = \frac{\partial}{\partial x_j}(A_{ijkl}p_la_k^0) + A_{ijkl}p_j\frac{\partial a_k^0}{\partial x_l},$$

For the general term with $n > 1$:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^n = \frac{\partial}{\partial x_j}(A_{ijkl}p_la_k^{n-1}) + A_{ijkl}p_j\frac{\partial a_k^{n-1}}{\partial x_l} - \frac{\partial}{\partial x_j}(A_{ijkl}\frac{\partial a_k^{n-2}}{\partial x_l})$$

First equation determines **p**, while the later ones successively determine the polarisation vectors at different orders

Where have we made an approximation?

A short cut

In this course we will consider only the equation derived from the **highest-order** term in the expansion

If that is all that we required, we can systematically drop all but the highest order terms within the previous derivation:

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left(\frac{\partial a_k^n}{\partial x_l} - i\omega p_l a_k^n \right) e^{-i\omega T} \approx -i\omega p_l a_k^0 e^{-i\omega T}$$

This saves a lot of time!

You can practice these steps in Example sheet 3

Highest order condition

From the highest order coefficient, we have:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^0 = 0,$$

This motivates the definition of a local Christoffel matrix:

$$\Gamma_{ik}(\mathbf{x}, \mathbf{p}) = \frac{1}{\rho(\mathbf{x})} A_{ijkl}(\mathbf{x}) p_j p_l.$$

And so write the above condition as:

$$\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} \mathbf{a}^0(\mathbf{x}) = \mathbf{0},$$

Locally, things look identical to plane waves!

The eikonal equation

The equation

$$\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} \mathbf{a}^0(\mathbf{x}) = \mathbf{0},$$

has non-trivial solutions for the zeroth-order polarisation vector if and only if:

$$\det\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} = 0,$$

This is a non-linear first order PDE for the travel time called the **eikonal equation**

If the local slowness surfaces are everywhere non-degenerate:

$$||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0$$

The method of characteristic

Ray Hamiltonian

Recall the eikonal equation is:

$$||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0$$

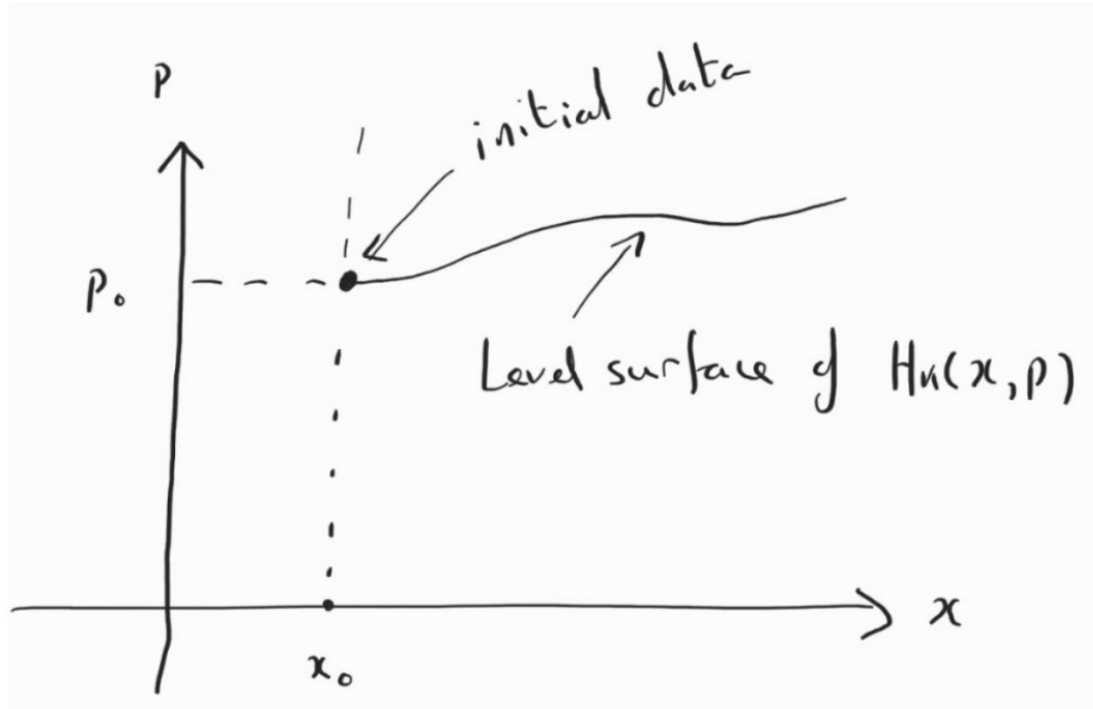
If we defined the **ray Hamiltonian** for the k th sheet of the slowness surface:

$$H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2} ||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2,$$

We can write the eikonal equation in the form:

$$H_k[\mathbf{x}, \mathbf{p}(\mathbf{x})] = \frac{1}{2}, \quad \mathbf{p}(\mathbf{x}) = (\nabla T)(\mathbf{x}).$$

Geometric interpretation



Solution of eikonal equation is a 3D level surface of the Hamiltonian in a 6D phase space

Initial data specifies a 2D level surface in the phase space

The problem is how to extend this 2D surface into the appropriate 3D level surface

This is where Hamilton's equations enter into the method

Hamilton's equations

Initial data extended through integration of Hamilton's equations:

$$\frac{dx_i}{d\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{dp_i}{d\sigma} = -\frac{\partial H_k}{\partial x_i}$$
$$x_i(0) = x_i^0, \quad p_i(0) = p_i^0.$$

This works because the Hamiltonian is constant along the integral curves:

$$\frac{d}{d\sigma} H[\mathbf{x}(\sigma), \mathbf{p}(\sigma)] = \frac{\partial H}{\partial x_i} \frac{dx_i}{d\sigma} + \frac{\partial H}{\partial p_i} \frac{dp_i}{d\sigma} = -\frac{dp_i}{d\sigma} \frac{dx_i}{d\sigma} + \frac{dx_i}{d\sigma} \frac{dp_i}{d\sigma} = 0,$$

There are a few subtler points to the method that are being ignored

Recovering the travel time

Rays are the projection of integral curves into the **position-domain**

Rate of change of the travel time along a ray:

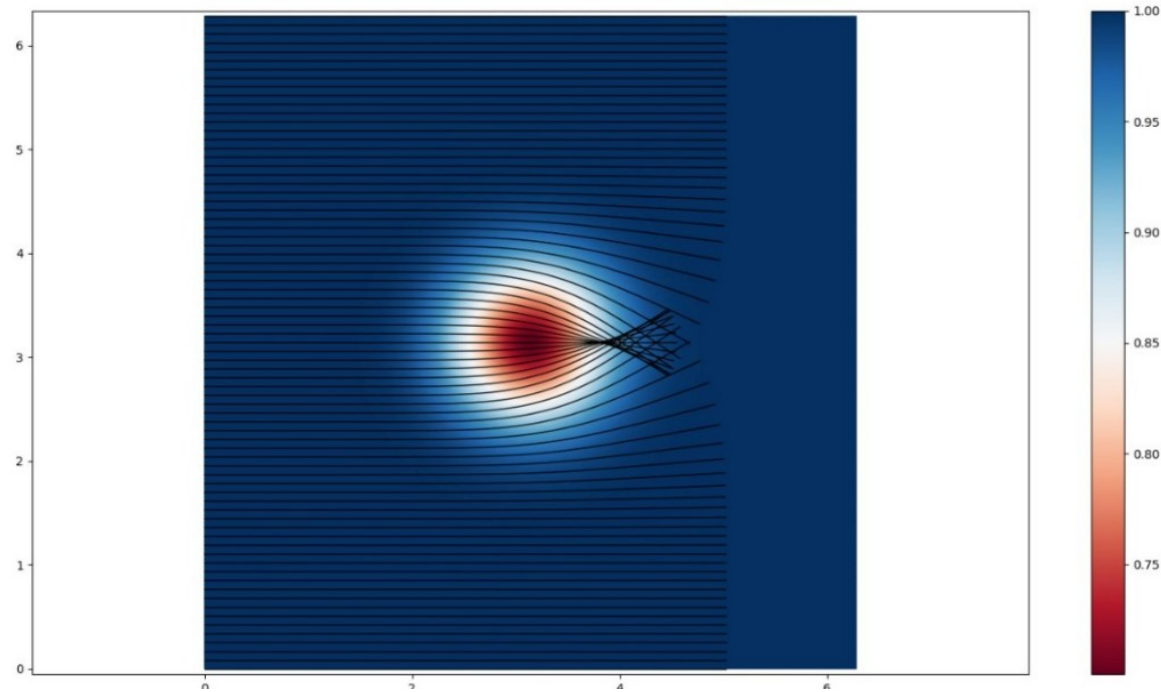
$$\frac{d}{d\sigma} T[\mathbf{x}(\sigma)] = \frac{\partial T}{\partial x_i} \frac{dx_i}{d\sigma} = p_i \frac{\partial H}{\partial p_i}$$

Right hand side simplified using Euler's homogeneous function theorem;

$$\frac{d}{d\sigma} T[\mathbf{x}(\sigma)] = 1.$$

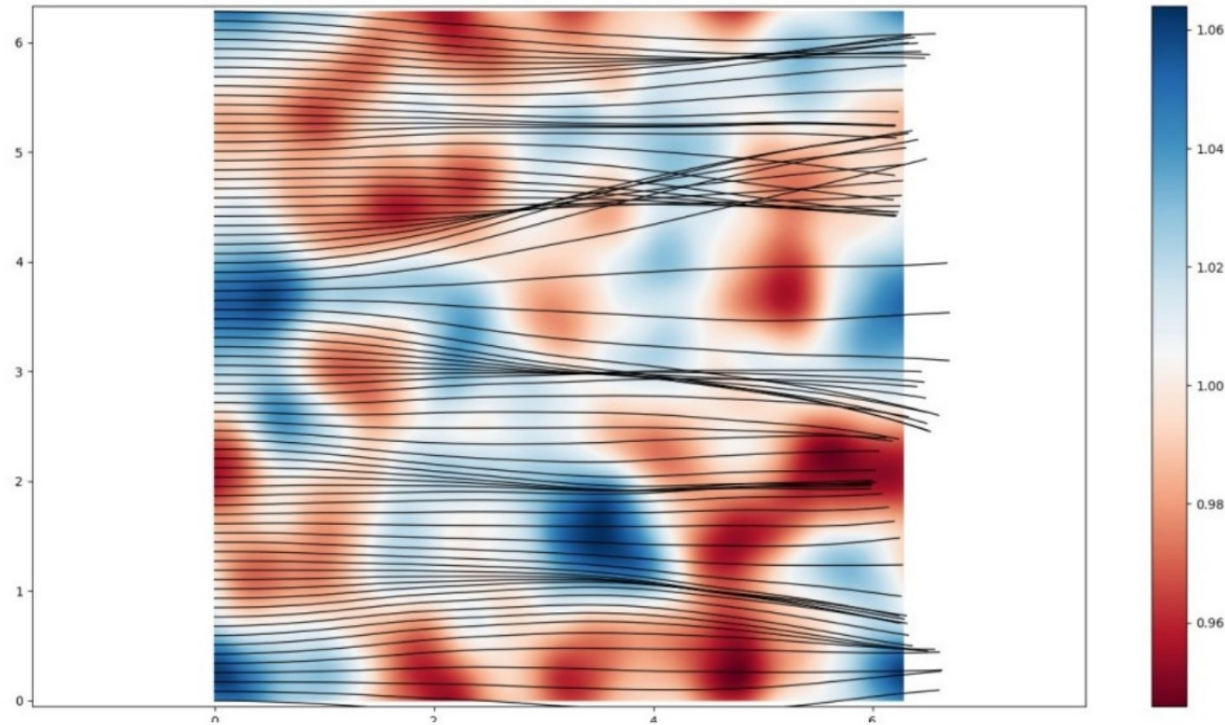
The generating parameter along the ray is equal to the travel time

Caustics



If rays for two different integral curves touch, then the travel time need not be well-defined. These points are known as **caustics**. The **ray theoretic amplitude is infinite** at such points, while the true amplitude is large but finite

Caustics



Within heterogeneous materials, caustics occur frequently – c.f. Penrose Hawking singularity theorems in general relativity

The spherical Earth

From last lecture

Ray Hamiltonian defined by:

$$H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2} \|\mathbf{p}(\mathbf{x})\|^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2,$$

Eikonal equation:

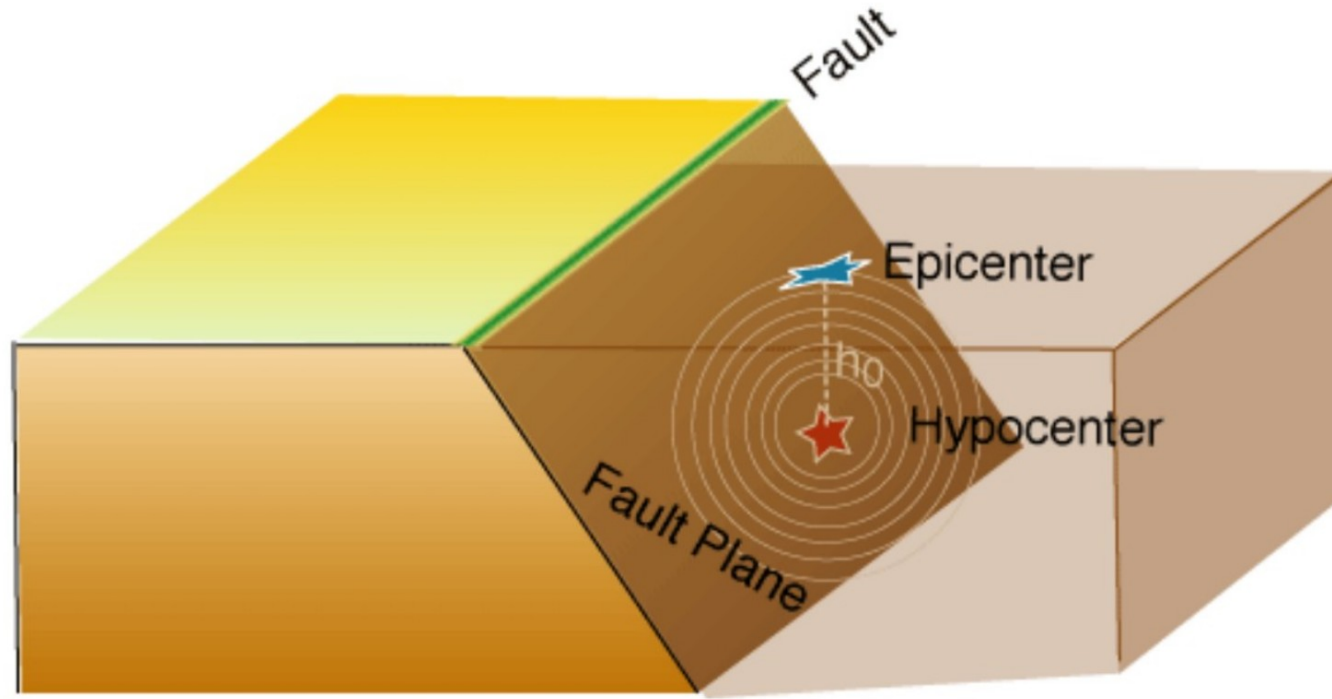
$$H_k[\mathbf{x}, \mathbf{p}(\mathbf{x})] = \frac{1}{2}, \quad \mathbf{p}(\mathbf{x}) = (\nabla T)(\mathbf{x}).$$

Solved through integration of Hamilton's equations:

$$\frac{dx_i}{d\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{dp_i}{d\sigma} = -\frac{\partial H_k}{\partial x_i}$$

Earthquake location

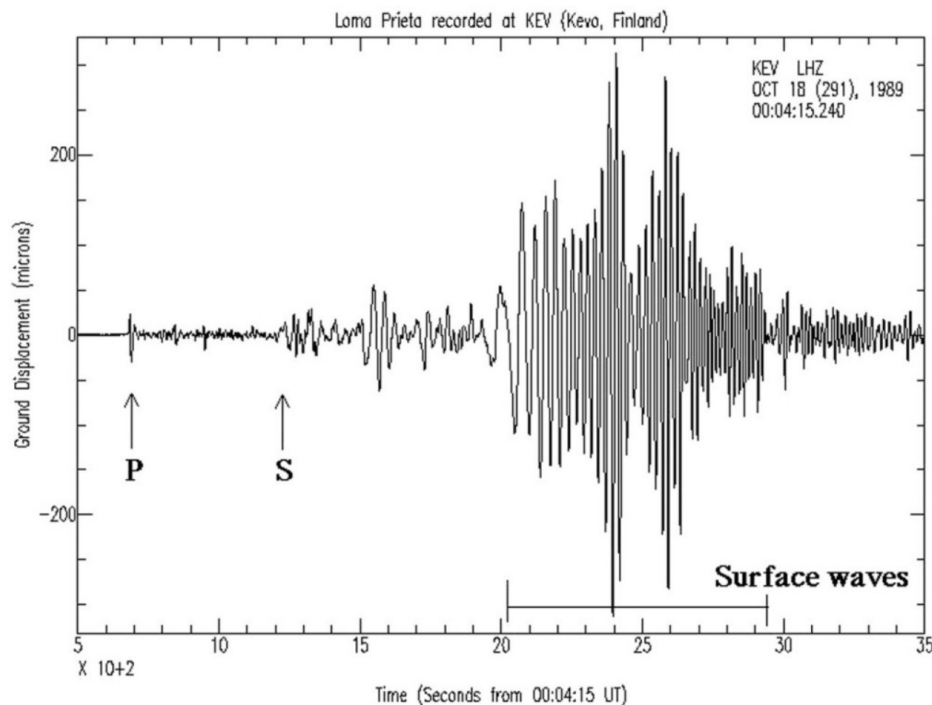
Hypocentre location



The **hypocentre** is where the rupture initiates

It is the origin of the **first waves** recorded on seismometers

Arrival time measurements



The **arrival time** is the actual time (e.g., noon on Wednesday) at which a seismic phase arrives at a seismometer. Measured from **high-frequency** recordings to allow for precise measurements

Forward modelling arrival times

Arrival time at the i th seismometer for a given phase takes the form:

$$t_i = t_h + T(\mathbf{x}_i, \mathbf{x}_h) + e_i$$

Travel time can be calculated **exactly** using ray theory. But to do this, we need to know the Earth's velocity structure

Forward problem depends on just four model parameters:

- Three components of the hypocentre position
- The hypocentral time

Inversion of arrival times

Misfit between data and proposed model parameters can be quantified:

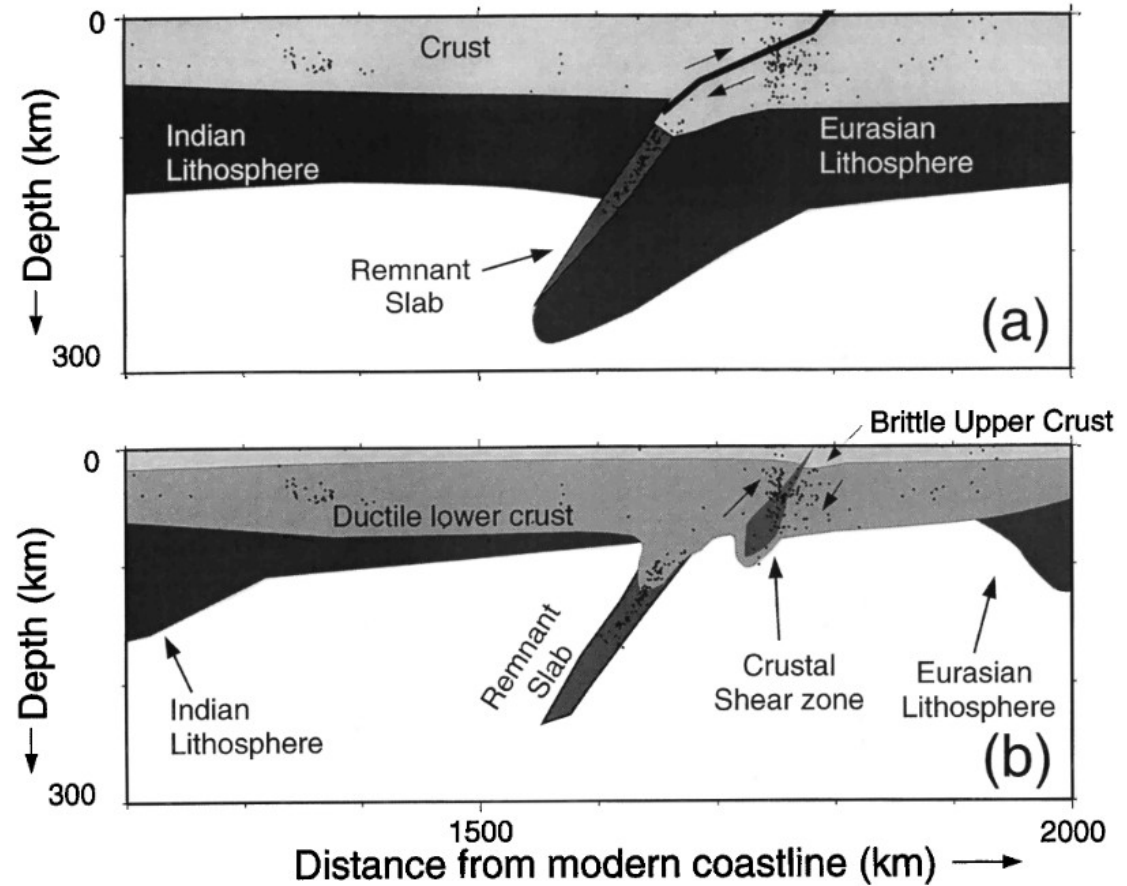
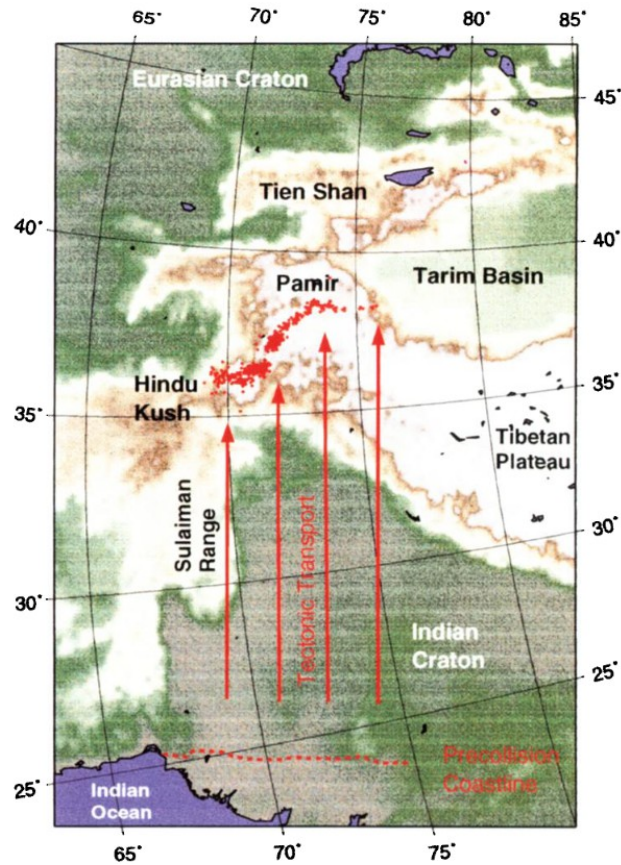
$$J(\mathbf{x}_h, t_h) = \sum_{i=1}^{N_s} \frac{[t_h + T(\mathbf{x}_i, \mathbf{x}_h) - t_i]^2}{\sigma_i^2}$$

Optimise J with respect to the four model parameters to find a solution

This can be done iteratively using gradient-based methods or through a brute-force grid search

Uncertainties can be estimated by considering a range of model parameters that fit the data sufficiently well

Accuracy ultimately limited by the **dominant wavelength** of observed waves



Application of hypocentral locations to test competing tectonic models for the formation of the Pamir-Hindu Kush mountains (from Palvis & Das 2000)

Rays in spherical Earth models

P-wave eikonal equation

We focus on an isotropic model, and look at P-waves for definiteness

Eikonal equation takes the form:

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2},$$

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2} \alpha(\mathbf{x})^2 \|\mathbf{p}\|^2,$$

We assume further that the P-wave velocity **varies only with radius**

Hamilton's equations

General form of Hamilton's equations:

$$\frac{d\mathbf{x}}{d\sigma} = \frac{\partial H}{\partial \mathbf{p}}, \quad \frac{d\mathbf{p}}{d\sigma} = -\frac{\partial H}{\partial \mathbf{x}},$$

For the spherically symmetric P-wave problem, they become:

$$\frac{d\mathbf{x}}{d\sigma} = \alpha \hat{\mathbf{p}}, \quad \frac{d\mathbf{p}}{d\sigma} = -\frac{1}{\alpha} \frac{d\alpha}{dr} \hat{\mathbf{x}},$$

Here hats denote unit vectors

A conserved quantity

We define the vector:

$$\mathbf{q} = \mathbf{x} \times \mathbf{p}$$

Using Hamilton's equations, we find that:

$$\frac{d}{d\sigma}(\mathbf{x} \times \mathbf{p}) = \frac{d\mathbf{x}}{d\sigma} \times \mathbf{p} + \mathbf{x} \times \frac{d\mathbf{p}}{d\sigma} = \alpha \hat{\mathbf{p}} \times \mathbf{p} - \frac{1}{\alpha} \frac{d\alpha}{dr} \mathbf{x} \times \hat{\mathbf{x}} = \mathbf{0}.$$

The form of this conserved quantity follows from the rotational symmetry via Noether's theorem (but you don't have to know this!)

Reduction of the problem

Conservation of

$$\mathbf{q} = \mathbf{x} \times \mathbf{p}$$

Implies that the ray stays in the plane spanned by initial values of $\mathbf{x}$ and

Dynamics **reduced to a plane** – c.f. two body problem

Magnitude of the conserved quantity is:

$$q = \frac{r \sin \theta}{\alpha}$$

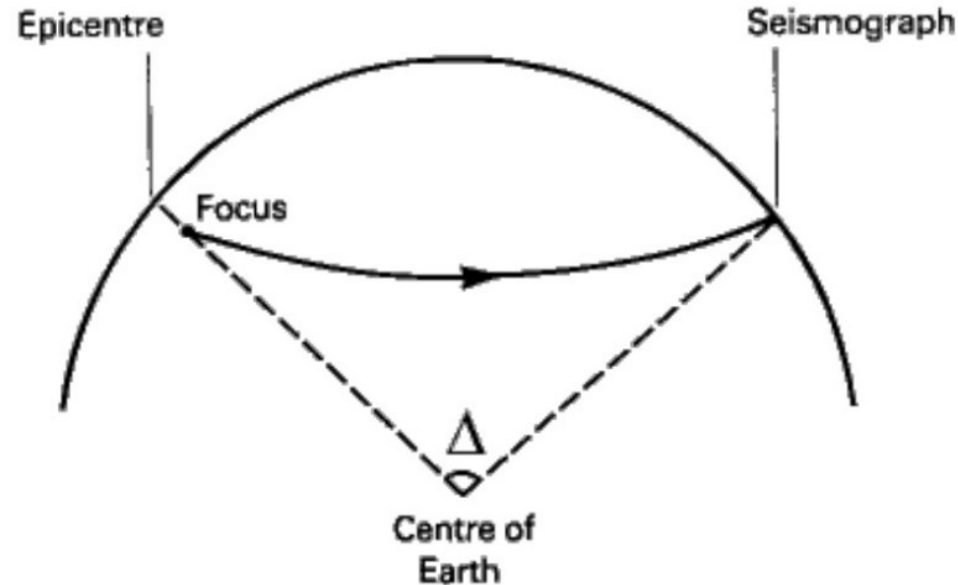
where the angle, theta, is that between $\mathbf{x}$ and $\mathbf{p}$

This is **Snell's law** expressed in spherical geometry

Reduction of the problem

Along the ray, we can write:

$$\hat{\mathbf{p}} = \cos[\theta(\mathbf{x})]\hat{\mathbf{x}} + \sin[\theta(\mathbf{x})]\hat{\Delta}(\mathbf{x}),$$



Reduction of the problem

From the conserved quantity, we can write:

$$\sin \theta = \frac{\alpha q}{r}, \quad \cos \theta = \pm \sqrt{1 - \frac{\alpha^2 q^2}{r^2}},$$

showing that the ray's direction is fixed by its radius. Plus or minus sign in the cosine term corresponds to up- or down-going rays

Using these results in Hamilton's equations, we arrive at:

$$\frac{d\mathbf{x}}{d\sigma} = -\alpha \sqrt{1 - \frac{\alpha^2 q^2}{r^2}} \hat{\mathbf{x}} + \frac{\alpha^2 q}{r} \hat{\Delta}.$$

The slowness vector has now been eliminated from the problem

Solving the reduced problem

Using the reduced ODE, we find that

$$\frac{d}{d\sigma} r[\mathbf{x}(\sigma)] = \frac{\partial r}{\partial x_i} \frac{dx_i}{d\sigma} = -\alpha \sqrt{1 - \frac{\alpha^2 q^2}{r^2}},$$

$$\frac{d}{d\sigma} \Delta[\mathbf{x}(\sigma)] = \frac{\partial \Delta}{\partial x_i} \frac{dx_i}{d\sigma} = \frac{\alpha^2 q}{r^2},$$

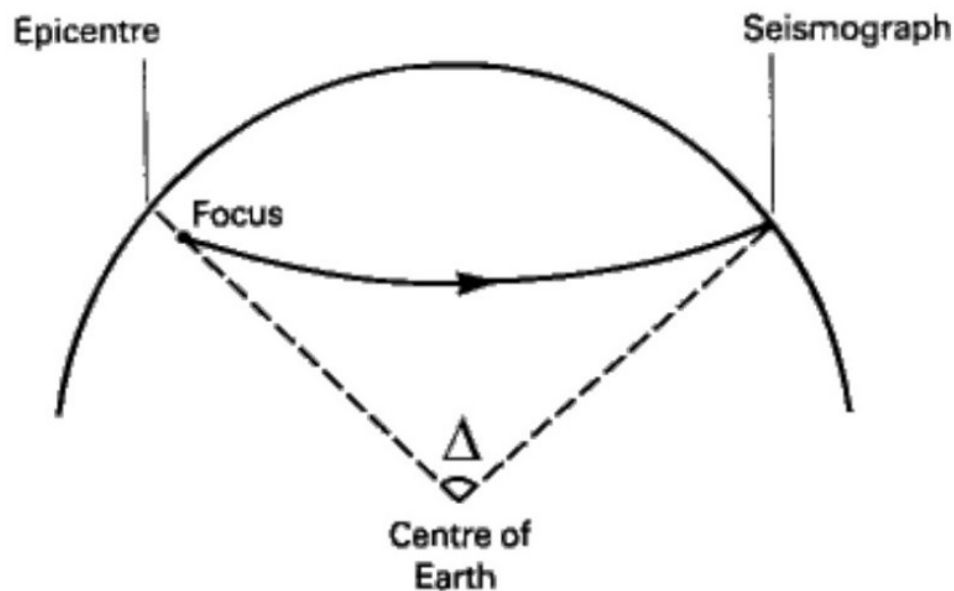
And hence, we have:

$$\frac{d\Delta}{dr} = \frac{d\Delta}{d\sigma} \left(\frac{dr}{d\sigma} \right)^{-1} = \frac{-\alpha q}{r \sqrt{r^2 - \alpha^2 q^2}}.$$

Solving the reduced problem

In a similar manner, we find:

$$\frac{dT}{dr} = \frac{dT}{d\sigma} \left(\frac{dr}{d\sigma} \right)^{-1} = \frac{-r}{\alpha \sqrt{r^2 - \alpha^2 q^2}}.$$



Solving the reduced problem

Assuming velocity decreases monotonically with radius, the ray reaches a **turning point** where it is travelling horizontally at:

$$q = \frac{r_t}{\alpha(r_t)}$$

Afterwards, it returns to the surface along a symmetric path

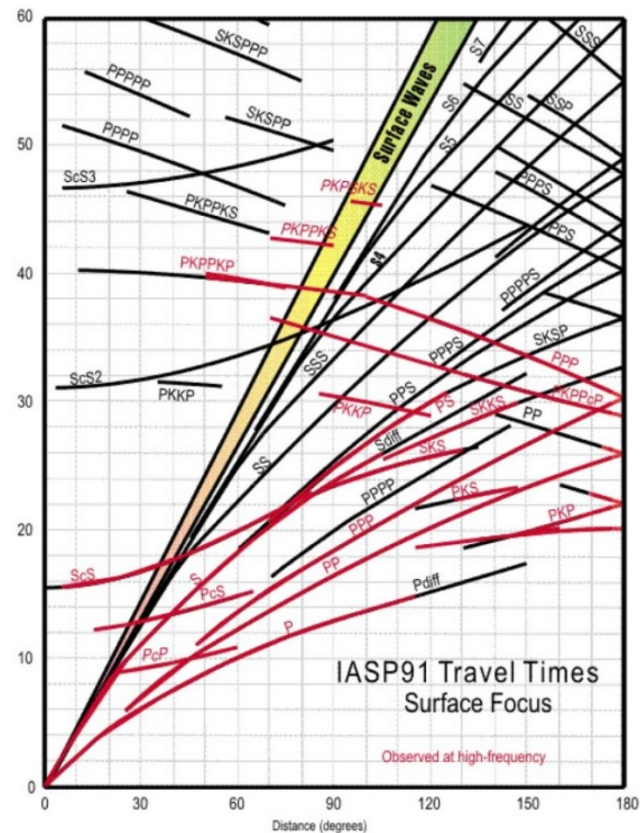
Angle and travel time expressed through the integrals:

$$\Delta = \int_{r_t}^a \frac{2\alpha(r)q}{r\sqrt{r^2 - \alpha(r)^2q^2}} dr; \quad T = \int_{r_t}^a \frac{2r}{\alpha(r)\sqrt{r^2 - \alpha(r)^2q^2}} dr.$$

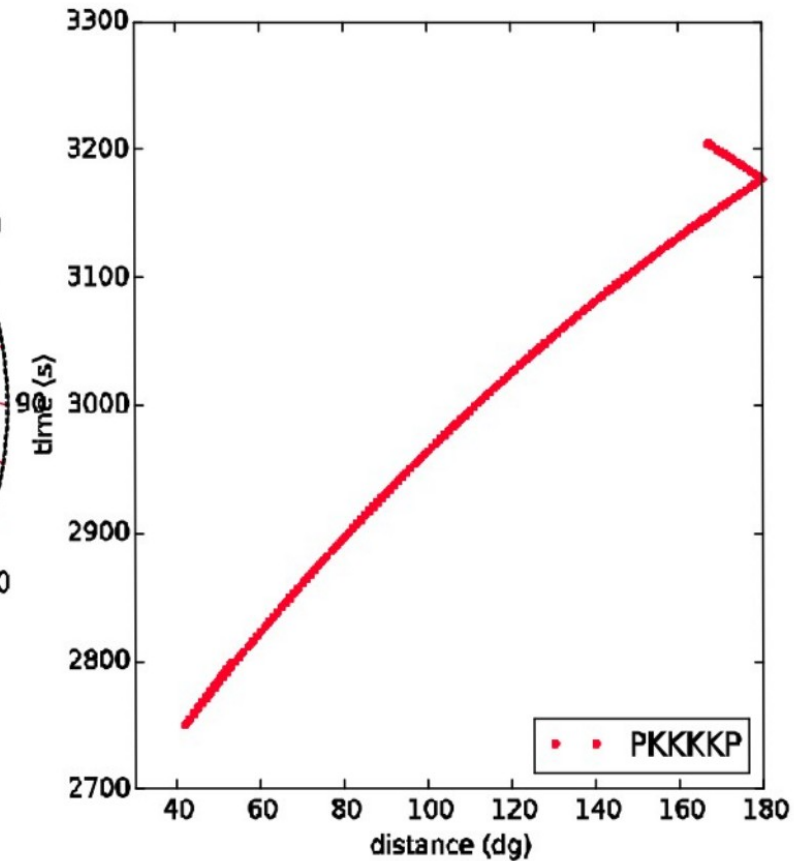
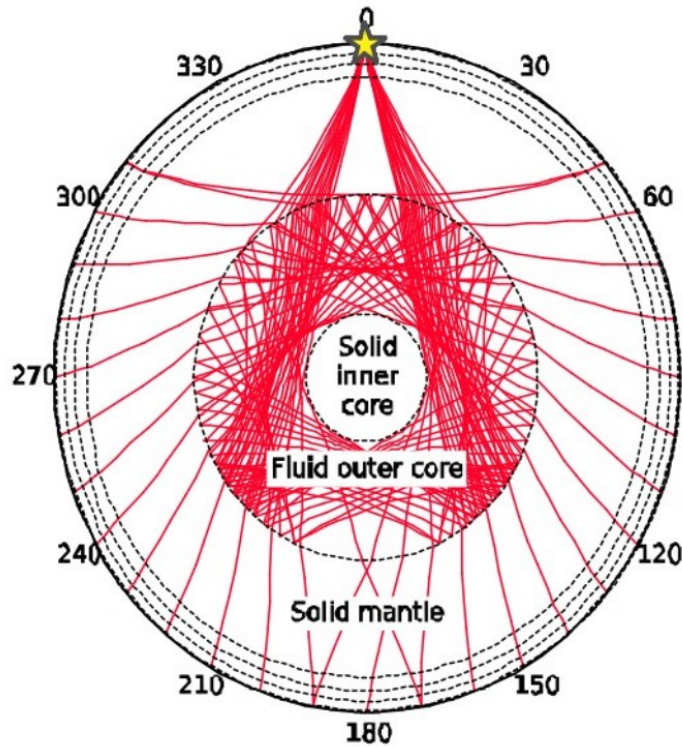
These expressions could be evaluated in the 20s and 30s using **mechanical calculating machines**

Spherical velocity structure

IRIS



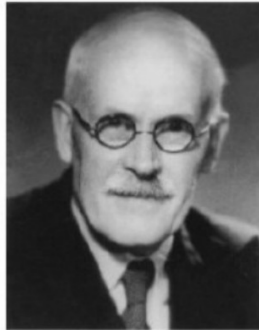
Rat theoretical predictions



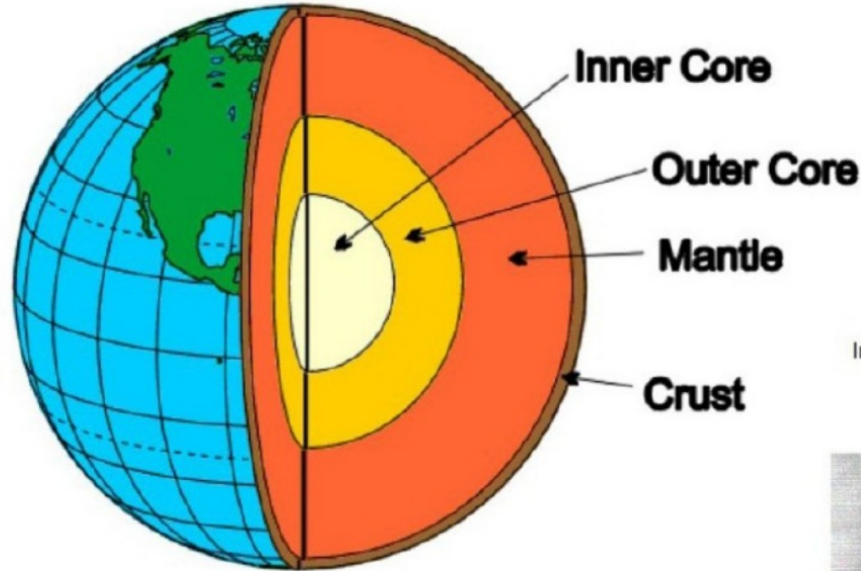
Early pioneers



Richard Oldham 1858 – 1936



Harold Jeffreys 1891 – 1989



Of all regions of the earth none invites speculation more than that which lies beneath our feet, and in none is speculation more dangerous; yet, apart from speculation, it is little that we can say regarding the constitution of the interior of the earth ... Many theories of the earth have been propounded at different times: the central substance of the earth has been supposed to be fiery, fluid, solid, and gaseous in turn, till geologists have turned in despair from the subject, and become inclined to confine their attention to the outermost crust of the earth, leaving its centre as a playground for mathematicians.

Richard Oldham 1906

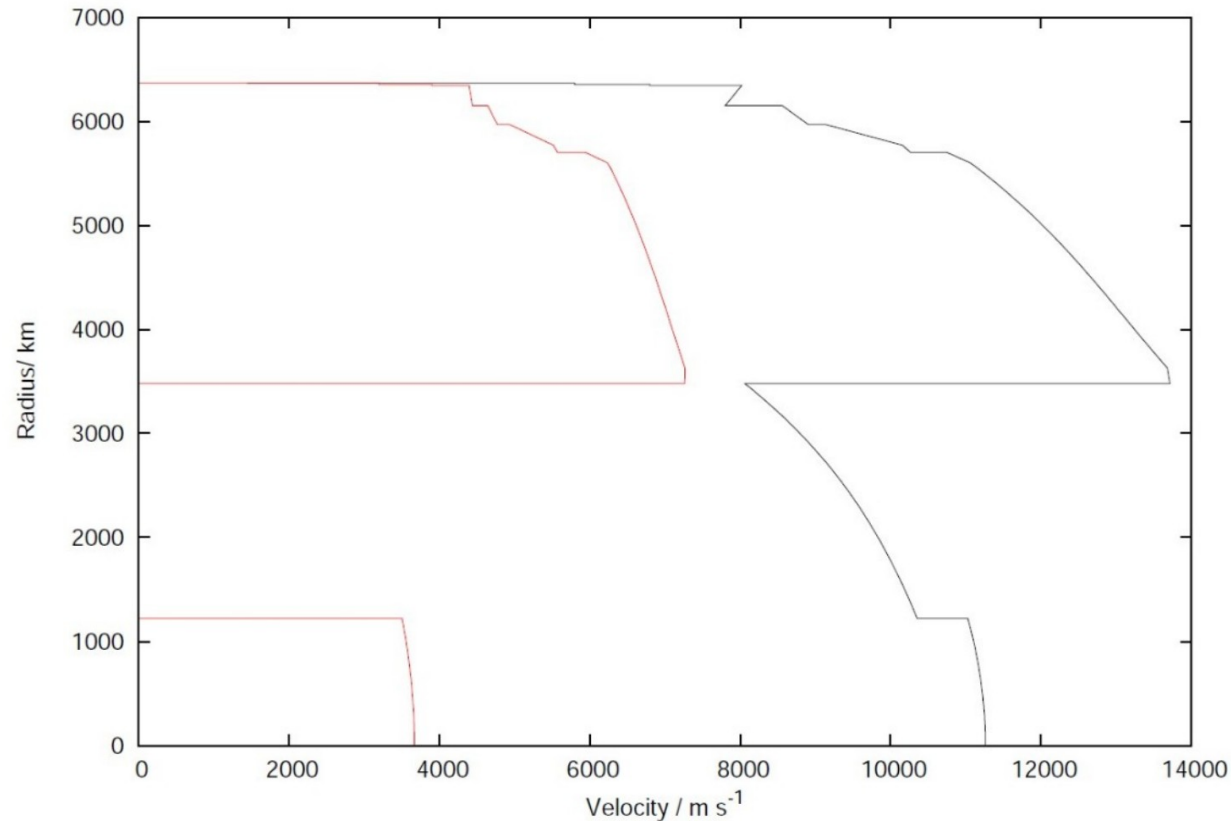


Inge Lehmann 1888 – 1993

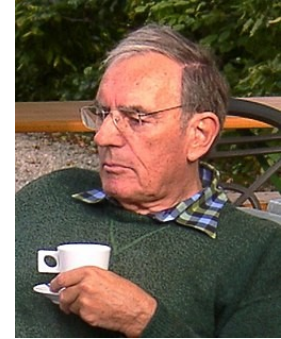


Beno Gutenberg 1889 – 1960

Spherical velocity structure



Preliminary Reference Earth Model (PREM) of Dziewonski & Anderson (1981)

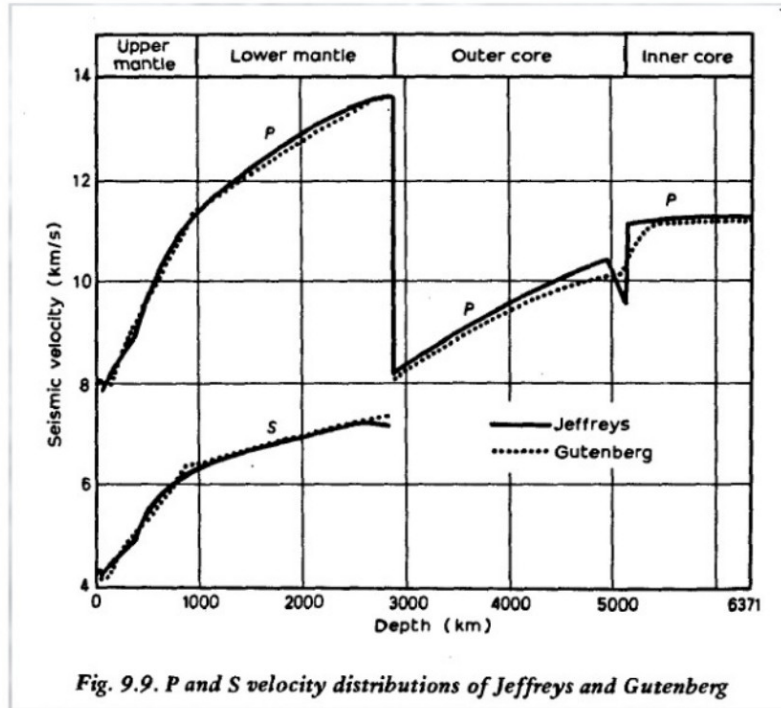


Adam Dziewonski (1936-2016)

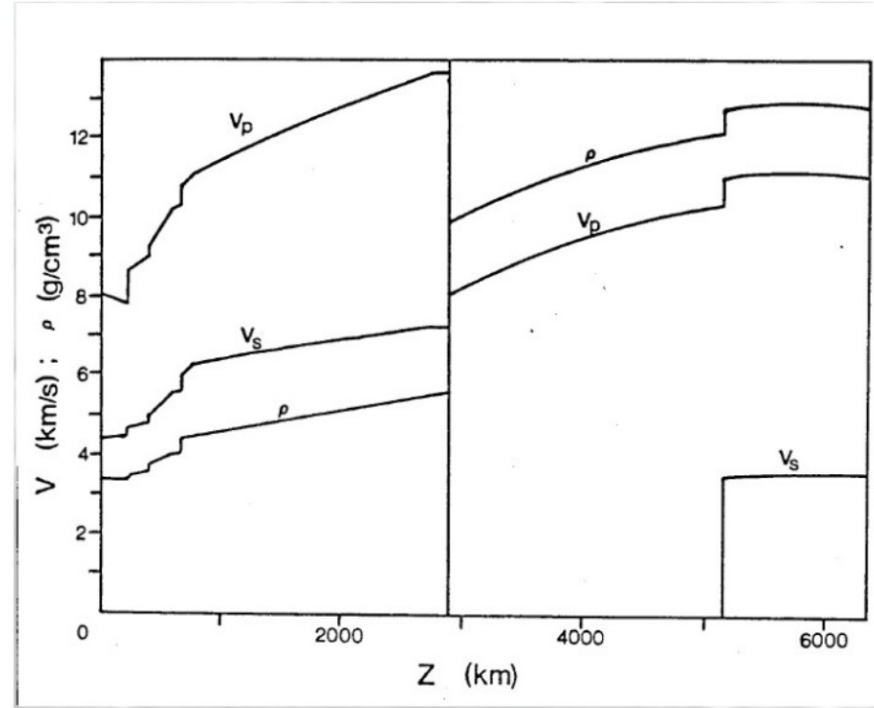


Don Anderson (1933-2014)

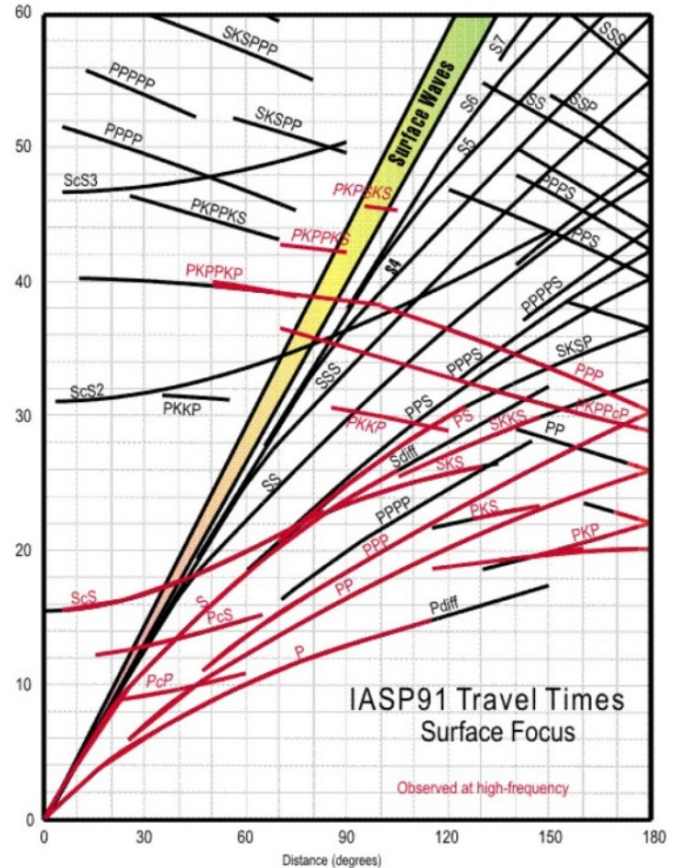
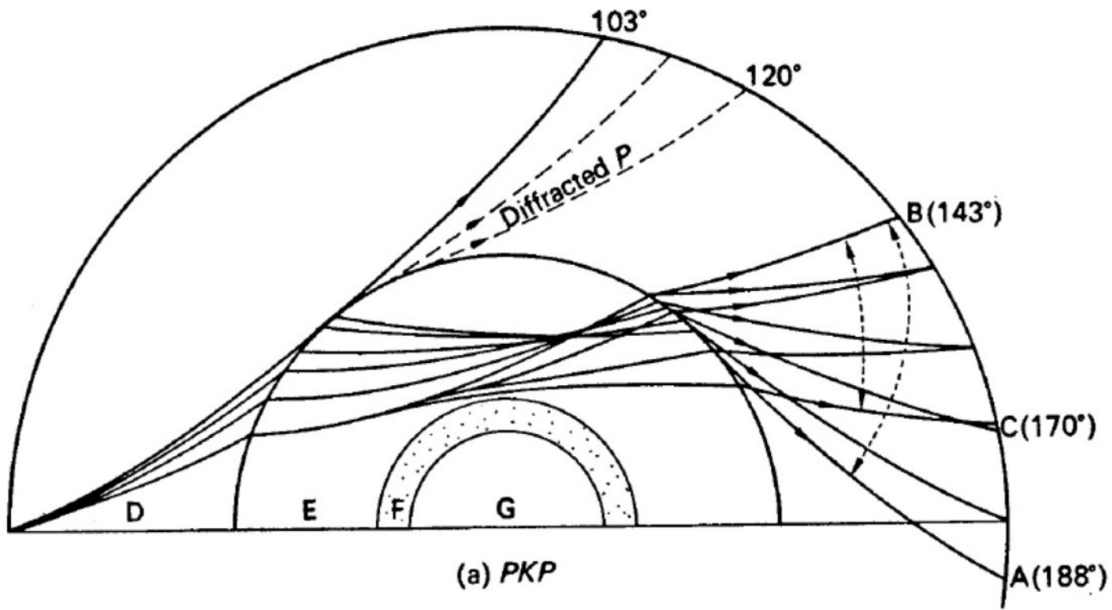
Comparison with earlier studies



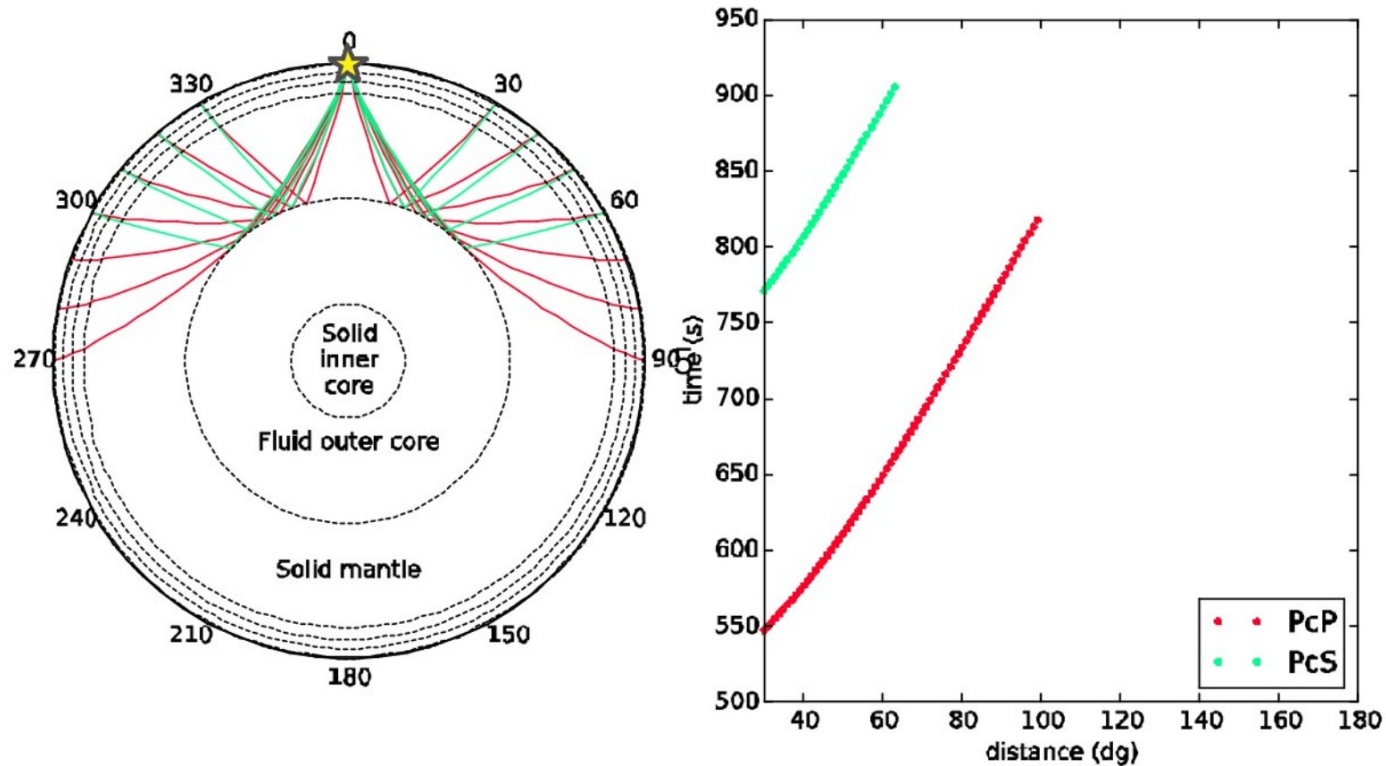
Models from the 1940s



Existence of the Earth's core

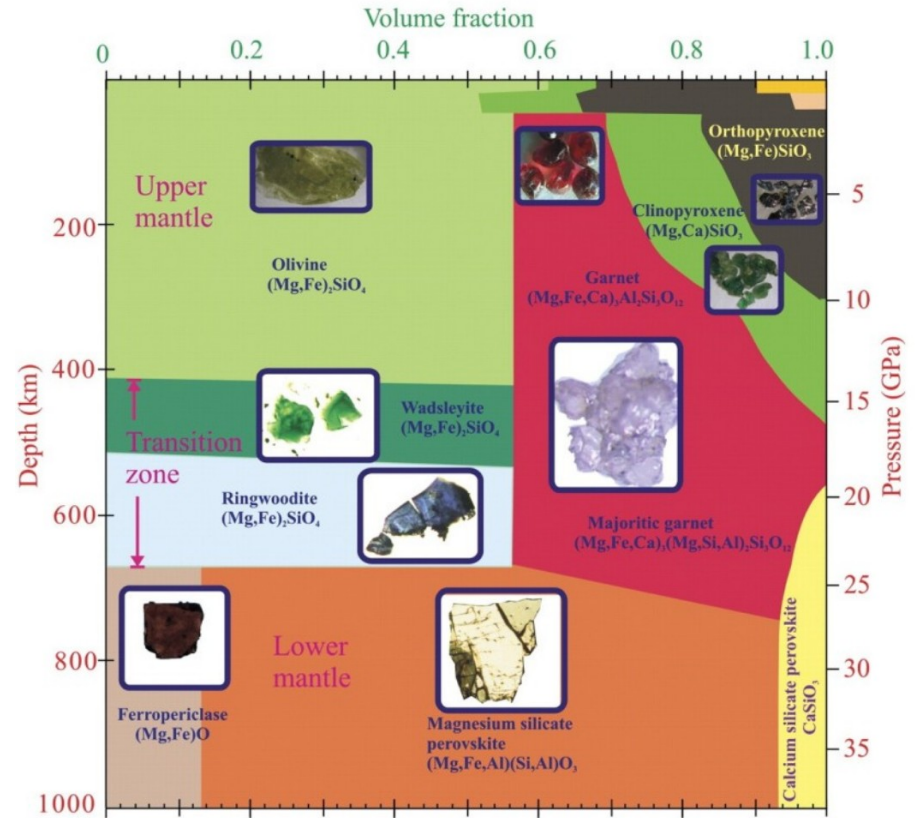
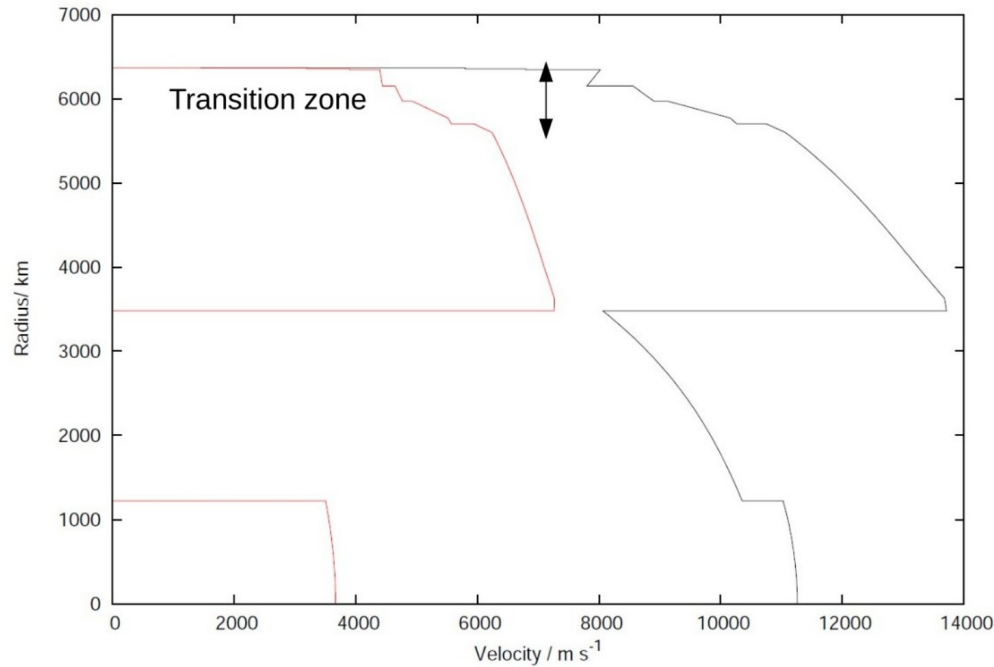


Determining internal boundaries



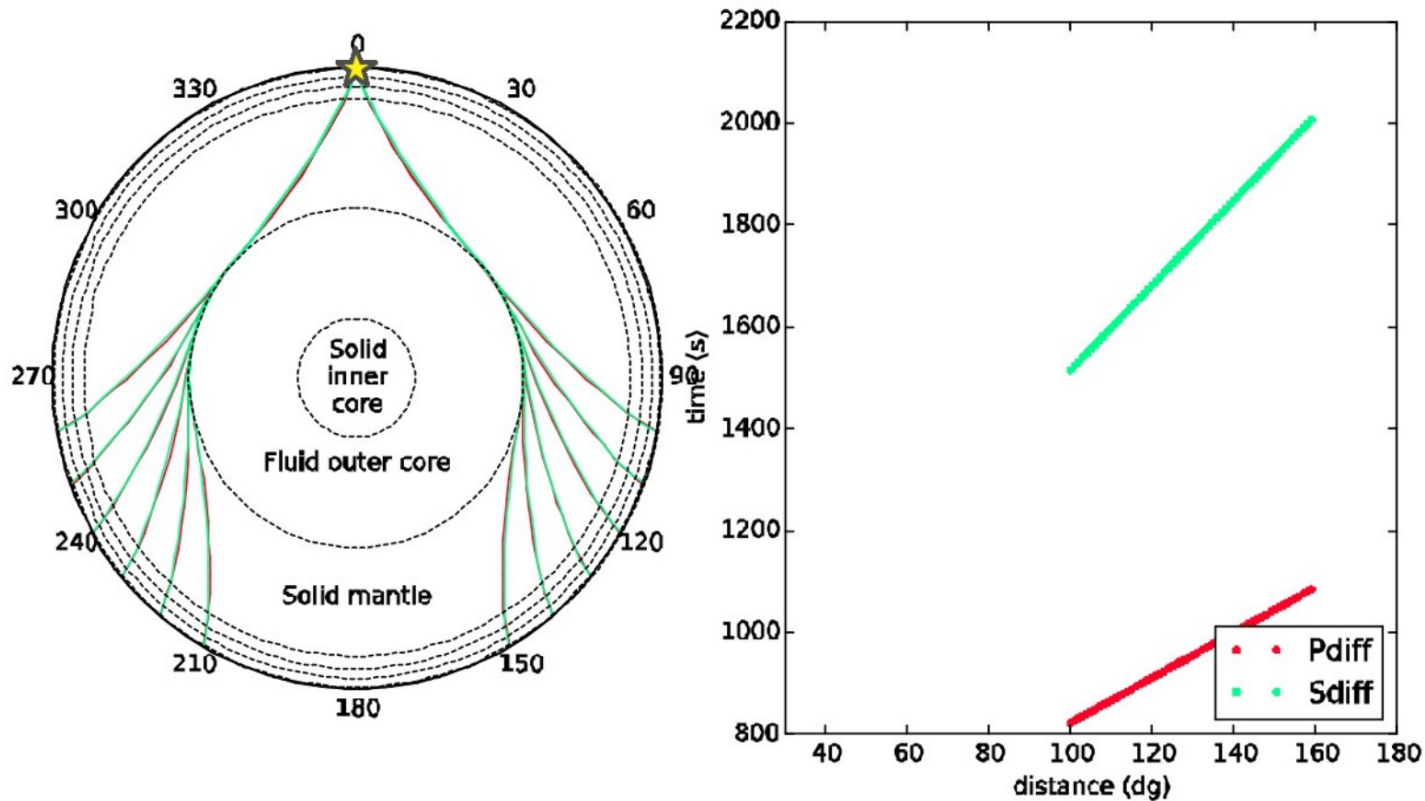
Constraints on boundary locations come largely from reflected phases

The transition zone



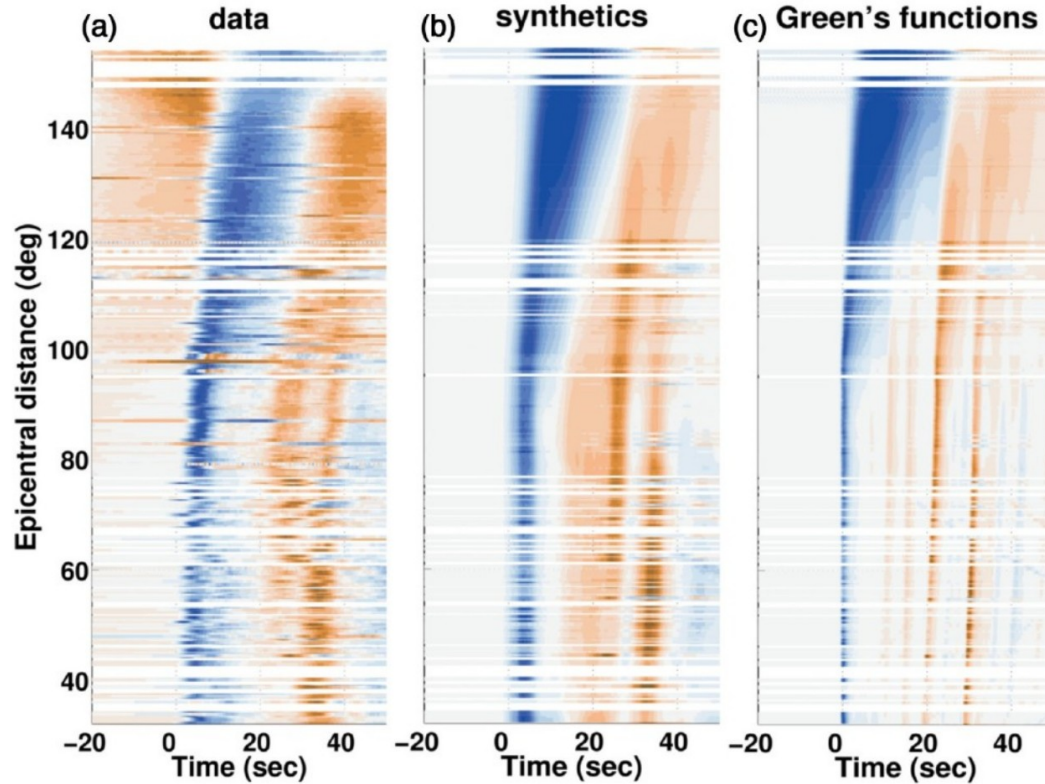
Jumps in velocity within the mid mantle believed to represent **solid-state phase changes** in silicate minerals – you don't need to know any details!

Non-geometric phases



There are also core diffracted phases that are not predicted by ray theory

Non-geometric phases



Observations and synthetics for core diffracted p-waves (normalised amplitudes)



Feedback form for the course

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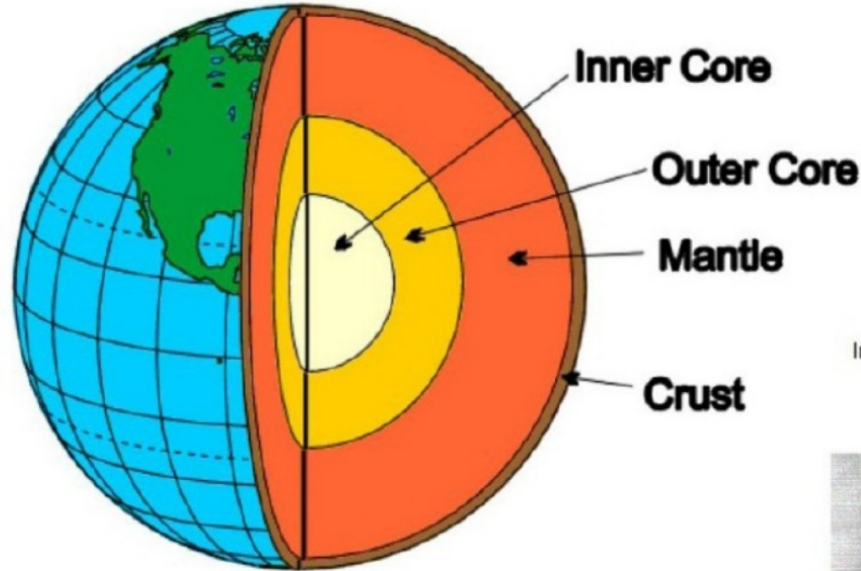
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Delay time tomography

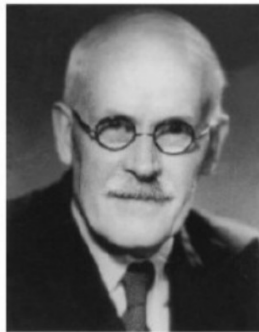
From last lecture



Richard Oldham 1858 – 1936



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Of all regions of the earth none invites speculation more than that which lies beneath our feet, and in none is speculation more dangerous; yet, apart from speculation, it is little that we can say regarding the constitution of the interior of the earth ... Many theories of the earth have been propounded at different times: the central substance of the earth has been supposed to be fiery, fluid, solid, and gaseous in turn, till geologists have turned in despair from the subject, and become inclined to confine their attention to the outermost crust of the earth, leaving its centre as a playground for mathematicians.

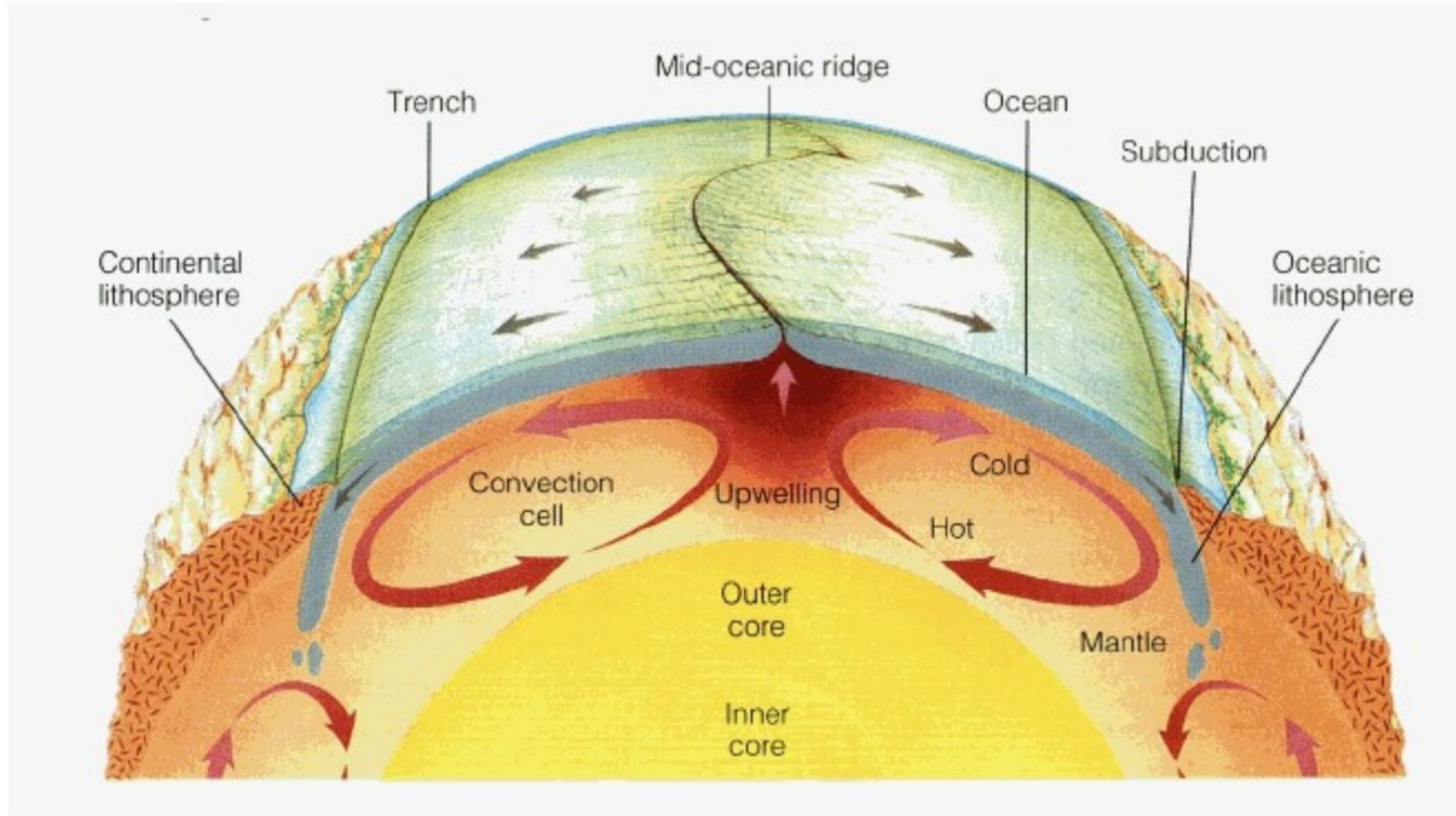
Richard Oldham 1906



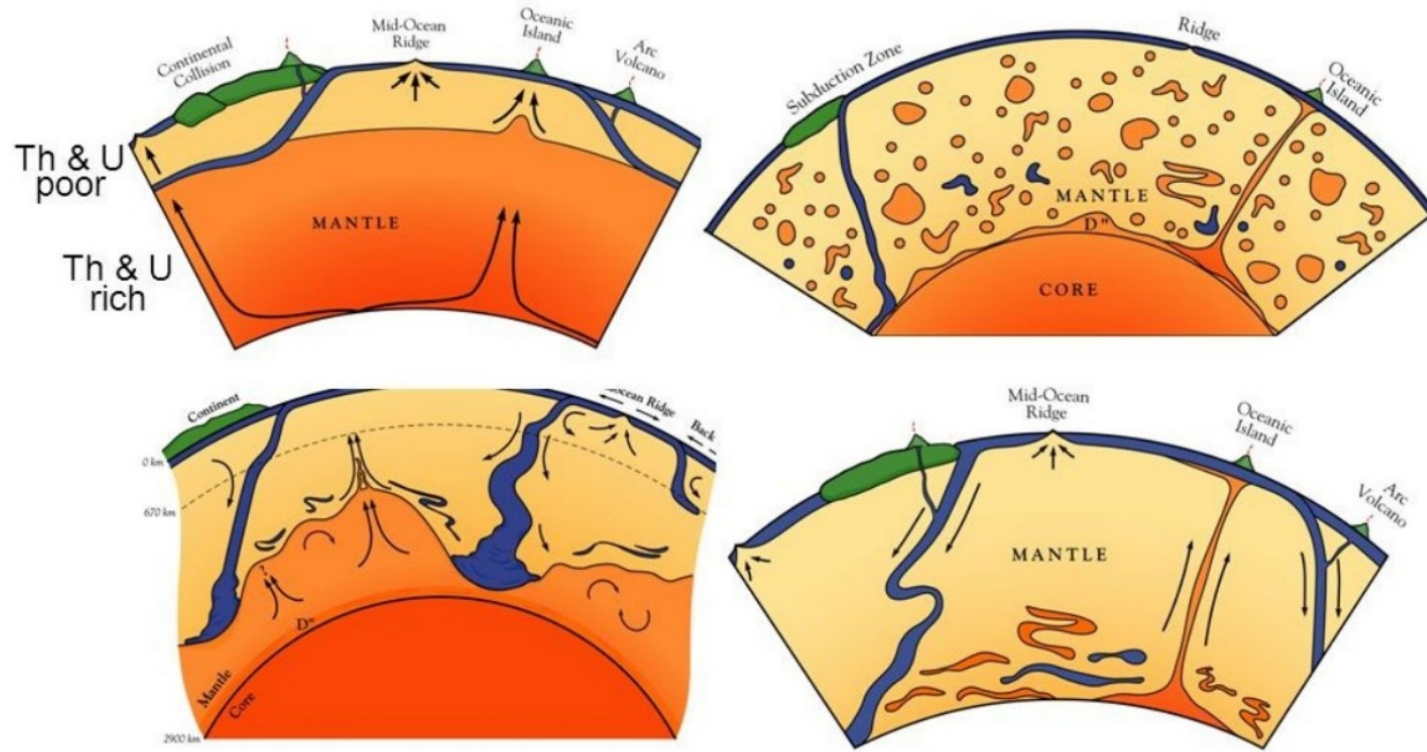
Beno Gutenberg 1889 – 1960

Lateral velocity variations

Mantle convection

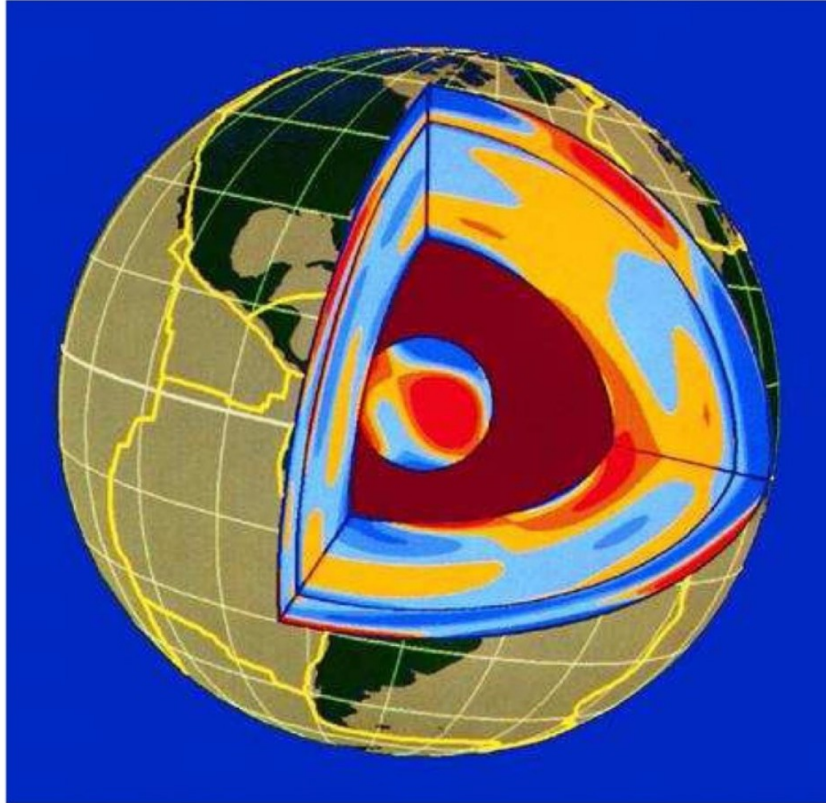


Mantle convection



Can we do better than drawing cartoons?

Seismic tomography



Dziewonski & Woodhouse (1989)

Quantitative models for lateral variations within the Earth derived from seismological observations

Most models show perturbations in elastic wave speeds relative to a spherically symmetric reference model

Other parameters, (e.g., density, attenuation can be sought with more difficulty)

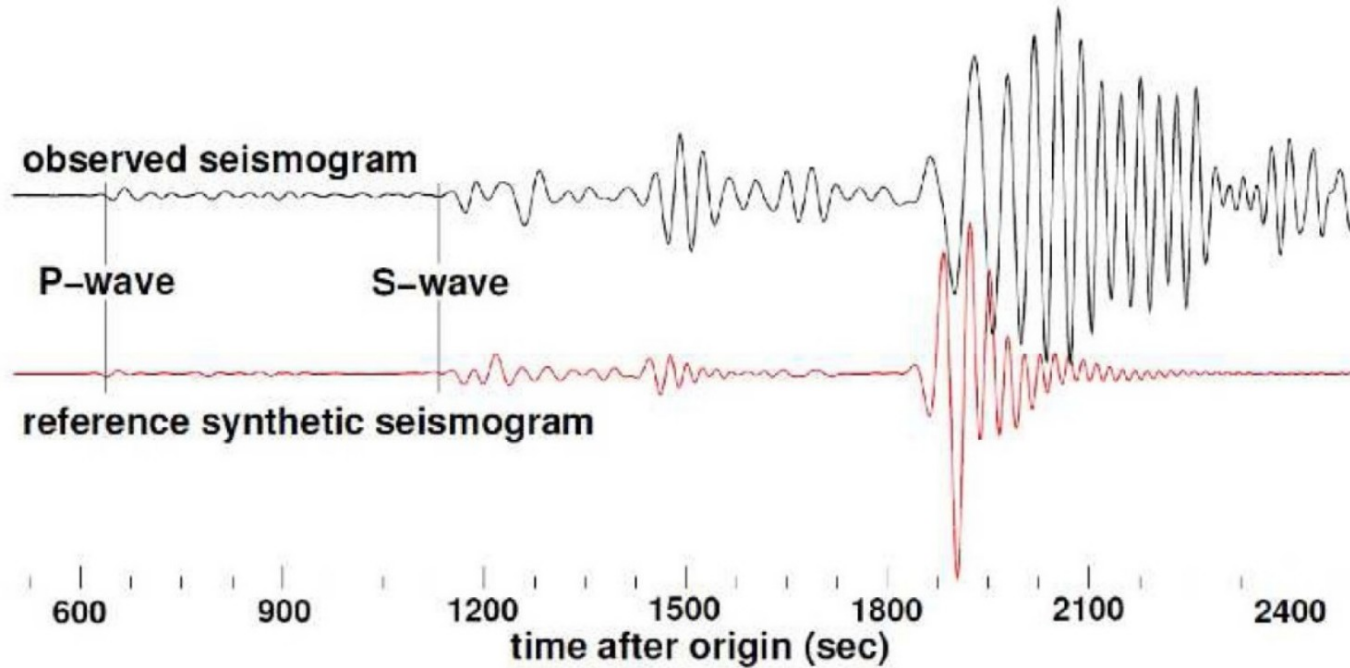
Simple interpretation:

Red = slow = hot

Blue = fast = cold

Setting up the inverse problem

Delay time measurements



$$\delta T = T - T_0.$$

Ray theoretic delay times

Rays determined through integration of Hamilton's equations:

$$\frac{dx_i}{d\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{dp_i}{d\sigma} = -\frac{\partial H_k}{\partial x_i}$$

Once the ray path is known, the travel time can be written:

$$T = \int_{\text{ray}} \frac{1}{v} ds,$$

If velocity variations are sufficiently small, we can seek to linearise:

$$T(v + \delta v) = T(v) + \langle DT(v) | \delta v \rangle + O(\delta v^2).$$

Ray theoretic delay times

Using Fermat's principle, we find that:

$$\delta T = - \int_{\text{ray}} \frac{\delta v}{v^2} ds,$$

First-order contribution from varying ray geometry vanishes, and so we need only integrate the velocity perturbation along the reference ray path

Functional derivative identified by using the standard inner product:

$$\langle u | w \rangle = \int_M u w d^3 \mathbf{x},$$

Note that the functional derivative is singular on the ray path and zero elsewhere

Finite-dimensional parametrisation

For the i th delay time measurement:

$$\delta T_i = - \int_{\text{ray}_i} \frac{\delta \alpha}{\alpha^2} \text{d}s + e_i.$$

Expand the velocity in a finite-dimensional basis:

$$\delta \alpha(\mathbf{x}) = \sum_{j=1}^m \delta \alpha_j \varphi_j(\mathbf{x}).$$

Choice of basis functions determined by assumptions about the Earth and the potential resolution of the data as set by its dominant frequencies

Reduction to linear algebra

Using the model parametrisation, we now have:

$$\delta T_i = \sum_{j=1}^m \left(- \int_{\text{ray}_i} \frac{\varphi_j}{\alpha^2} \mathrm{d}s \right) \delta \alpha_j + e_i,$$

We can now use a familiar matrix-vector notation:

$$\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{e},$$

where we have set:

$$A_{ij} = - \int_{\text{ray}_i} \frac{\varphi_j}{\alpha^2} \mathrm{d}s,$$

Solving the inverse problem

The null space

Tomographic problem with perfect data:

$$\mathbf{d} = \mathbf{A}\mathbf{m}.$$

Is there unique model that solves this problem?

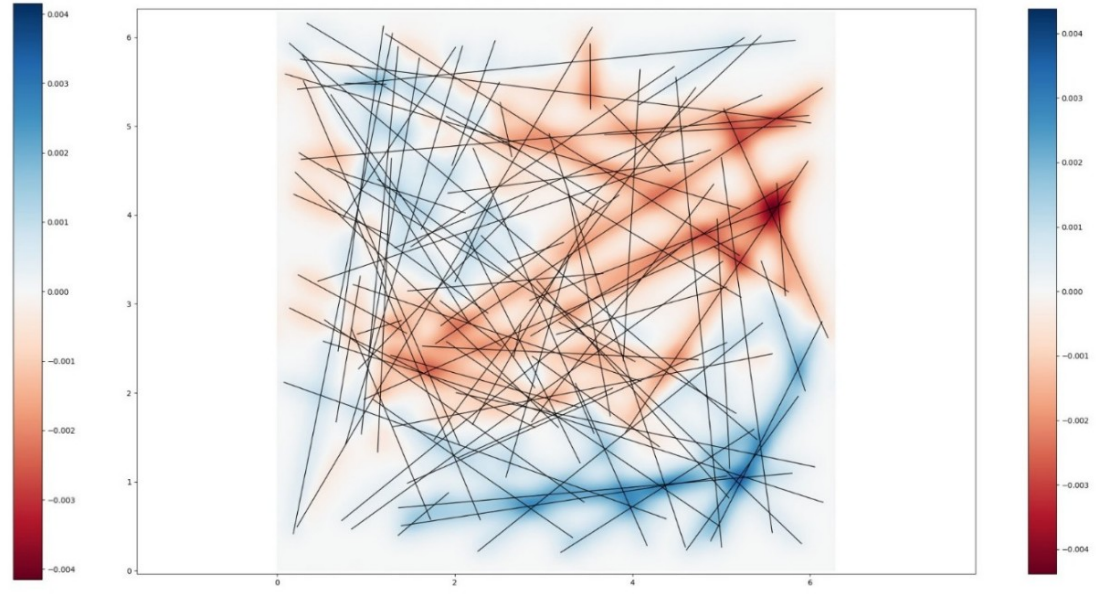
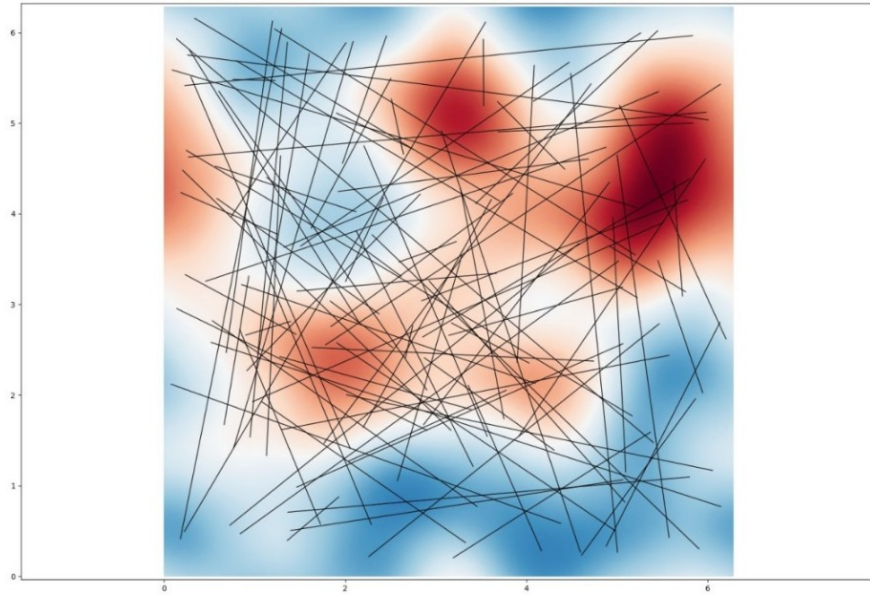
The **null space** for the matrix **A** comprises those vectors satisfying:

$$\mathbf{A}\mathbf{m}_0 = \mathbf{0}$$

This is a linear subspace. If non-trivial (not just the zero-vector), then the linear system cannot have unique solutions

The null space is always non-trivial for under-determined problems $M > N$

The null space



Lack of ray coverage is the most obvious ray null spaces are generated

Least squares solutions

An obvious way to proceed is through a **least-squares solution**

We define the least-squares misfit through:

$$J(\mathbf{m}) = \frac{1}{2} \sum_{i=1}^n \frac{[(\mathbf{A}\mathbf{m})_i - d_i]^2}{\sigma_i^2}$$

In matrix notation, this becomes:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}).$$

If there is a non-trivial null-space there is **no unique minimum!**

Regularisation

The idea is to penalise models with undesirable properties within the inversion

Suppose we prefer models with a small norm:

$$\int_M \delta\alpha^2 d^3\mathbf{x}.$$

Relative to our model parametrisation, this becomes:

$$\int_M \delta\alpha^2 d^3\mathbf{x} = \mathbf{m}^T \mathbf{B} \mathbf{m},$$

$$B_{jk} = \int_M \varphi_j \varphi_k d^3\mathbf{x}.$$

Regularised least squares

We modify the least squares misfit to incorporate regularisation:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}) + \frac{\lambda}{2}\mathbf{m}^T \mathbf{B}\mathbf{m},$$

Models with a small value of J now represent a balance between fitting the data well and keeping the chosen model property small

This balance is reflected in the choice of the **trade-off parameter**

Many choices of regularisation possible such as **norm or gradient damping**

Solving the optimisation problem

Misfit is quadratic, and so has at most one minimum value when:

$$\nabla J(\mathbf{m}) = \mathbf{0}$$

Through a calculation (detailed in the handout) this implies:

$$(\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B}) \mathbf{m} = \mathbf{A}^T \mathbf{C}^{-1} \mathbf{d}$$

If the matrix on the LHS is invertible, there is a unique solution to the problem

Uniqueness of the solution

Consider the associated eigenvalue problem:

$$(\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B}) \mathbf{m} = \mu \mathbf{m}.$$

If the eigenvalues are all non-zero, then the matrix is invertible

Take inner product with a normalised eigenvector:

$$(\mathbf{A} \mathbf{m})^T \mathbf{C}^{-1} (\mathbf{A} \mathbf{m}) + \lambda \mathbf{m}^T \mathbf{B} \mathbf{m} = \mu,$$

The LHS is positive so long as the trade-off parameter is positive

Regularised least squares is a means of calculating solutions to the inverse problem that also have desirable physical properties. But there are a **lot of arbitrary choices**

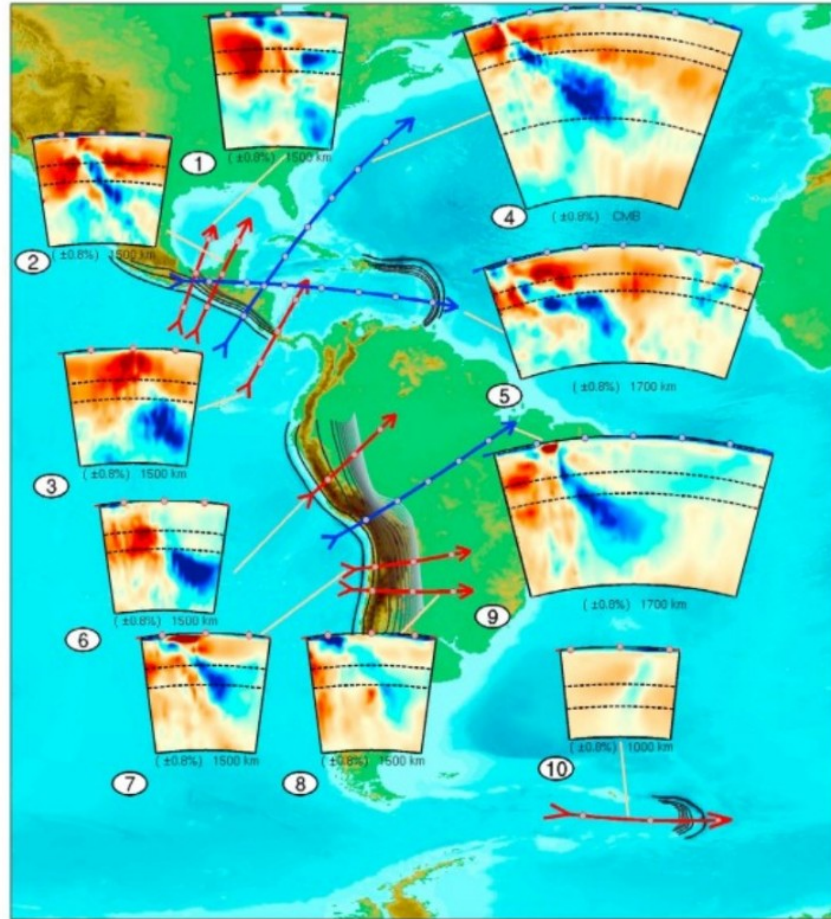
Generalised regularisation

We can also ask that the model stays “close” to a preferred point:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}) + \frac{\lambda}{2}(\mathbf{m} - \mathbf{m}_0)^T \mathbf{B}(\mathbf{m} - \mathbf{m}_0),$$

For positive trade-off parameter, we obtain the unique solution:

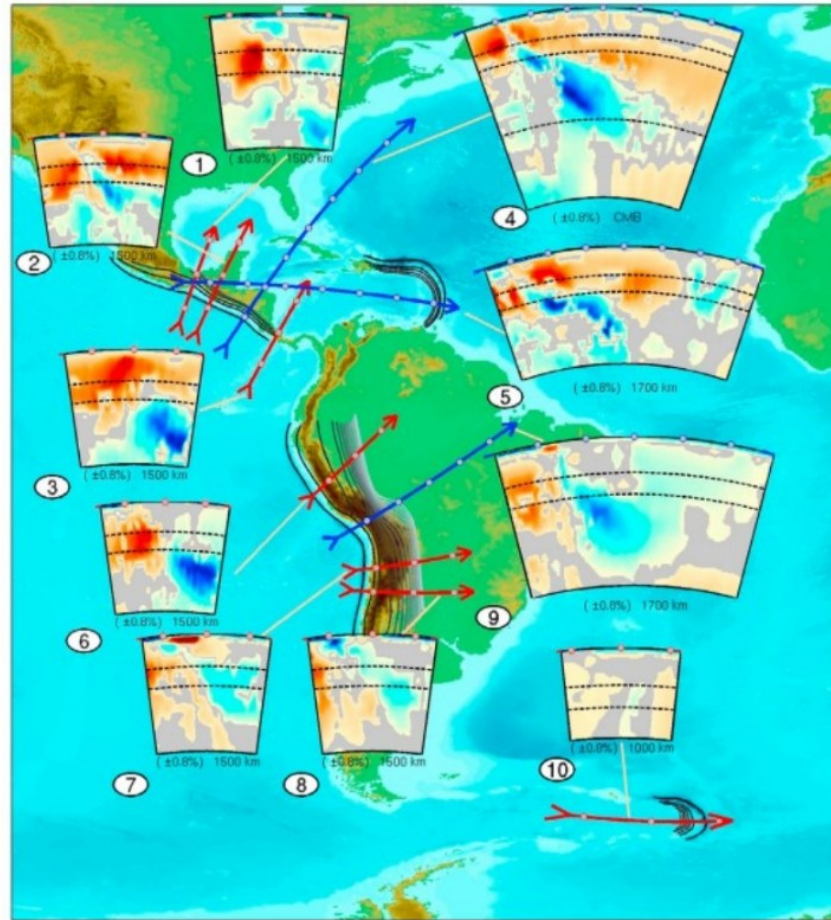
$$\mathbf{m} = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B})^{-1} (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{d} + \lambda \mathbf{B} \mathbf{m}_0).$$



De Witt et al. 2012



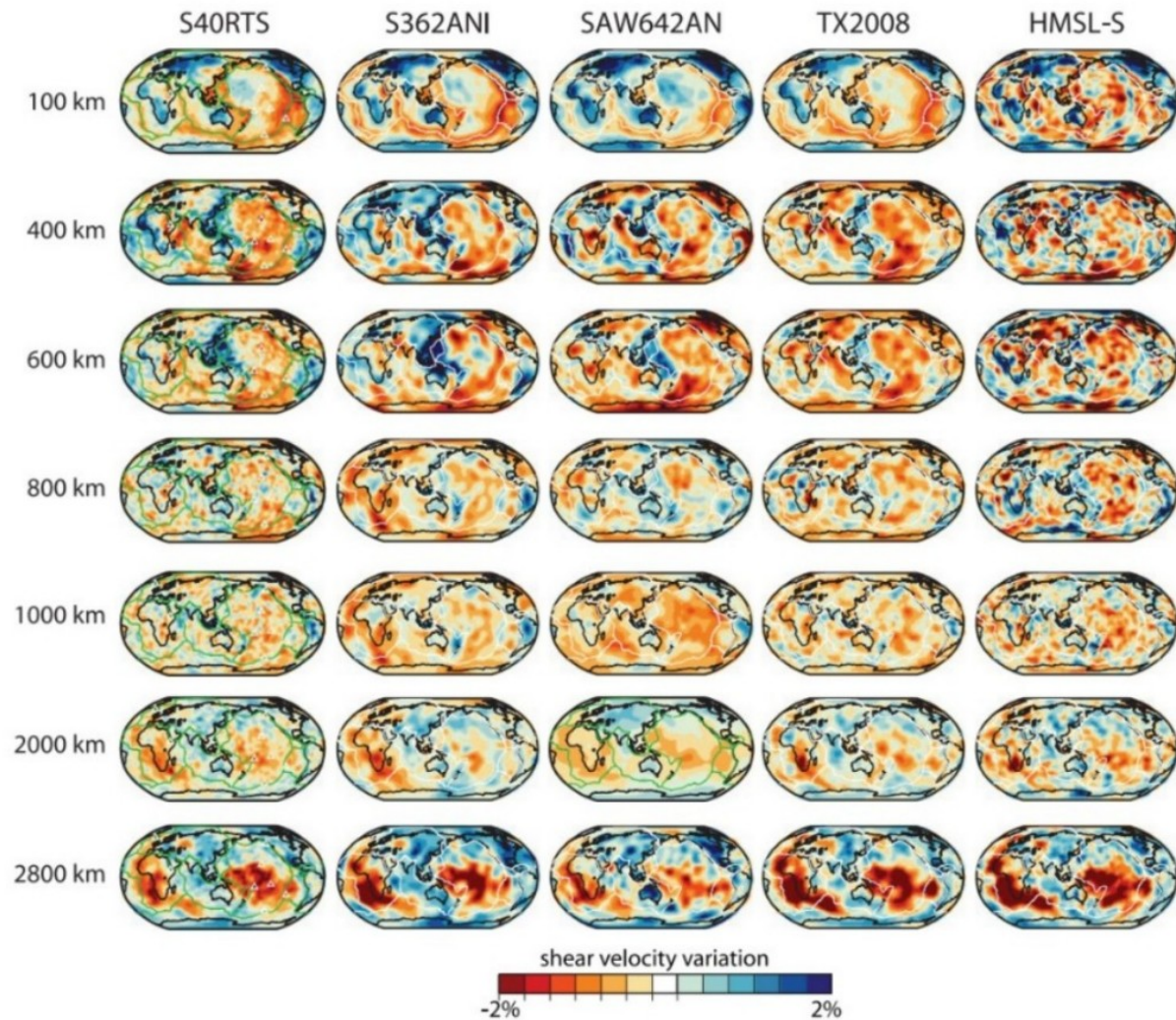
Which features in the model can be believed? Not an easy question!



De Witt et al. 2012

slow  fast

Can vary arbitrary parameters to see what has to be there to fit the data



Different data sets and methods lead to broadly consistent results

A Bayesian perspective

Bayes theorem

Bayes theorem expressed in terms of conditional probability density functions:

$$p(\mathbf{m}|\mathbf{d}) = \frac{p(\mathbf{d}|\mathbf{m})p(\mathbf{m})}{\int p(\mathbf{d}|\mathbf{m})p(\mathbf{m}) d\mathbf{m}'},$$

This is a complete recipe to update prior beliefs given data

But what do probabilities mean in this context?

And how do you arrive at the “correct prior”

Now any decision that depends on the data that is being used in making the inference only requires from the data the posterior distribution. Consequently the problem of inference is effectively solved by stating the posteriori distribution. This is the reason I feel that the basic problem of inference is solved.

D.V. Lindley, *Foundations of Statistical Inference*

I am often asked if the method gives the right answer; or, more particularly, how do you know if you have got the right prior. My reply is that I don't know what is meant by "right" in this context. The Bayesian theory is about coherence, not about right or wrong.

D.V. Lindley. *Bayesian Statistics*

Linear problems with Gaussian priors

Prior and likelihood take the form:

$$p(\mathbf{m}) = \frac{1}{\sqrt{(2\pi)^m \det \mathbf{C}_m}} \exp \left[-\frac{1}{2} (\mathbf{m} - \mathbf{m}_0)^T \mathbf{C}_m^{-1} (\mathbf{m} - \mathbf{m}_0) \right]$$

$$p(\mathbf{d}|\mathbf{m}) = \frac{1}{\sqrt{(2\pi)^n \det \mathbf{C}}} \exp \left[-\frac{1}{2} (\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1} (\mathbf{A}\mathbf{m} - \mathbf{d}) \right].$$

Bayes theorem results in a posterior PDF that is also Gaussian

Posterior mean and covariance

Posterior mean has the same form as a regularised least-squares solution:

$$\mathbf{m}_p = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \mathbf{C}_m^{-1})^{-1} (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{d} + \mathbf{C}_m^{-1} \mathbf{m}_0),$$

Method also provides the posterior covariance:

$$\mathbf{C}_p = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \mathbf{C}_m^{-1})^{-1}.$$

Crucially, this quantifies uncertainty on the model – a combination of propagated data errors and the limitations imposed by the physics of the problem

Bayesian methods have a lot going for them, but they are only as good as the prior



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https://cambridge.eu.qualtrics.com/jfe/form/SV_3z0oG7tq3O9xtpc

Waveform tomography

From last time

Using Fermat's principle, we find that:

$$\delta T = - \int_{\text{ray}} \frac{\delta v}{v^2} ds,$$

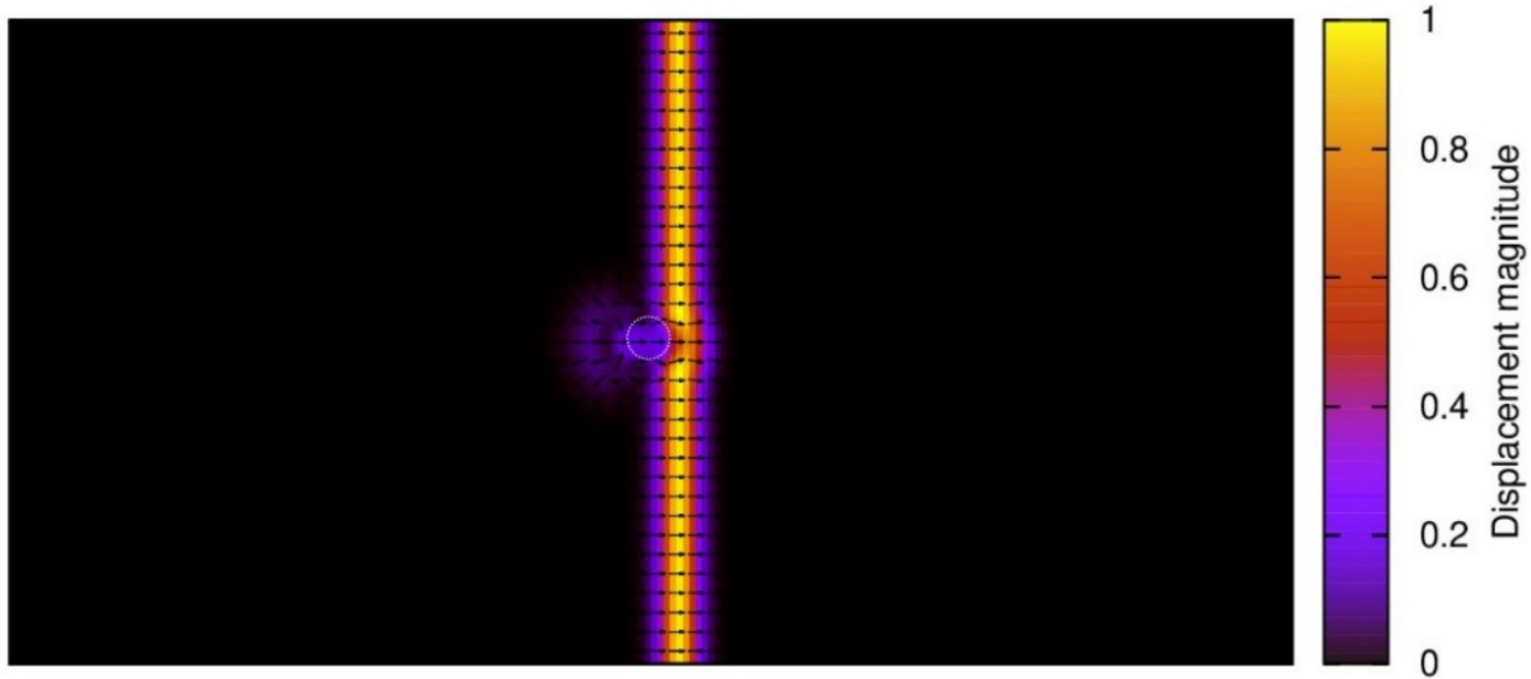
Expand the velocity in a finite-dimensional basis:

$$\delta \alpha(\mathbf{x}) = \sum_{j=1}^m \delta \alpha_j \varphi_j(\mathbf{x}).$$

Formulation as a regularised least squares problem:

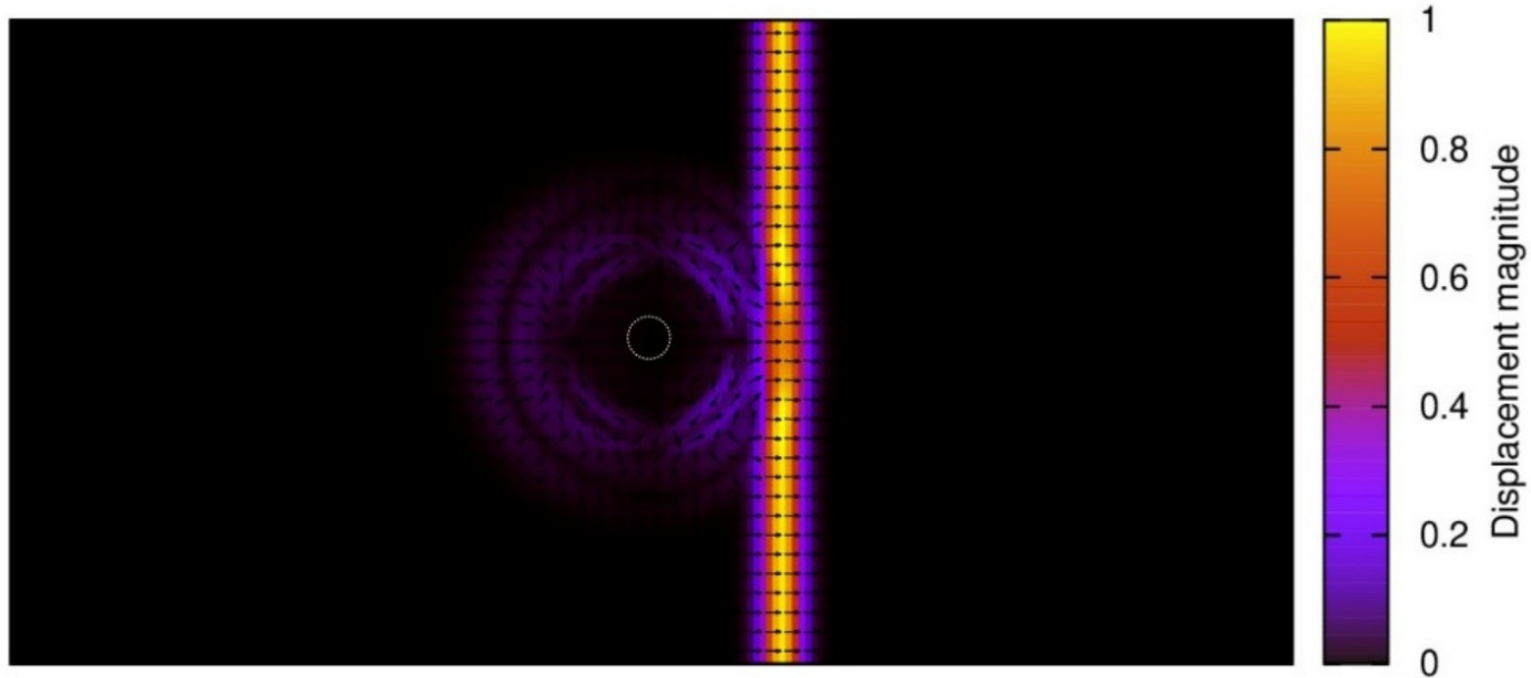
$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}) + \frac{\lambda}{2}(\mathbf{m} - \mathbf{m}_0)^T \mathbf{B}(\mathbf{m} - \mathbf{m}_0),$$

Wavefront healing



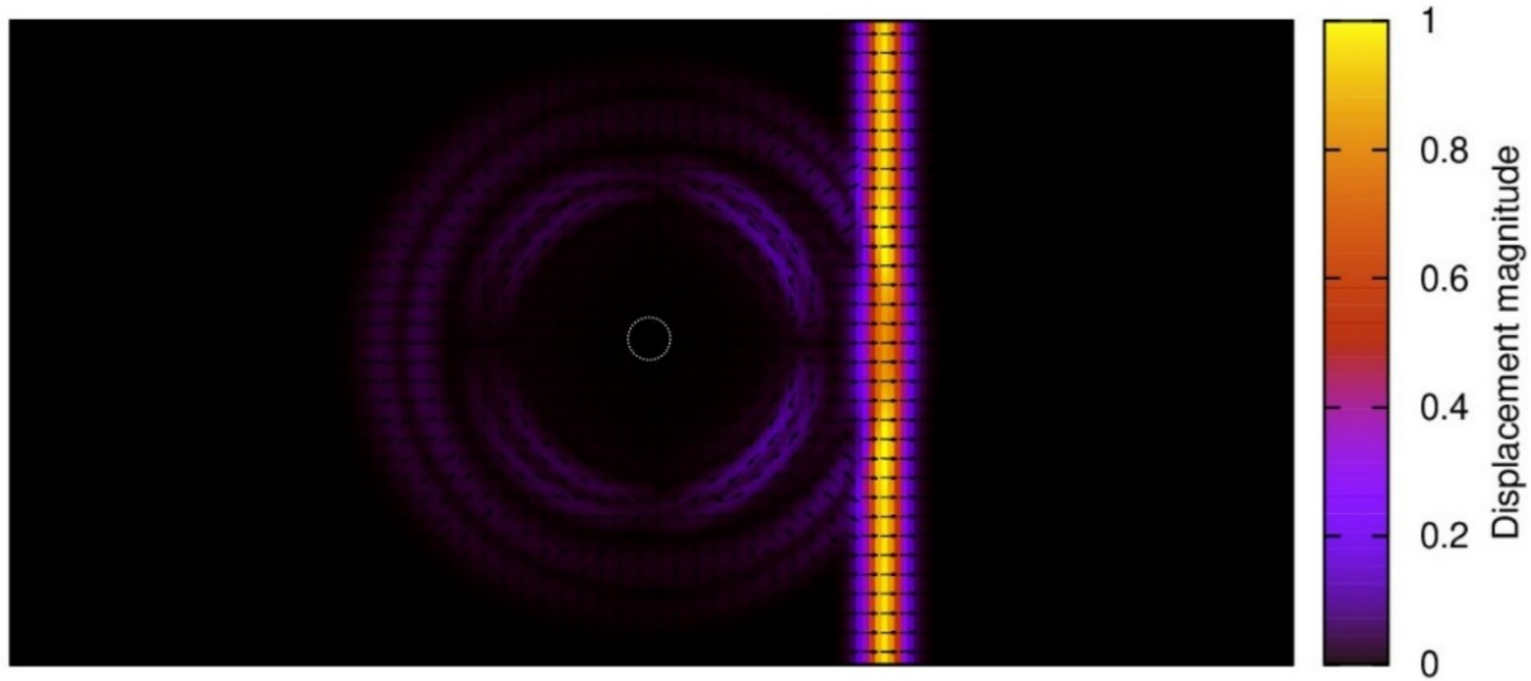
Scattering regime: $\frac{L}{\lambda} \approx 1$ Geometric optics regime: $\frac{L}{\lambda} \gg 1$

Wavefront healing



Scattering regime: $\frac{L}{\lambda} \approx 1$ Geometric optics regime: $\frac{L}{\lambda} \gg 1$

Wavefront healing

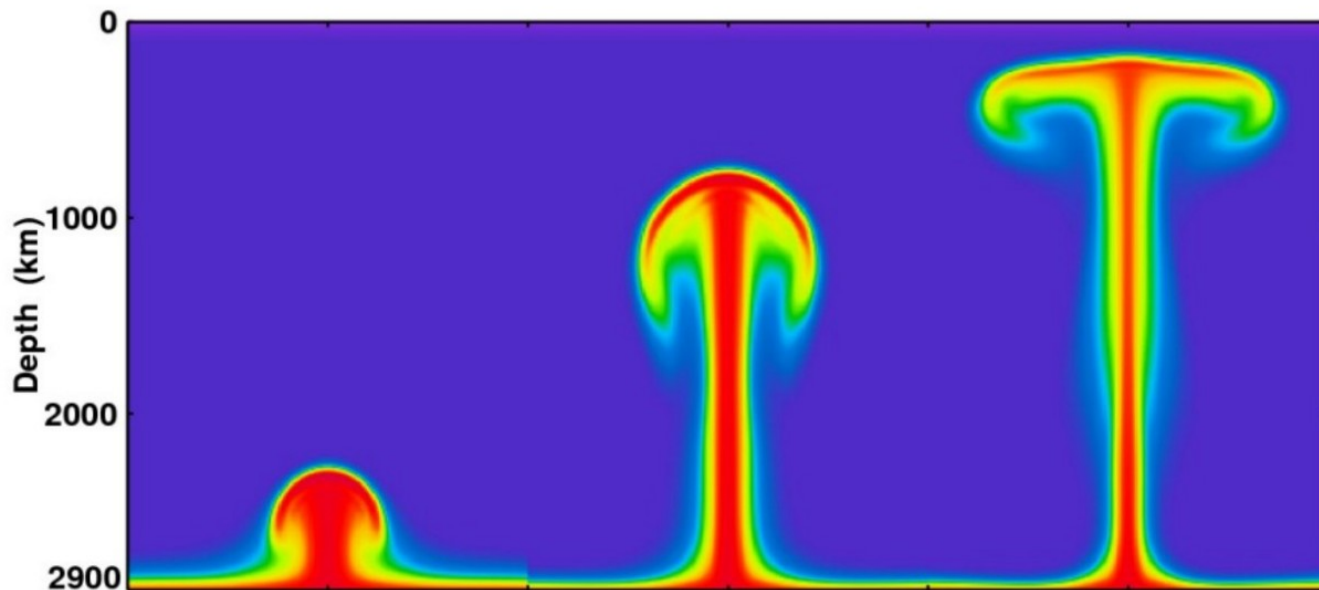


Scattering regime: $\frac{L}{\lambda} \approx 1$ Geometric optics regime: $\frac{L}{\lambda} \gg 1$

Lengths scales in the mantle

Typical periods for global seismology: 1-40s

Corresponding wavelengths: 10-400km



Farnetani 1997

Waveform inversion

Equations of motion

Equations of motion:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j},$$

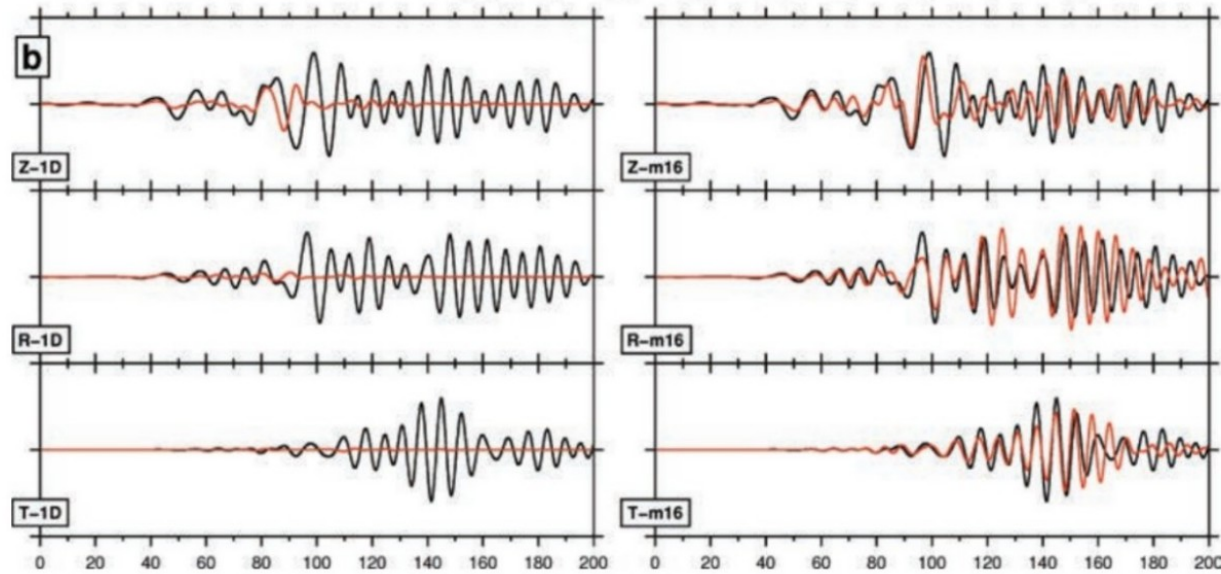
$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

These can be solved numerically in realistic laterally heterogeneous earth models using techniques like the **finite element methods**

These are demanding calculations, run on HPC systems using parallel computing over hundreds or thousands of cores. Applications of GPU computing is common

Each simulation take **tens of minutes to hours** depending on resources

Waveform misfits



Tape et al. 2010

$$J = \frac{1}{2} \int_0^T ||\mathbf{u}(\mathbf{x}_r, t) - \mathbf{u}^{obs}(t)||^2 dt,$$

Formulation of the problem

Misfit is an implicit function of the parameters in the elastic wave equation:

- 1) **Density** – piecewise constant function
- 2) **Elastic tensor** – piecewise constant tensor field (21 independent components)
- 3) **Stress glut** – space and time dependent tensor field
- 4) **Geometry** of the model (e.g., topography on internal boundaries)

Relation can be condensed to:

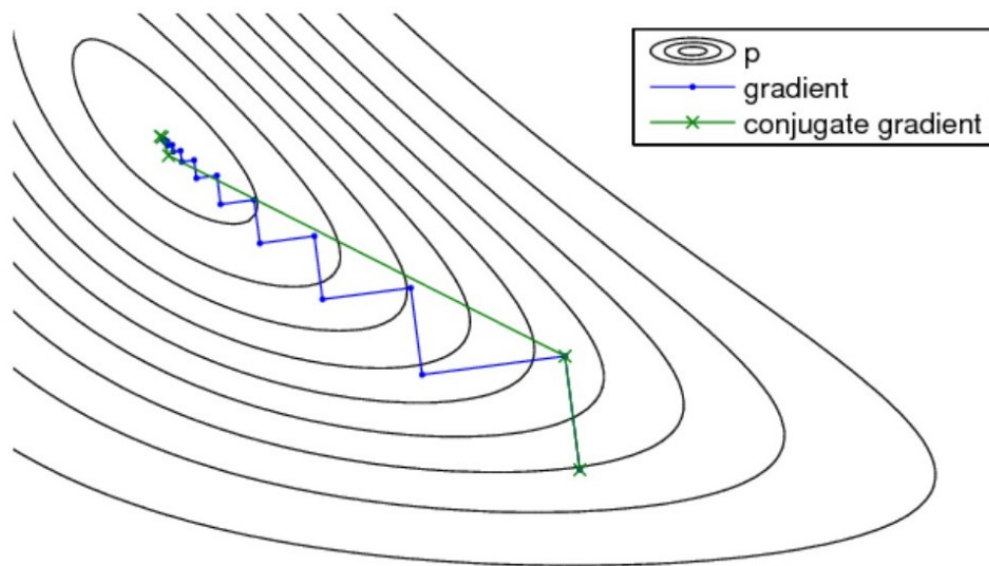
$$\mathbf{m} \mapsto J(\mathbf{m})$$

The aim is to **minimise** this functional

Strictly defined on an **infinite-dimensional space**. In practice reduced to a finite but high-dimensional space linked to the numerical discretisation of the equations of motion

Millions or billions of degrees of freedom would be typical

Gradient based optimisation



Iterative reduction of the misfit using the local gradient to take steps downhill

For large-scale optimisation problems, this is often the only viable approach

Bayesian sampling methods are not typically an option, though this is active research topic

To proceed, we need a way of evaluating the gradients

Functional derivatives

Functional derivative defined through a first-order Taylor expansion:

$$J(\mathbf{m} + \delta\mathbf{m}) = J(\mathbf{m}) + \langle DJ(\mathbf{m}) | \delta\mathbf{m} \rangle + O(||\delta\mathbf{m}||^2),$$

The terms **gradient** and **sensitivity kernel** are equivalent and widely used

Focusing on S-wave velocity variations alone, we can write to first order:

$$\langle DJ(\mathbf{m}) | \delta\mathbf{m} \rangle = \int_M K_\beta \delta\beta d^3\mathbf{x}$$

Methods for calculation will be discussed later

Gradient based optimisation

Condition for a minimum – global and then local:

$$J(\overline{\mathbf{m}} + \delta \mathbf{m}) \geq J(\overline{\mathbf{m}}),$$

$$DJ(\overline{\mathbf{m}}) = \mathbf{0}.$$

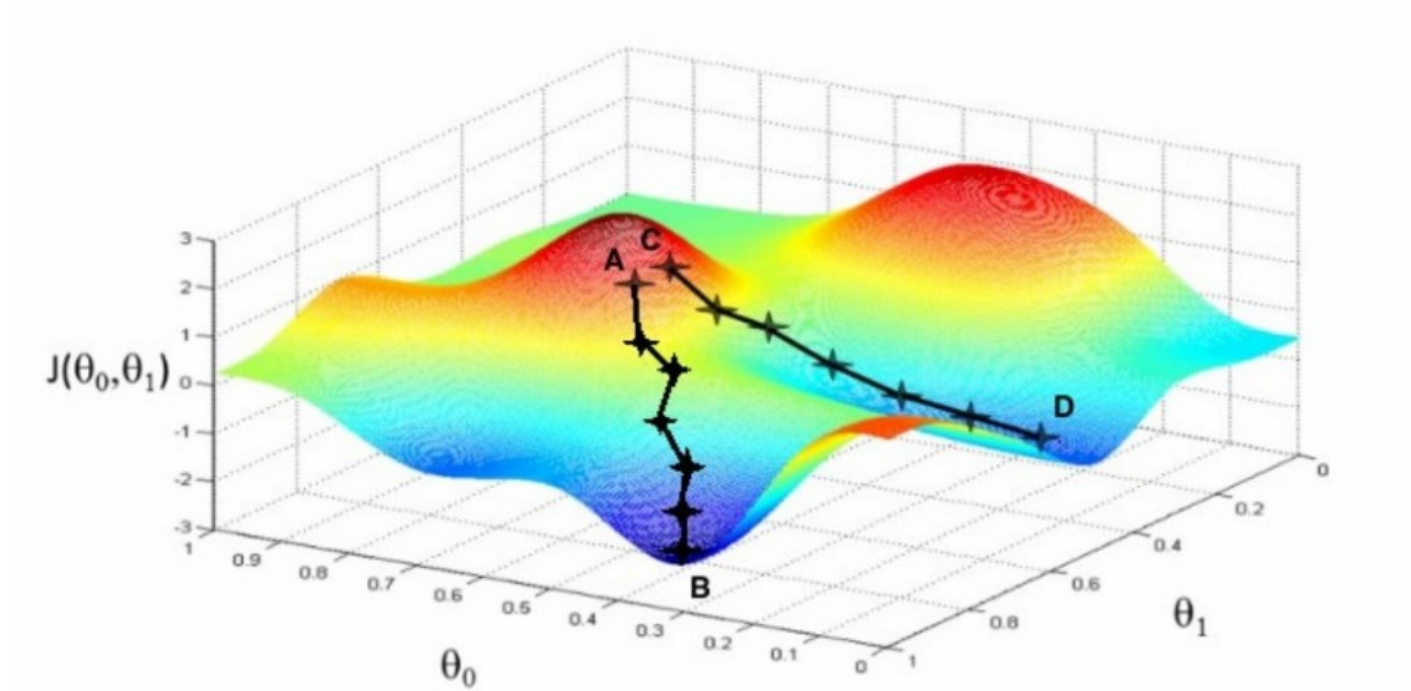
Starting an an initial point, we consider the update:

$$\mathbf{m}_1 = \mathbf{m}_0 - \lambda DJ(\mathbf{m}_0)$$

Step length determined by approximately solving a 1D minimisation problem

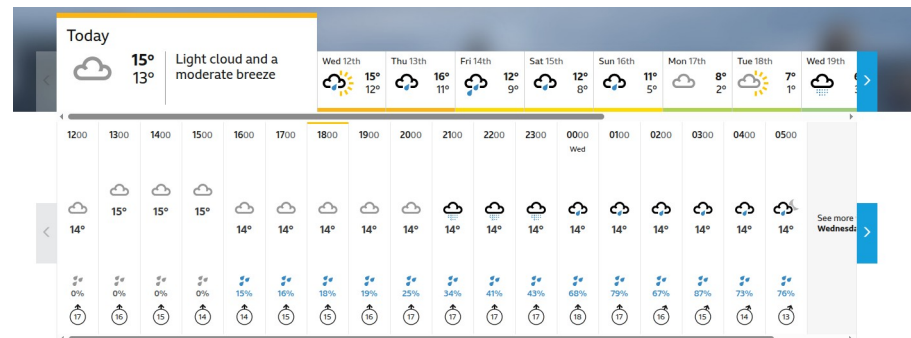
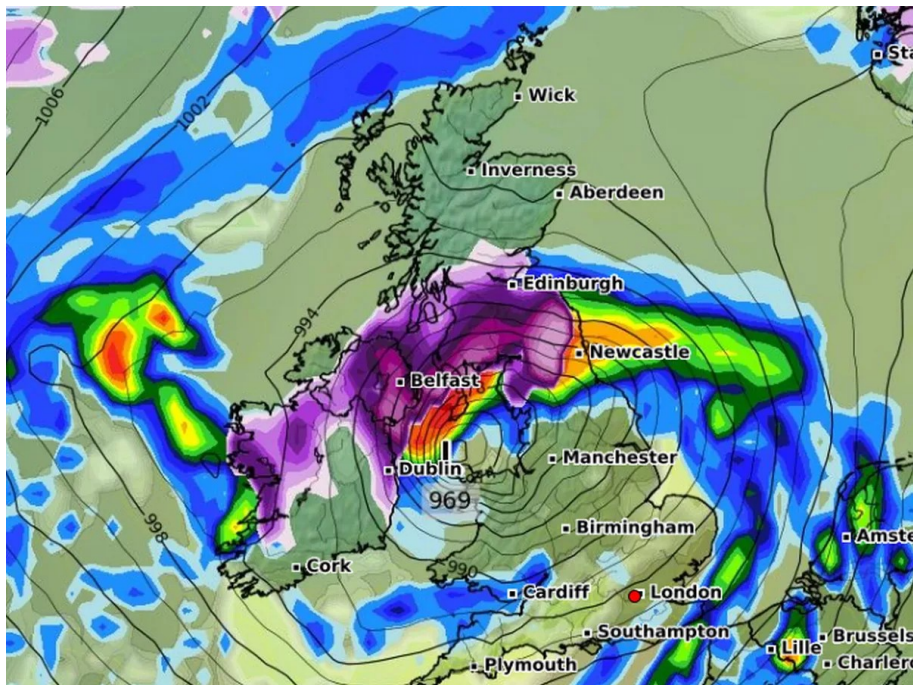
Process is repeated until convergence or the misfit is sufficiently small

Local minima



Gradient based methods produce only local minima

Uncertainty quantification possible to a limit extent, but adds significantly to the cost



Essentially the same methods underlie weather forecasting along with a host of other important applications

Calculating the gradients

How not to do it

Finite difference approximation:

$$\langle DJ(\mathbf{m}) | \delta \mathbf{m} \rangle \approx J(\mathbf{m} + \delta \mathbf{m}) - J(\mathbf{m}).$$

Approximation of derivative relative to an **n-dimensional model parametrisation** requires **n+1 forward calculations**

One gradient calculation using this method for a realistic problem would take of order hundreds of years on a massive HPC system!

The **adjoint method** provides the exact gradient at a cost of **two forward simulations**

PDE constrained optimisation

We want to minimise:

$$J = \frac{1}{2} \int_0^T ||\mathbf{u}(\mathbf{x}_r, t) - \mathbf{u}^{obs}(t)||^2 dt,$$

subject to the constraints:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = -\frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

This can be done using the **method of Lagrange multipliers**

PDE constrained optimisation

We start by defining the **Lagrangian**:

$$L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = J(\mathbf{u}) - \int_0^T \int_M \left[\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \right] u'_i d^3 \mathbf{x} dt \\ - \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \hat{n}_j w'_i dS dt.$$

Each constraint is associated with a **Lagrange multiplier field**

Zero initial conditions on displacement and velocity are being imposed directly

Within the Lagrangian, all fields are regarded as being independent

Lagrange multiplier theorem

The following equality holds:

$$DJ(\mathbf{m}) = D_{\mathbf{m}}L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}),$$

so long as the following conditions are met:

$$D_{\mathbf{u}}L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = 0,$$

$$D_{\mathbf{u}'}L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = 0,$$

$$D_{\mathbf{w}'}L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = 0.$$

Subscripts used for partial functional derivatives

You only need to know how to apply this method **not to prove** that it works – details provided in an appendix to the handout for anyone interested

The forward problem

Recall that the Lagrangian is:

$$\begin{aligned} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = & J(\mathbf{u}) - \int_0^T \int_M \left[\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \right] u'_i d^3 \mathbf{x} dt \\ & - \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \hat{n}_j w'_i dS dt. \end{aligned}$$

Stationarity condition wrt to $\mathbf{u}'$ gives the equations of motion

Stationary conditions wrt to $\mathbf{w}'$ give the boundary conditions

This is how the forward problem emerges within the method

The adjoint problem

Varying wrt to $\mathbf{u}$ gives:

$$\begin{aligned} \langle D_{\mathbf{u}}L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) | \delta \mathbf{u} \rangle &= \langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle - \int_0^T \int_M \left[\rho \frac{\partial^2 \delta u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \right] u'_i \, d^3 \mathbf{x} dt \\ &\quad - \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j w'_i \, dS \, dt, \end{aligned}$$

From the chosen form for J we can write:

$$\langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle = \int_0^T \int_{\partial M} h'_i \delta u_i \, dS \, dt,$$

$$h'_i(\mathbf{x}, t) = [u_i(\mathbf{x}_r, t) - u_i^{obs}(t)] \delta(\mathbf{x} - \mathbf{x}_r).$$

The adjoint problem

We need the following condition to hold:

$$\begin{aligned} \int_0^T \int_{\partial M} h'_i \delta u_i \, dS \, dt - \int_0^T \int_M \left[\rho \frac{\partial^2 \delta u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \right] u'_i \, d^3 \mathbf{x} \, dt \\ - \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j w'_i \, dS \, dt = 0, \end{aligned}$$

For all variations of the displacement vector

To see the consequence, we need to pass all the derivatives to the other terms using integration by parts in time and the divergence theorem spatially

Full details are in the handout

The adjoint problem

The result of the long calculation is that:

$$\begin{aligned}
 & - \int_0^T \int_M \left[\rho \frac{\partial^2 u'_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) \right] \delta u_i \, d^3 \mathbf{x} \, dt \\
 & - \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j (w'_i - u'_i) \, dS \, dt \\
 & + \int_0^T \int_{\partial M} \left[h'_i - \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) \hat{n}_j \right] \delta u_i \, dS \, dt \\
 & - \int_M \rho \left\{ \left[\frac{\partial \delta u_i}{\partial t} u'_i \right]_{t=T} - \left[\delta u_i \frac{\partial u'_i}{\partial t} \right]_{t=T} \right\} d^3 \mathbf{x} = 0,
 \end{aligned}$$

From this we can read off the necessary adjoint equations

The adjoint problem

We have shown that the Lagrange multipliers must satisfy the equations:

$$\rho \frac{\partial^2 u'_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) = 0.$$

$$\left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) \hat{n}_j = h'_i, \quad u'_i(\mathbf{x}, T) = \frac{\partial u'_i}{\partial t}(\mathbf{x}, T) = 0.$$

along with the equality $\mathbf{u}' = \mathbf{w}'$ on the boundary

This is a **terminal-value problem** for $\mathbf{u}'$ driven by the boundary traction $\mathbf{h}'$:

$$h'_i(\mathbf{x}, t) = [u_i(\mathbf{x}_r, t) - u_i^{obs}(t)] \delta(\mathbf{x} - \mathbf{x}_r).$$

Sensitivity kernels

Using the multiplier theorem and integrating by parts, we find that:

$$\langle DJ(\mathbf{m})|\delta\mathbf{m}\rangle = \int_0^T \int_M \left[\delta\rho \frac{\partial u_i}{\partial t} \frac{\partial u'_i}{\partial t} - \delta A_{ijkl} \frac{\partial u'_i}{\partial x_j} \frac{\partial u_k}{\partial x_l} + \delta \bar{S}_{ij} \frac{\partial u'_i}{\partial x_j} \right] d^3\mathbf{x} dt,$$

We can then read-off the sensitivity kernels:

$$\begin{aligned} K_\rho &= \int_0^T \frac{\partial u_i}{\partial t} \frac{\partial u'_i}{\partial t} dt, \\ K_{A_{ijkl}} &= - \int_0^T \frac{\partial u'_i}{\partial x_j} \frac{\partial u_k}{\partial x_l} dt, \\ K_{\bar{S}_{ij}} &= \frac{\partial u'_i}{\partial x_j}. \end{aligned}$$

Summary of the adjoint method

For each gradient calculation we must:

- 1) Solve the forward problem for $\mathbf{u}$
- 2) From this solution, compute J , and form the adjoint traction $\mathbf{h}'$
- 3) Solve the adjoint problem for $\mathbf{u}'$
- 4) Combine the forward and adjoint fields to form the sensitivity kernels

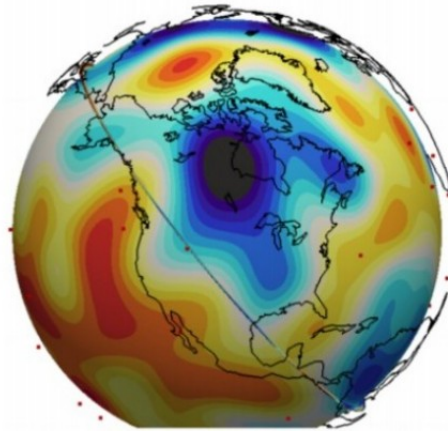
Such a calculation has the cost of two forward calculations

Adjoint equation near identical to the forward one, and so the same numerical code can be used with only minor modification

Choice of misfit enters into the problem only through the form of the adjoint traction

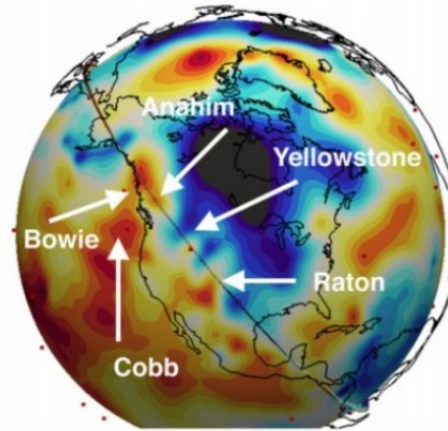
Calculations that would take hundred of years reduced to minutes or hours

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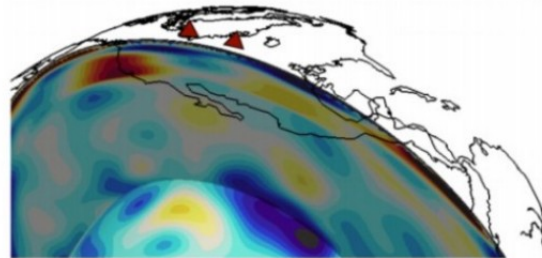
$d\ln V_{sv}$ (2%)
min max

GLAD-M15



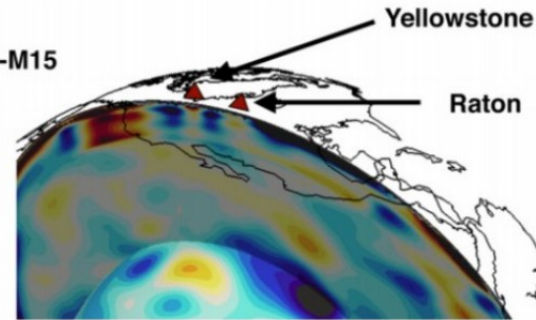
$d\ln V_{sv}$ (2%)
min max

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$d\ln V_{sv}$ (2%)
min max

GLAD-M15



$d\ln V_{sv}$ (2%)
min max

Banana doughnut kernels

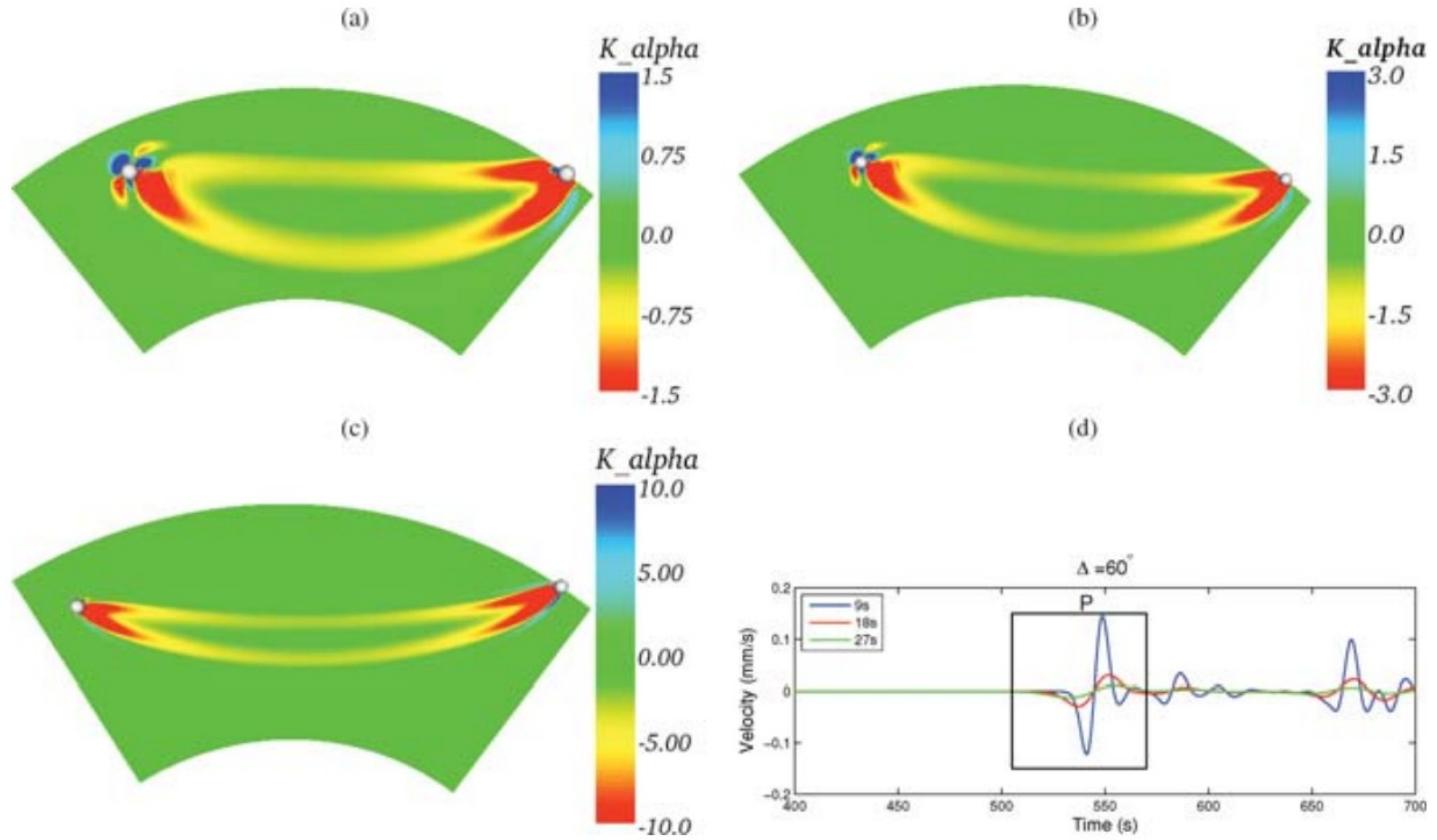
In practice, delay times are measured using **cross-correlations**:

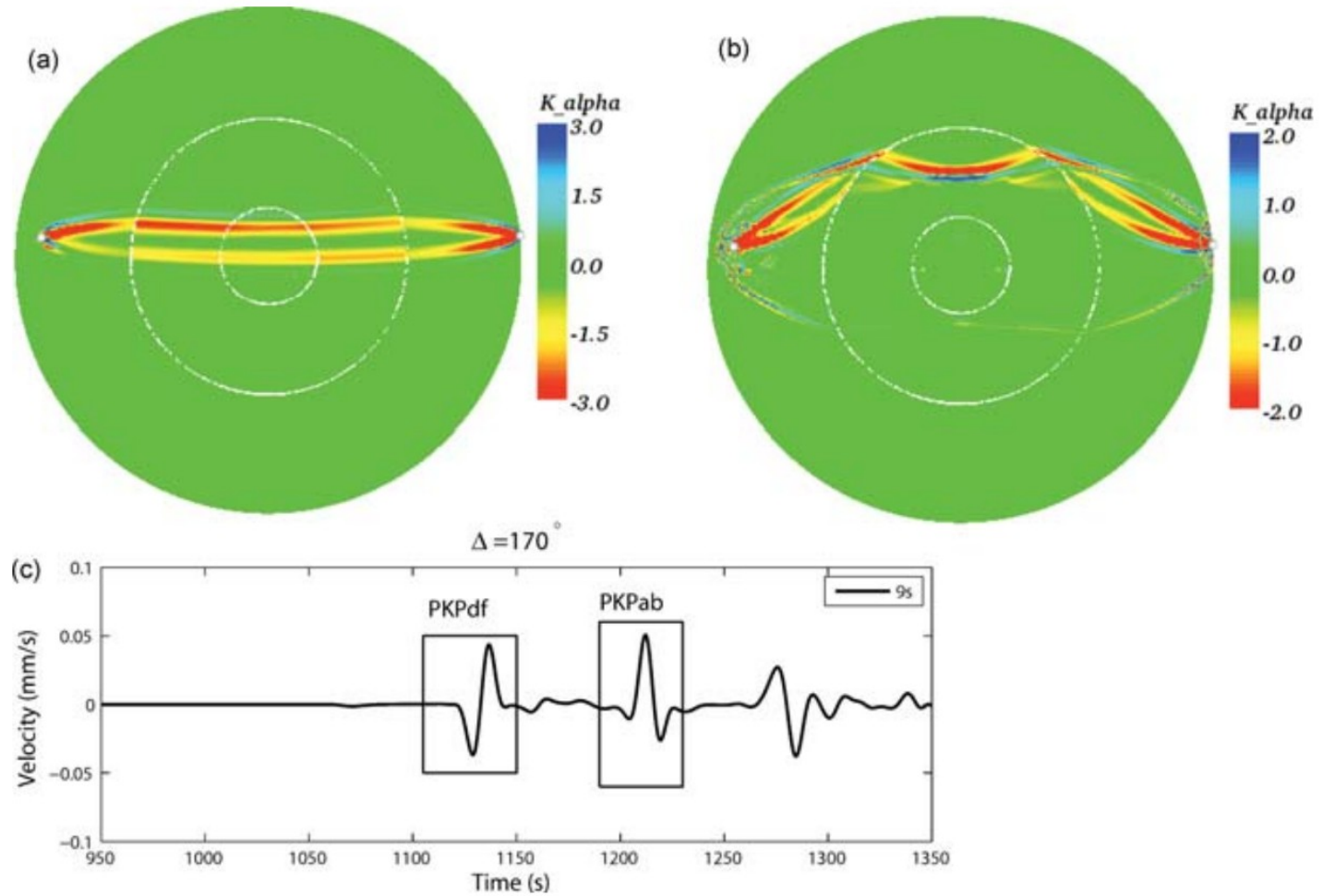
$$C(\tau) = \int_{-\infty}^{\infty} s^{obs}(t - \tau) s(\mathbf{x}_s, t) dt.$$

Time-shift that maximises the **cross-correlation** taken as the delay time

This defines a functional of the synthetic waveform, and hence we can derive the associated sensitivity kernels using the adjoint method

The only step we haven't covered is determining the adjoint traction for this choice of functional – this is an easy calculation, but not one you need to know







Feedback form for the course

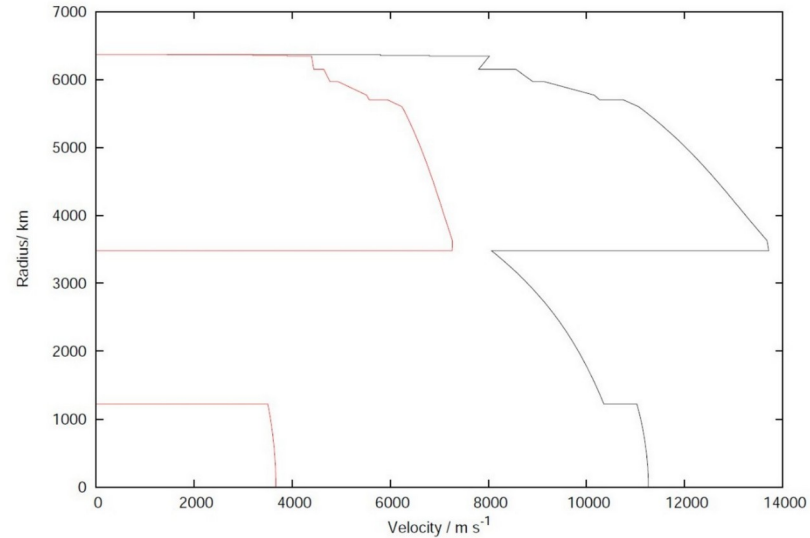
Please do fill this in before the end of term

We do take comments seriously, and they help to make the course better over time

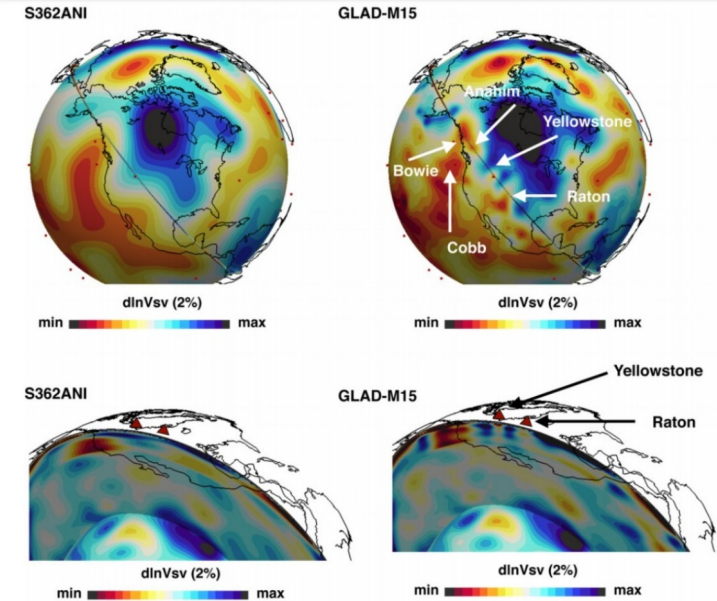
https://cambridge.eu.qualtrics.com/jfe/form/SV_3z0oG7tq3O9xtpc

Self gravitation and rotation

From earlier lectures



Preliminary Reference Earth Model (PREM) of Dziewonski & Anderson (1981)



Bozdag *et al.* 2016 – first global waveform inverse model

Newtonian gravity

Poisson's equation

Gravitational potential satisfies:

$$\nabla^2 \phi = 4\pi G \rho.$$

Associated gravitational acceleration:

$$g_i = -\frac{\partial \phi}{\partial x_i}$$

Equation subject to continuity conditions:

$$[\phi]_{-}^{+} = 0, \quad [g_i \hat{n}_i]_{-}^{+} = 0$$

across any jumps in density, while the potential also tends to zero at infinity

Poisson's equation

Solution to the problem can be expressed as a familiar integral:

$$\phi(\mathbf{y}, t) = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)}{[(y_i - y'_i)(y_i - y'_i)]^{\frac{1}{2}}} d^3\mathbf{y}',$$

and differentiating, the gravitational acceleration can be similarly written:

$$g_i(\mathbf{y}, t) = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)(y_i - y'_i)}{[(y_j - y'_j)(y_j - y'_j)]^{\frac{3}{2}}} d^3\mathbf{y}'.$$

Transformation to the reference body

Using the motion of the body, we can define a **referential gravitational potential**:

$$\zeta(\mathbf{x}, t) = \phi[\varphi(\mathbf{x}, t), t],$$

Using conservation of mass, we can transform the integral solution to obtain:

$$\zeta(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d\mathbf{x}',$$

Similarly for the referential gravitational acceleration:

$$\gamma_i(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]}{\{[\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)][\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)]\}^{\frac{3}{2}}} d\mathbf{x}',$$

Gravitational binding energy

This is the potential energy of a body associated with gravitational forces

Can be calculated by assuming the body is assembled bit-by-bit, summing up the work required to move particles in from infinity within the potential field already built up

Calculation identical to that in 1B Electrodynamics for the energy of an electrostatic field:

$$\mathcal{V}_g(t) = \frac{1}{2} \int_{M_t} \varrho(\mathbf{y}, t) \phi(\mathbf{y}, t) d^3\mathbf{y},$$

Full details in the notes, but you won't be asked to recall the calculation

Can express referentially:

$$\mathcal{V}_g(t) = \frac{1}{2} \int_M \rho(\mathbf{x}) \zeta(\mathbf{x}, t) d\mathbf{x}.$$

Equations of motion

Hamilton's principle

Potential energy is now the sum of elastic and gravitational contributions

Modified Lagrangian density:

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x}, t),$$

Statement of Hamilton's principle is unchanged

Derivation of Euler-Lagrange equations is harder because we have a **non-local** term in the Lagrangian – this reflects the **action at a distance** property of Newtonian gravity

First variation of the action

Functional derivatives still defined through linear term in Taylor expansion

The result can be expressed in the form

$$\langle D\mathcal{S}(\varphi)|\delta\varphi\rangle = \int_0^T \int_M \left(\frac{\delta\mathcal{L}}{\delta\varphi_i} \delta\varphi_i + \frac{\delta\mathcal{L}}{\delta v_i} \delta v_i + \frac{\delta\mathcal{L}}{\delta F_{ij}} \delta F_{ij} \right) d\mathbf{x} dt.$$

The “derivative” terms here are, by definition, just the quantities multiplying the first-order perturbations to the fields

In the case of local Lagrangians, these are just normal partial derivatives

It is only the variation with respect to the motion that needs additional work

Euler-Lagrange equations

Follow the standard integrations by parts, we obtain:

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} - \frac{\partial}{\partial t} \left(\frac{\delta \mathcal{L}}{\delta v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\delta \mathcal{L}}{\delta F_{ij}} \right) = 0,$$

$$\frac{\delta \mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0$$

As previously, we have:

$$\frac{\delta \mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta \mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}$$

In example sheet 4 you will show that:

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i$$

Euler-Lagrange equations

The equations of motion can be written in the form:

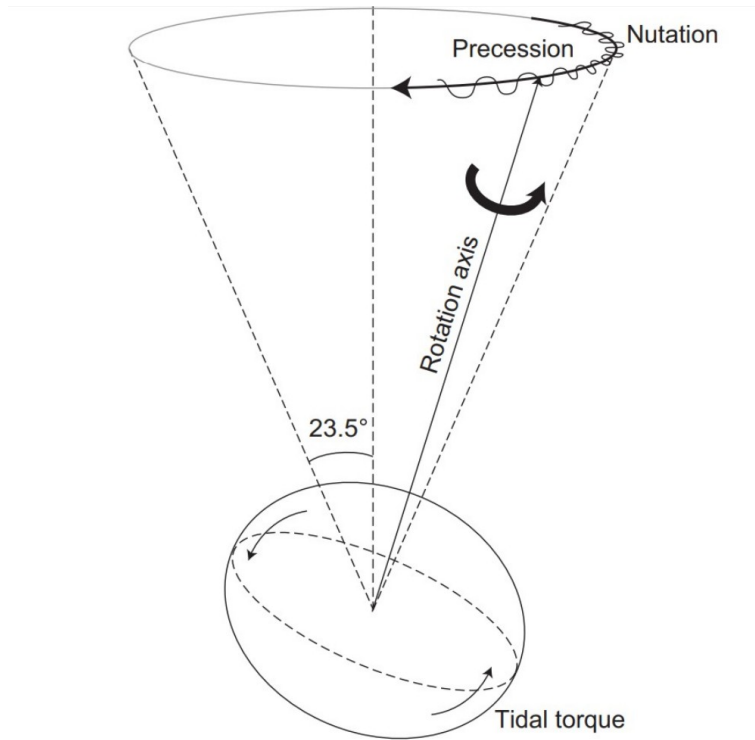
$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0,$$

As expected, we have just added a gravitational body force to the balance of linear momentum

This force is expressed in terms of the motion through an integral over the whole body

These are now **integro partial differential equations**

Co-rotating reference frames



We have described the motion relative to an inertial frame defined by the “fixed stars”

For applications to the Earth, it is more natural to use a rotating frame

The Earth’s rotation is **not steady**, varying due to both external and internal processes

Within seismology, it is common to use an **steady and approximately co-rotating reference frame**

This will introduce the usual **fictitious forces** into the equations of motion

Decomposition of the motion

The motion relative to inertial space can be written:

$$\varphi_i(\mathbf{x}, t) = \Phi_i(t) + R_{ij}(t)\bar{\varphi}_j(\mathbf{x}, t),$$

Translation vector represents motion of the origin of the reference frame

Rotation matrix represents rotation of the reference frame

On the right hand side we see the **relative motion** as seen from the non-inertial frame

Because of material frame indifference, the choice of frame has no effect on the potential energy associated with the motion

Relative velocity

Differentiating the motion in time, we find:

$$v_i = \frac{d\Phi_i}{dt} + R_{ij} \left(\frac{\partial \bar{\varphi}_j}{\partial t} + R_{kj} \frac{dR_{kl}}{dt} \bar{\varphi}_l \right)$$

Introduction of the angular velocity as seen from the non-inertial frame:

$$R_{kj} \frac{dR_{kl}}{dt} = \epsilon_{jkl} \Omega_k$$

The velocity can then be expressed in the form:

$$v_i = \frac{d\Phi_i}{dt} + R_{ij} (\bar{v}_j + \epsilon_{jkl} \Omega_k \bar{\varphi}_l).$$

Relative acceleration and choice of frame

Differentiating again, we can write the acceleration:

$$\frac{\partial v_i}{\partial t} = \frac{d^2 \Phi_i}{dt^2} + R_{ij} \left(\frac{\partial \bar{v}_j}{\partial t} + 2\epsilon_{jkl} \Omega_k \bar{v}_l + \epsilon_{jkl} \frac{d\Omega_k}{dt} \bar{\varphi}_l + \epsilon_{jkl} \epsilon_{lmn} \Omega_k \Omega_m \bar{\varphi}_n \right)$$

At this point, we define an approximately co-rotating frame by setting:

$$\frac{d\Phi}{dt} = 0, \quad \frac{d\Omega}{dt} = 0,$$

This approach can be generalised to allow for interactions between different planets (e.g., Earth-Moon system)

Stress and gravity in the non-inertial frame

Differentiating the motion spatially, we find:

$$F_{ij} = R_{ik} \bar{F}_{kj}$$

From the definition of the first Piola-Kirchhoff stress and frame indifference, we find:

$$T_{ij} = R_{ik} \bar{T}_{kj}$$

Similarly, for the referential gravitational acceleration:

$$\gamma_i = R_{ij} \bar{\gamma}_i$$

New equations of motion

Putting things together and cancelling common rotation factors:

$$\rho \left(\frac{\partial \bar{v}_i}{\partial t} + 2\epsilon_{ijk}\Omega_j\bar{v}_k + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l\bar{\varphi}_m \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho\bar{\gamma}_i = 0.$$

All terms defined relative to the rotating reference frame

Identical to the inertial case except for the addition of **Coriolis and centrifugal forces**

From here on we will always work in the rotating frame and drop the overbars

Linearisation

Same procedure as ever for linearisation on application of a small stress glut

At zeroth-order we have the equilibrium equations:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 = 0,$$

Stress-free equilibrium solutions are not now possible

At first order we have the linearised equations for the displacement vector:

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho \gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j},$$

Perturbation to referential gravitational acceleration given as an integral of the displacement vector – see handout for full expression

Scaling analysis

Order of magnitude estimates

Derivatives can be readily estimated in terms of typical time and length scales

$$\begin{aligned}\rho \frac{\partial^2 u_i}{\partial t^2} &\sim \frac{\bar{\rho} U}{T^2}, \\ \rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} &\sim \frac{\bar{\rho} \Omega U}{T}; \\ \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m &\sim 2\bar{\rho} \Omega^2 U, \\ \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) &\sim \frac{\bar{\mu} U}{L^2},\end{aligned}$$

Estimating the magnitude of the gravitational acceleration is a bit more work due to integral definition (see handout):

$$\rho \gamma_i^1 \sim 4\pi G \bar{\rho}^2 U$$

Relative importance of terms

Ratio of gravitational to elastic forces:

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \bar{\rho}^2 U}{\frac{\bar{\mu} U}{L^2}} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2},$$

This shows that gravity is negligible at sufficient small wavelengths and hence at correspondingly high frequencies

Taking L to be the radius of the Earth:

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3.$$

At the longest wavelengths gravity is just as important as elasticity within the equations of motion – this fact means that we can learn about density directly

Relative importance of terms

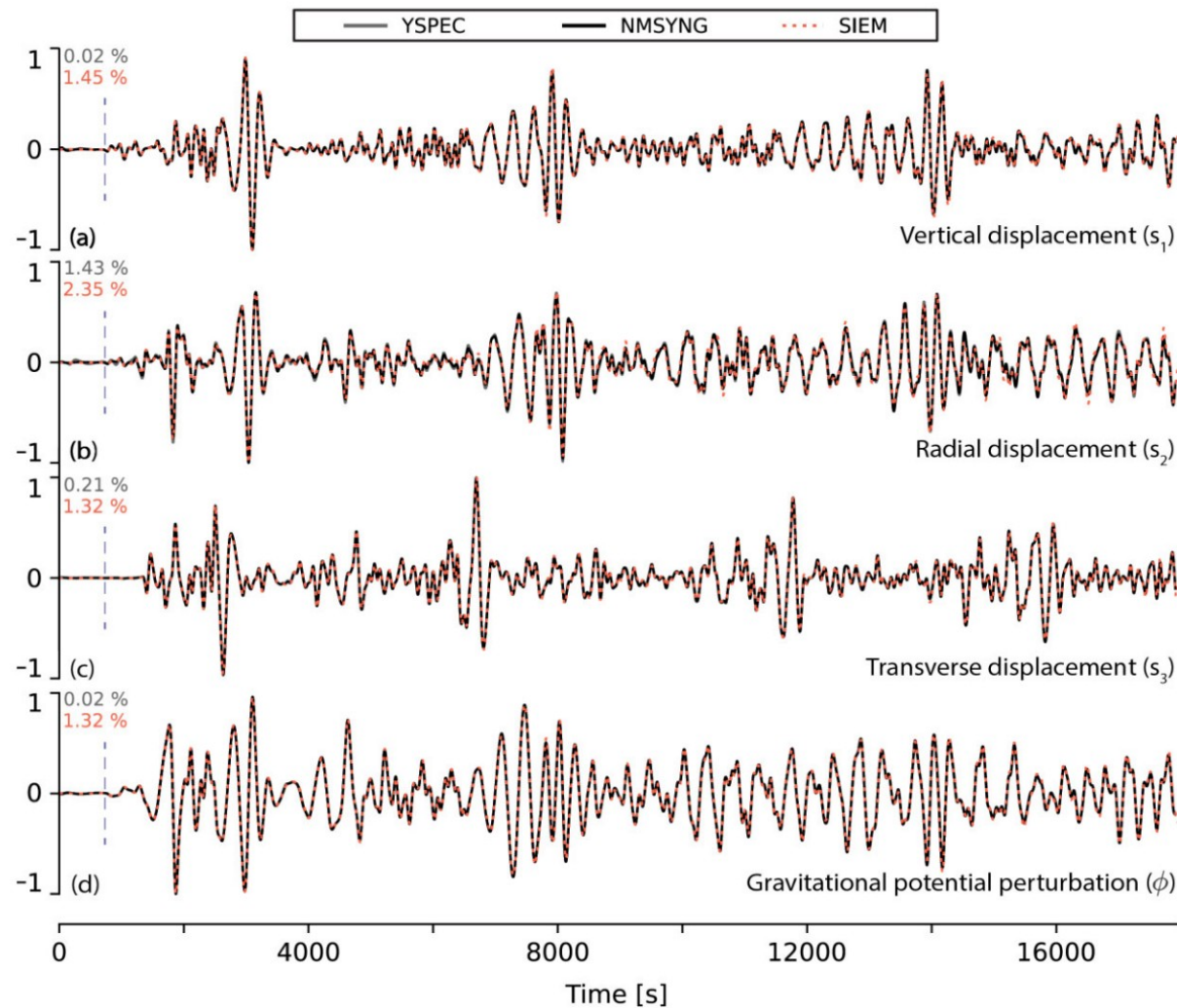
Ratio of rotational to inertial forces:

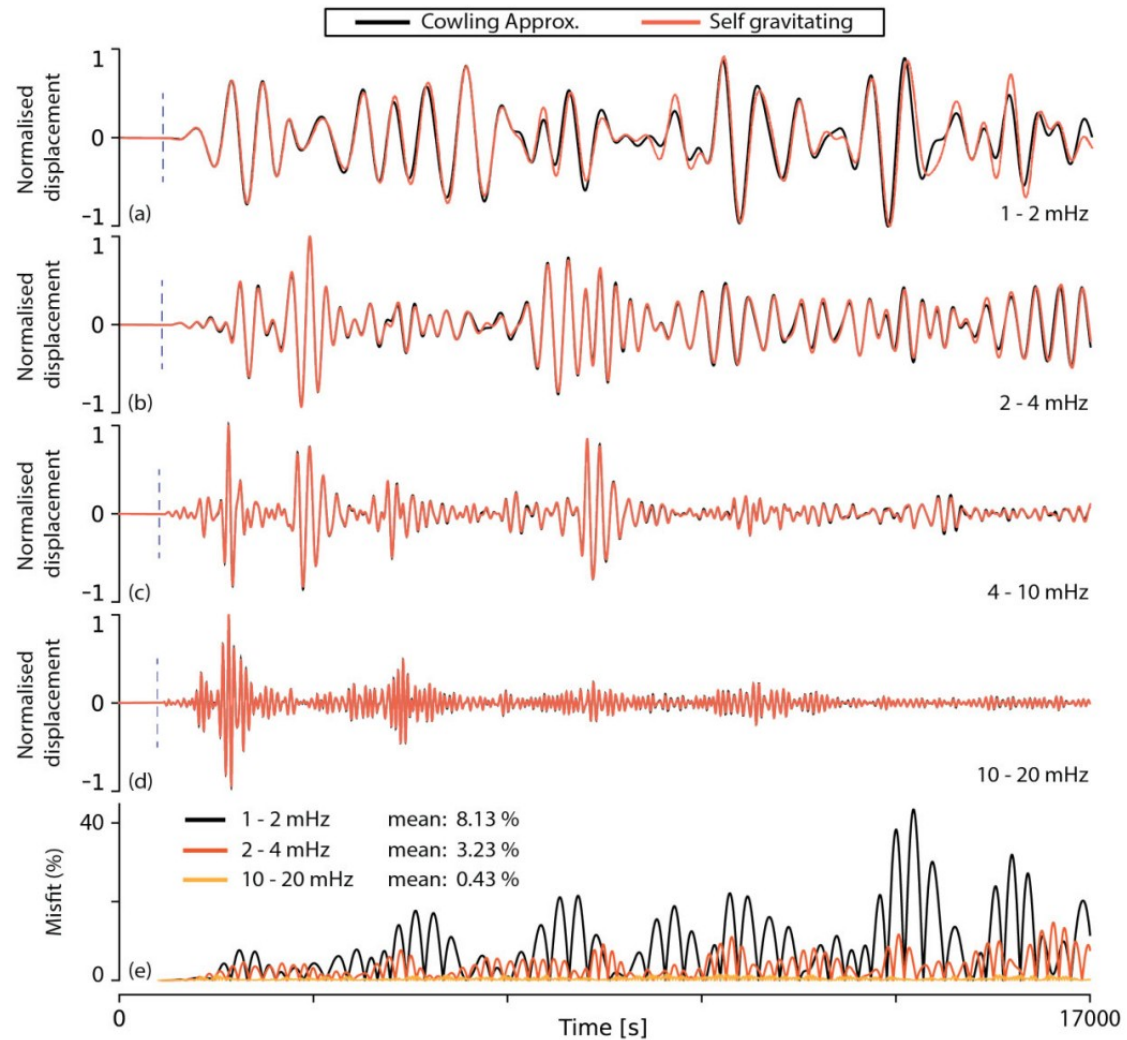
$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\bar{\rho}\Omega U}{T}}{\frac{\bar{\rho}U}{T^2}} = \Omega T$$

If time-scale is much less than a day, rotation will be negligible

Longest period seismic free oscillations have periods on one hour – rotation is not negligible, but is a relatively small effect

For longer period deformation such as that due to tidal forces, rotation becomes a dominant feature within the dynamics







Feedback form for the course

Please do fill this in before the end of term

We do take comments seriously, and they help to make the course better over time

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Equilibrium figures

From last lecture

Lagrangian in a self-gravitating planet:

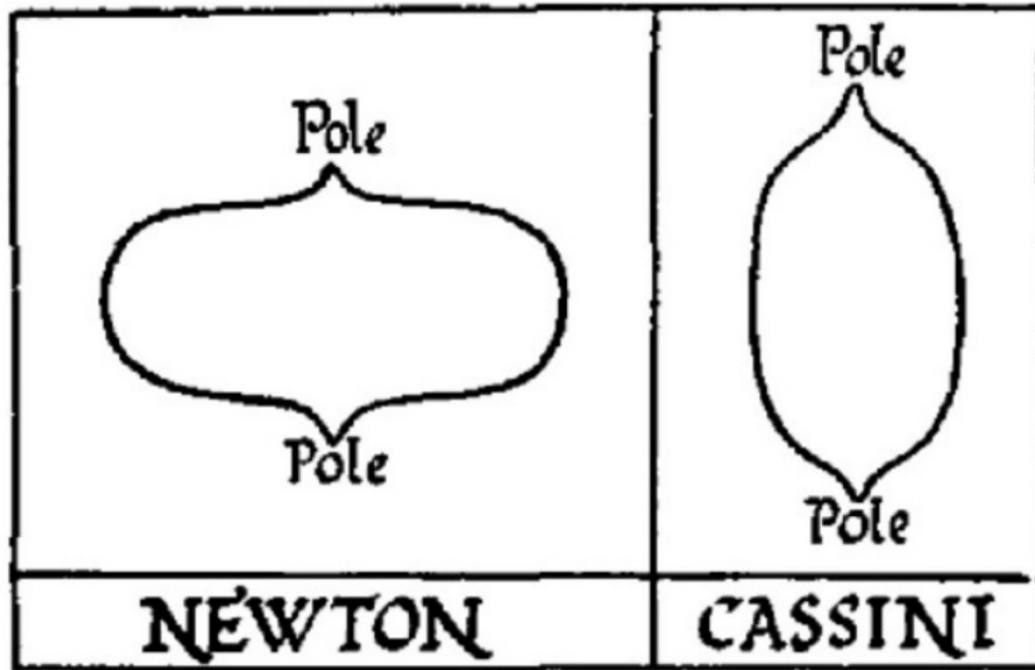
$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}||^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x}, t),$$

Decomposition of the motion relative to an approximately rotating frame:

$$\varphi_i(\mathbf{x}, t) = \Phi_i(t) + R_{ij}(t)\bar{\varphi}_j(\mathbf{x}, t),$$

Equilibrium equations:

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho\gamma_i^0 = 0,$$



“Vouz avez confirmé dans les lieux pleins d’ennui
Ce que Newton connut sans sortir de chez lui.”

Solutions of the equilibrium equations

Simplified notations

Notation from last lecture:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 = 0,$$

To simplify things we now write:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i = 0,$$

At equilibrium referential and spatial quantities coincide

The stress tensor is symmetric and so has six independent components

Under-determined system

Suppose the planet's shape and density are known

We then have **three** equations for the **six** components of stress:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i = 0,$$

This is an under-determined system and so we **cannot expect unique solutions**

More formally, there exists **non-trivial divergence-free symmetric second order tensor fields** that satisfy the correct boundary conditions

Compare with the existence of divergence-free vector fields

Hydrostatic equilibrium

Stress in a viscous fluid

The (Cauchy) stress in a compressible Newtonian fluid:

$$\sigma_{ij} = -p \delta_{ij} + \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3} \frac{\partial v_k}{\partial x_k} \delta_{ij} \right)$$

At equilibrium, the stress is **hydrostatic**:

$$\sigma_{ij} = -p \delta_{ij}$$

The same result holds in more general viscous fluids

Scaling estimate for deviatoric stress associated with convection:

$$\eta V/L \sim 10^7 \text{ Pa}$$

Pressures in the Earth's interior are of order 10^{10} Pa – deviation from hydrostatic state in Earth-like planets represents only a **small perturbation**

Hydrostatic equilibrium equations

Given the hydrostatic approximation, we have:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m + \frac{\partial p}{\partial x_i} - \rho g_i = 0.$$

This is now an **over-determined** set of equations – three equations for one unknown

Helpful to switch to a vector-calculus notation:

$$\nabla p + \rho \nabla \gamma = 0,$$

Here we have introduced the **gravity-potential**, this being the **sum of the gravitational and centrifugal potentials**

Consequences of the equations

Hydrostatic equilibrium equation:

$$\nabla p + \rho \nabla \gamma = 0,$$

Gradients of pressure and gravity potential are parallel, and so a level surface of one field is a level surface of the other

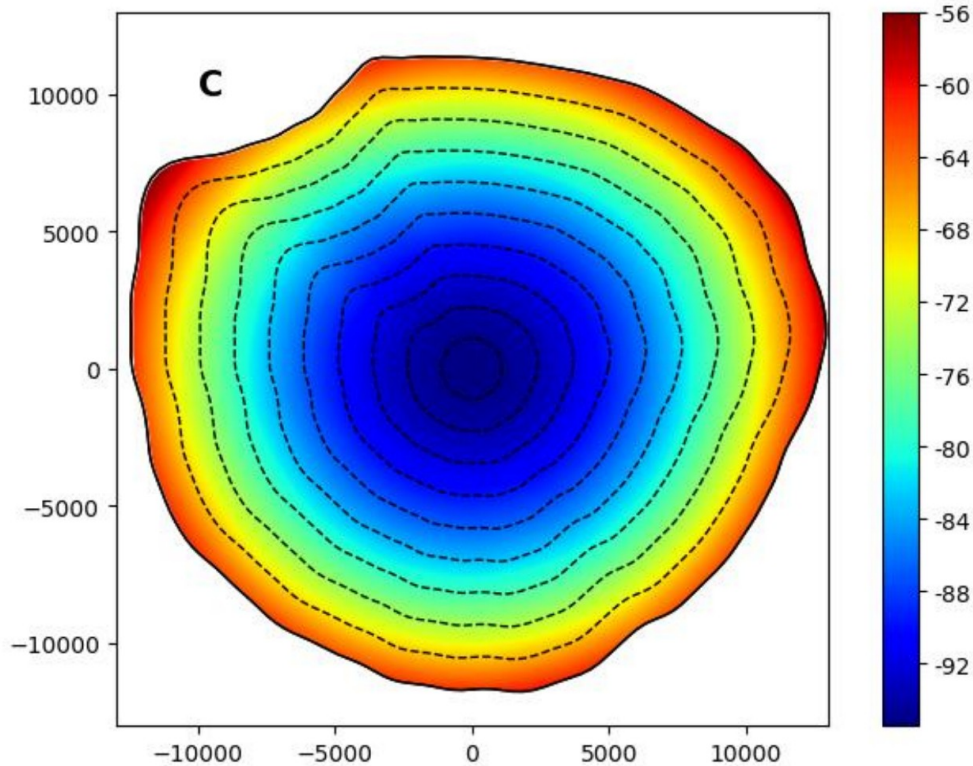
Ignoring the atmosphere, the pressure vanishes at the planet's surface, and hence this must be a level surface of the gravity potential

Taking the curl of the equation, we find:

$$\nabla \rho \times \nabla \gamma = 0,$$

This shows that level surfaces of density are also coincident with the other two fields

Gravitational potential for Phobos



Equatorial slice through the gravitational potential for Phobos calculated assuming a constant interior density

The potential field is non constant on the moon's surface, and hence this cannot be a hydrostatic equilibrium state

In general, the hydrostatic equilibrium condition imposes very strong conditions on a planet's shape, density, and rotation

Non-rotating planets

In the non-rotating case, suppose the planet is spherically symmetric

Equilibrium equation reduces to:

$$\nabla p + \rho \frac{d\phi}{dr} \hat{\mathbf{r}} = \mathbf{0},$$

There is just one non-trivial equation here, and so we can find an equilibrium solution

It can be shown that this is the **only** solution in the non-rotating case

Slowly rotating planets

By definition, **slowly rotating** means that the centrifugal potential is small relative to the gravitational potential:

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim \frac{3\Omega^2}{8\pi G\bar{\rho}}$$

For the Earth, we find:

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim 2 \times 10^{-3},$$

In such cases, we can apply first-order perturbation theory

Higher-order theories exist and are sometimes necessary

Perturbation expansions

Expand about the non-rotating solution:

$$\begin{aligned}\rho &= \rho_0 + s\rho_1 + \cdots, \\ p &= p_0 + s p_1 + \cdots, \\ \phi &= \phi_0 + s \phi_1 + \cdots, \\ \psi &= s \psi_1,\end{aligned}$$

Similarly, the free-surface radius is expanded as:

$$r(\theta, \varphi) = b + s h_1(\theta, \varphi) + \cdots,$$

Topography on internal boundaries expanded similarly, but we neglect this detail for simplicity in what follows

First-order equilibrium equations

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0},$$

Exact boundary condition on the perturbed planetary surface:

$$p[b + s h_1(\theta, \varphi) + \dots, \theta, \varphi] = 0.$$

Expanded to first-order, this reduces to:

$$p_1 + \frac{dp_0}{dr} h_1 = 0$$

$$p_1 - \rho_0 g_0 h_1 = 0.$$

Analysis of the equations

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0},$$

Take cross-product with radial unit vector:

$$\hat{\mathbf{r}} \times \nabla p_1 + \hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 = \mathbf{0}.$$

$$\hat{\mathbf{r}} \times \nabla (p_1 + \rho_0 \gamma_1) = \mathbf{0}.$$

From this, we conclude that:

$$p_1 = -\rho_0 \gamma_1$$

Analysis of the equations

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0},$$

Taking the curl of this equation, we arrive at:

$$\hat{\mathbf{r}} \times \nabla \left(\frac{d\rho_0}{dr} \gamma_1 - g_0 \rho_1 \right) = \mathbf{0},$$

And hence, we see that:

$$\rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1$$

Analysis of the equations

We have now shown that the following identities hold for the first-order perturbations:

$$p_1 = -\rho_0 \gamma_1, \quad h_1 = -\frac{\gamma_1}{g_0}, \quad \rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1.$$

All other terms can be expressed through the gravity potential perturbation

Linearised Poisson equation

Linearisation of Poisson's equation is trivial:

$$\nabla^2 \phi_1 = \begin{cases} 4\pi G \rho_1 & r < b \\ 0 & r > b \end{cases}$$

Linearisation of continuity conditions requires more work:

$$[\phi_1]_{-}^{+} = 0, \quad [\hat{\mathbf{r}} \cdot \nabla \phi_1 + 4\pi G \rho_0 h_1]_{-}^{+} = 0$$

Derivation done within the final example sheet

Modified Poisson equation

Expressing other perturbations in terms of the gravity potential:

$$\nabla^2 \phi_1 = \begin{cases} 4\pi G \frac{1}{g_0} \frac{d\rho_0}{dr} (\phi_1 + \psi_1) & r < b \\ 0 & r > b \end{cases}$$

Expressing other perturbations in terms of the gravity potential:

$$[\phi_1]_{-}^{+} = 0, \quad \left[\hat{\mathbf{r}} \cdot \nabla \phi_1 - 4\pi G \rho_0 \frac{(\phi_1 + \psi_1)}{g_0} \right]_{-}^{+} = 0.$$

Linear boundary value problem for the perturbed gravitational potential, with the centrifugal potential providing the “force terms”

Theory applies also to equilibrium tides in gaseous planets

Ellipticity function

Separation of variables shows that:

$$\phi_1 + \psi_1 = \sqrt{\frac{4\pi}{5}} \frac{2}{3} r g_0 \epsilon Y_{20}.$$

which serves to define the **ellipticity function**

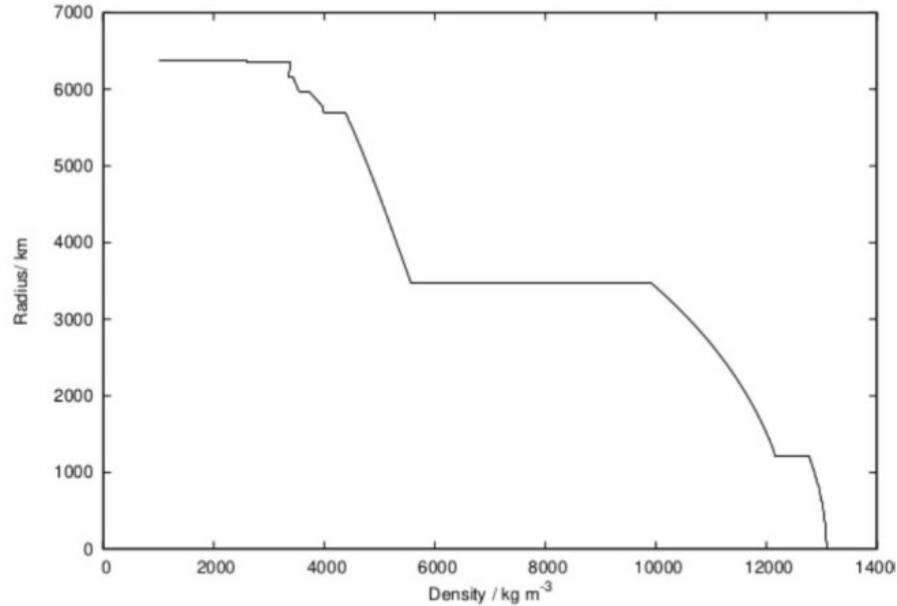
This function has the property:

$$\epsilon(b) = \frac{\text{equatorial radius} - \text{polar radius}}{\text{mean radius}}$$

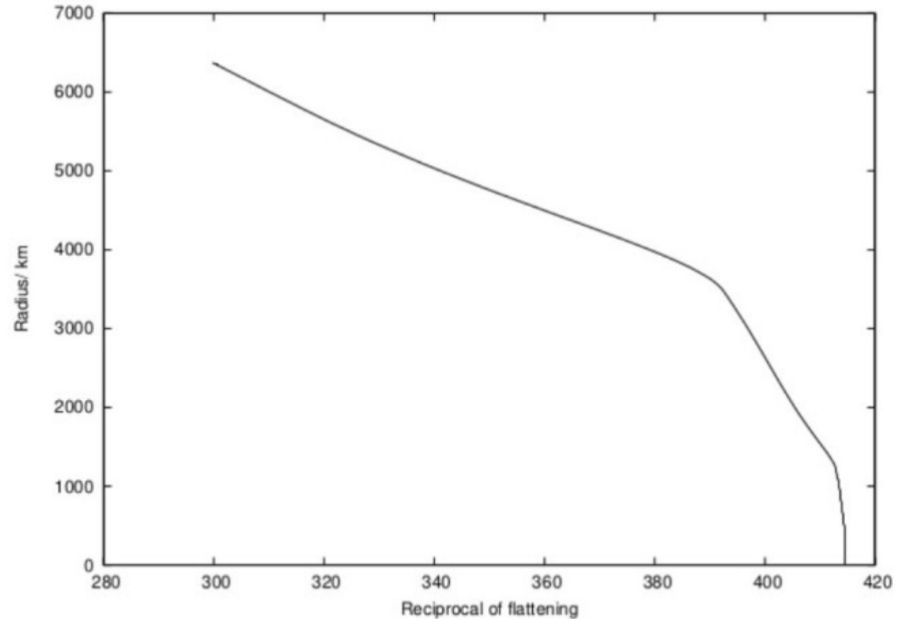
at the planets mean surface (and at internal ones too)

Surface ellipticity of the Earth is around 1/300

Numerical calculation for the Earth



Radial density profile obtained from analysis of the Earth's free oscillations – final two lectures



Corresponding value of ellipticity (or flattening) plotted here as a reciprocal

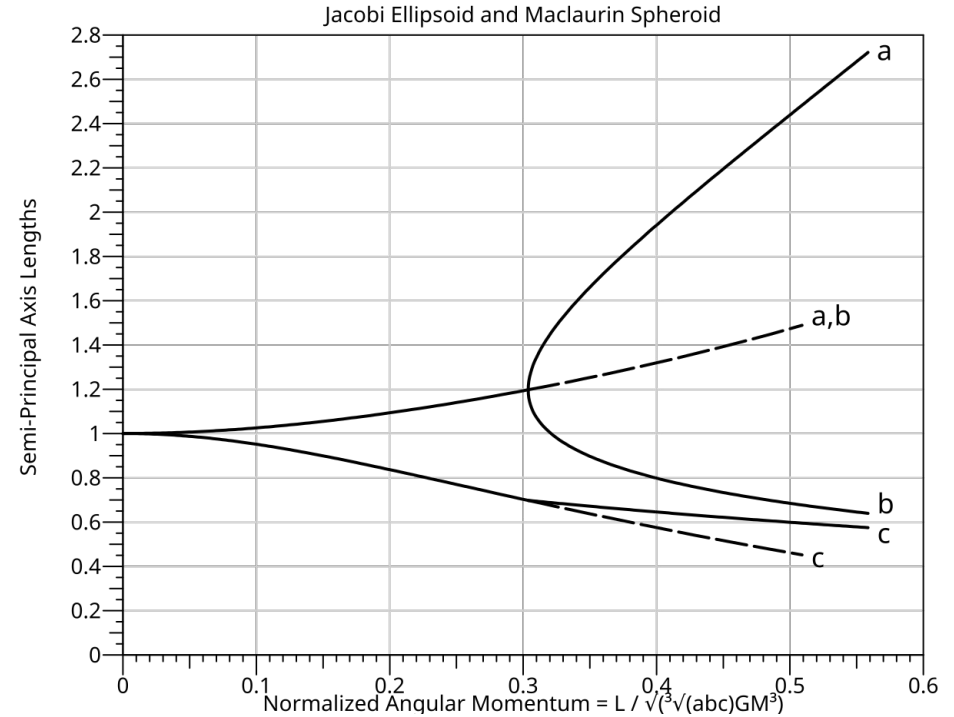
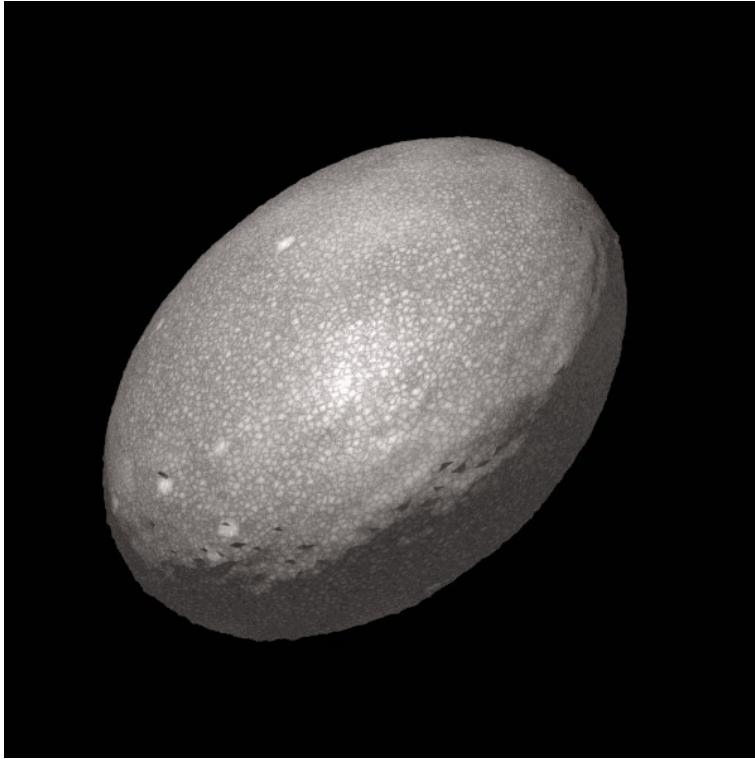
What is a planet?



A planet

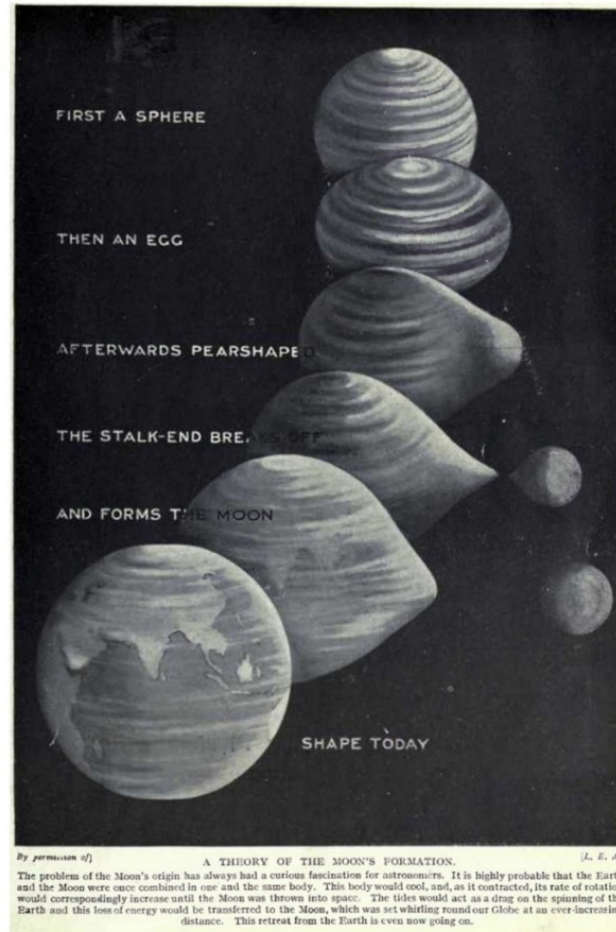


Not a planet



Equilibrium figures **do not have to be axisymmetric**

Haumea is a dwarf planet with rotation period of 4 hours and a triaxial figure



Fission theory for moon formation due to George Darwin. This idea is generally now dismissed, but it does have a real physical basis.



Feedback form for the course

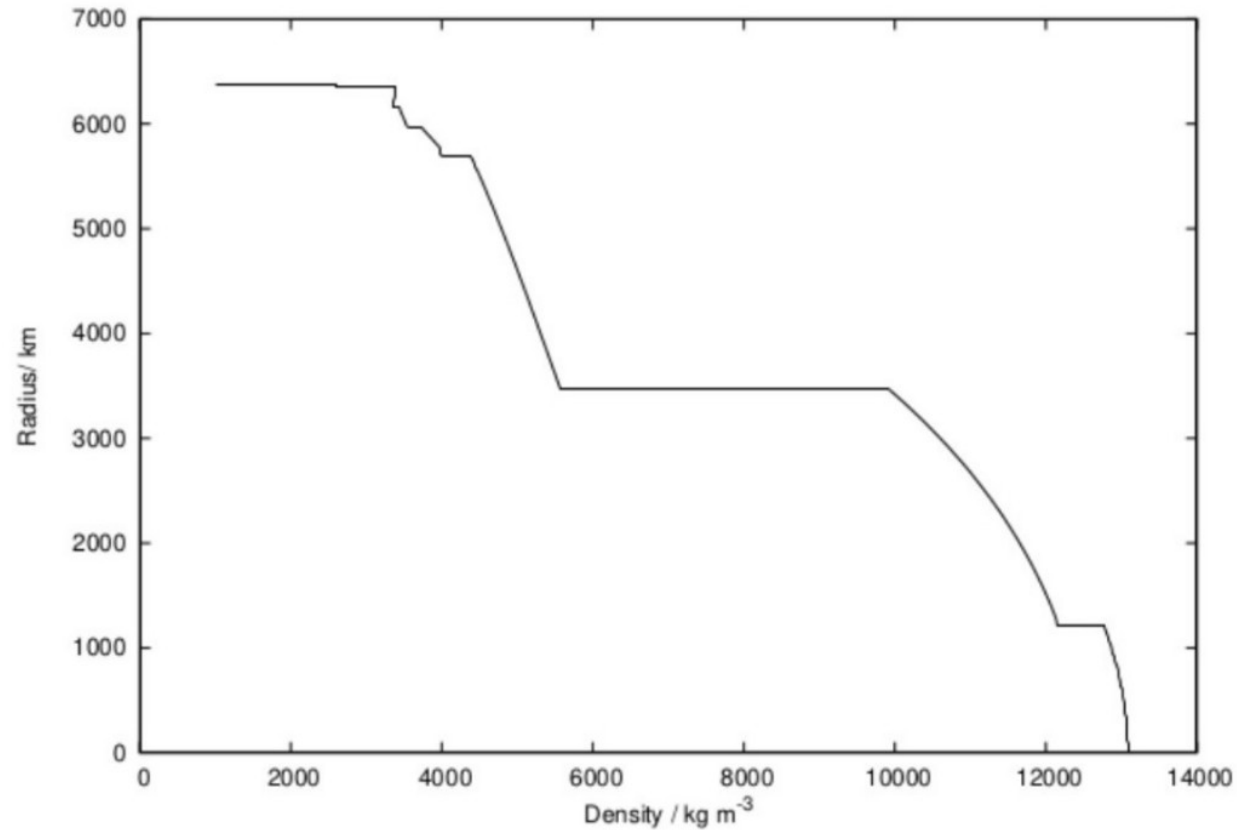
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Free oscillations of the Earth

From last lecture



Dziewonski & Anderson (1981)

Operator formalism

Equations of motion

Linearised equations with self-gravitation and rotation:

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho\gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3\mathbf{x}',$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

Simplifying notations will help a lot!

Weak form of the equations

For an arbitrary **test function**, we have:

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) - \rho\gamma_i^1 \right] w_i^* d^3\mathbf{x} = 0.$$

Integrating by parts and using the boundary conditions:

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m - \gamma_i^1 \right) w_i^* + \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \frac{\partial w_i^*}{\partial x_j} \right] d^3\mathbf{x} = 0.$$

If the latter identity holds for all test functions, then we can recover both the equations of motion and the boundary conditions

This is the **weak formulation** of the problem

Weak and strong formulations are equivalent up to technicalities

Kinetic energy form

We define the **kinetic energy form**:

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \int_M \rho w_i^* u_i \, d^3 \mathbf{x},$$

Acceleration term in the weak form can then be written:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle.$$

This **sesqui-linear form** is associated with the linear operator:

$$(P\mathbf{u})_i = \rho u_i$$

Coriolis form

We define the **Coriolis form**:

$$\langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, \mathrm{d}^3 \mathbf{x}$$

Associated with the linear operator:

$$(W\mathbf{u})_i = 2\rho \epsilon_{ijk} \Omega_j u_k$$

Potential energy form

We define the **potential energy form**:

$$\langle \mathbf{w} | H | \mathbf{u} \rangle = \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) d^3 \mathbf{x},$$

Along with the linear form linked to the stress glut:

$$\langle \mathbf{w} | \mathbf{f} \rangle = \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}.$$

These two terms incorporate **both** the equations of motion and the boundary conditions

Equations of motion

The weak form of the equations of motion can now be written:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle,$$

which is to hold for all test functions, $\mathbf{w}$

We can formally think of the problem as

$$P\ddot{\mathbf{u}} + W\dot{\mathbf{u}} + H\mathbf{u} = \mathbf{f}.$$

but the boundary conditions are then left implicit

Properties of the forms

The kinetic energy form is **self-adjoint**:

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \langle \mathbf{u} | P | \mathbf{w} \rangle^*,$$

The Coriolis form is **anti-self-adjoint**:

$$\langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, d^3\mathbf{x} = - \left(\int_M 2\rho \epsilon_{ijk} u_i^* \Omega_j w_k \, d^3\mathbf{x} \right)^* = - \langle \mathbf{u} | W | \mathbf{w} \rangle^*,$$

Properties of the forms

The potential energy form is **self-adjoint**:

$$\langle \mathbf{w} | H | \mathbf{u} \rangle = \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) d^3 \mathbf{x},$$

Elastic term is clear due to hyperelastic symmetric

Centrifugal term follows by a simple calculation – details in handout

Gravitational potential term takes more work but proceeds in the same manner – details are not in handout and are not examinable

Non-rotating eigenvalue problem

Free oscillations

For long-period seismology, rotation is a relatively small term that is often handled through perturbation theory

Weak form in a non-rotating planet:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle,$$

Free oscillations are time-harmonic solutions with no force term:

$$u_i(\mathbf{x}, t) = s_i(\mathbf{x}) e^{i\omega t}$$
$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

This is an eigenvalue problem, with squared eigenfrequency as the eigenvalue

Standard properties

You need to know how to show that:

- 1) Squared eigenfrequencies are real
- 2) Eigenfunctions in distinct eigenspaces are orthogonal:

$$\langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle = 0$$

Further properties to know but without proof:

- 1) Eigenspaces are all finite dimensional
- 2) Eigenfunctions form a complete basis for vector fields satisfying the traction-free boundary conditions

Eigenfunction expansions

We use k to label the eigenspaces, and m the eigenfunctions in each eigenspace

Within each eigenspace, we can choose an orthonormal basis, and hence:

$$\langle km|P|k'm'\rangle = \delta_{kk'}\delta_{mm'},$$

Completeness of the eigenfunctions means that we can write:

$$\mathbf{u} = \sum_{km} \langle km|P|\mathbf{u}\rangle |km\rangle,$$

This is the **eigenfunction expansion** of the vector field

Eigenfunction expansions

Equations of motion in the frequency-domain:

$$-\omega^2 \langle \mathbf{w} | P | \tilde{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \tilde{\mathbf{u}} \rangle = \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle.$$

Expanding the solution in the eigenfunction basis:

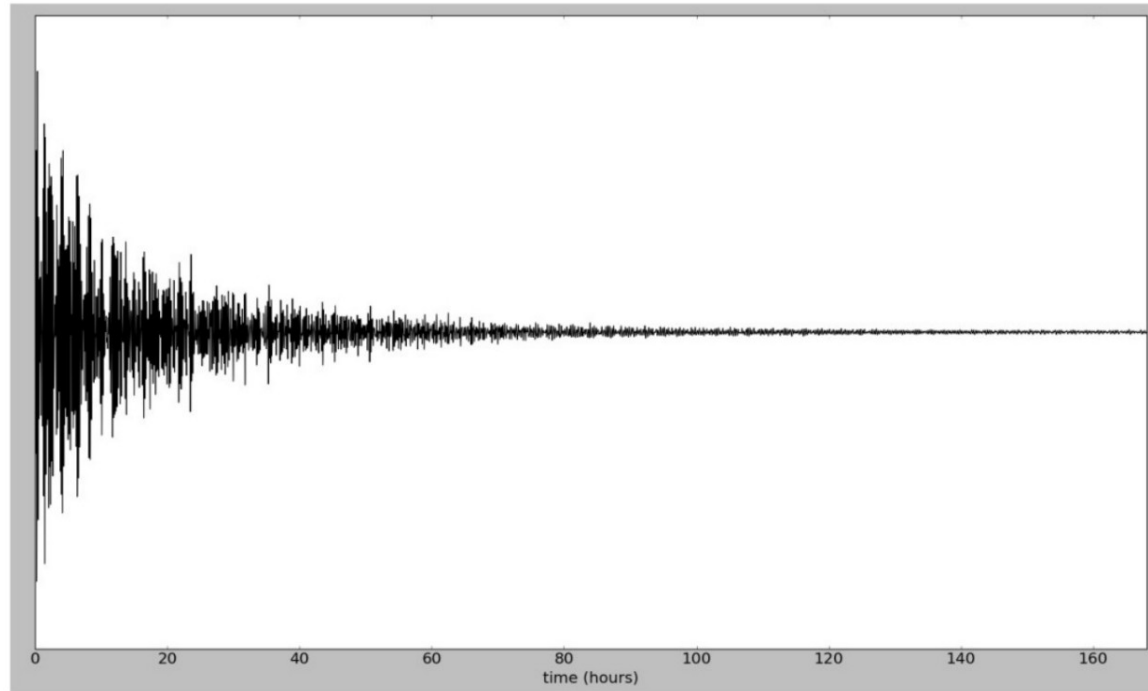
$$\tilde{\mathbf{u}} = \sum_{k'm'} \langle k'm' | P | \tilde{\mathbf{u}} \rangle |k'm'\rangle,$$

Frequency-domain solution readily derived:

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} |km\rangle,$$

Time-domain solution obtained by an inverse Fourier transformation – but details are not covered and not examinable

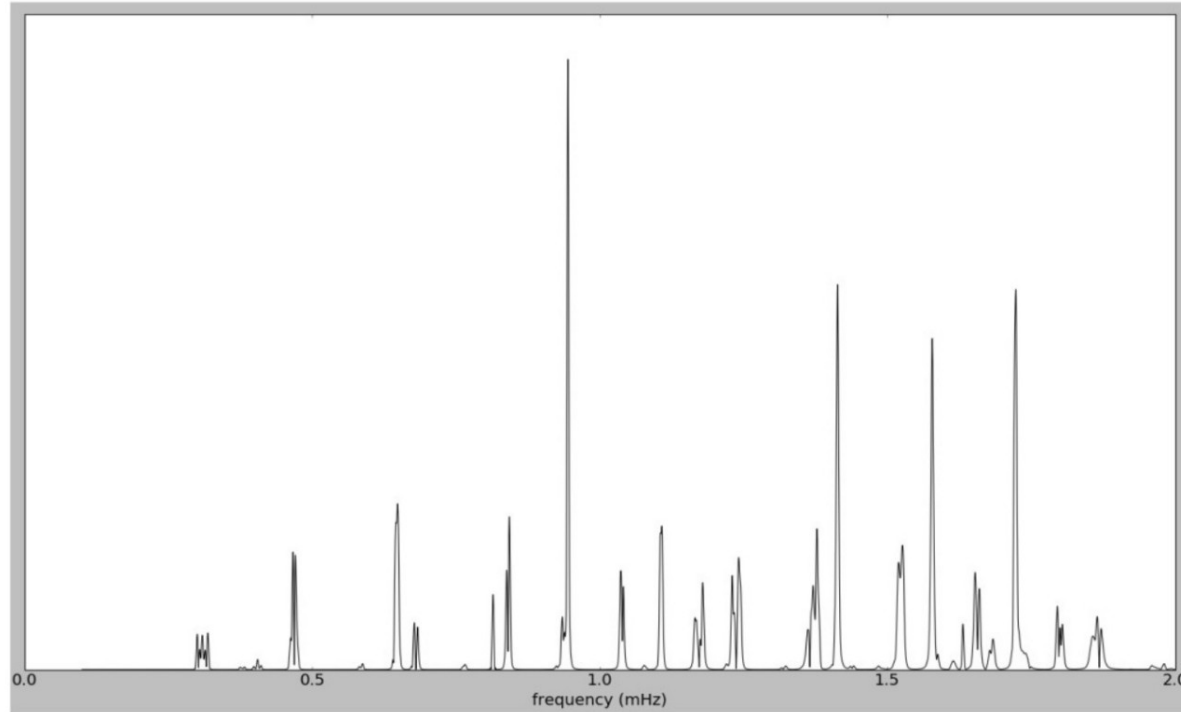
Long time-series



Several day long time-series following a large earthquake (Bolivia 1994)

The effects of attenuation are clearly seen within real data

Corresponding amplitude spectra



Resonance peaks (or poles) associated with each eigenfrequency

For good resolution in the frequency-domain, we need **long time series**

Gravitational stability

Rayleigh quotient

The non-rotating eigenvalue problem:

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

Taking the test function to equal the eigenfunction:

$$\omega_k^2 = \langle km | H | km \rangle.$$

Sign of the squared eigenfrequency determined by the potential energy form

Positive values imply **stable** oscillations about equilibrium

Negative values imply **unstable**, exponentially growing perturbations

Zero values correspond to **linear growth in time** (double pole at the origin)

Non-gravitating case

Potential energy form simplifies such that:

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x},$$

Vanishes for linearised rigid body motions:

$$u_i(\mathbf{x}) = a_i + A_{ij}x_j,$$

Six-dimensional eigenspace at zero-frequency, but not excited by earthquakes:

$$\langle \mathbf{w} | \mathbf{f} \rangle = \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x},$$

Non-gravitating case

Potential energy form simplifies such that:

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x},$$

Non-negative values so long as we have **uniform pointwise stability**:

$$A_{ijkl}(\mathbf{x}) e_{ij}^* e_{kl} \geq k e_{ij}^* e_{ij}$$

Implies elastic wave speeds are all positive

In isotropic case, implies P-wave speed greater than for S-waves

Gravitational contribution

Quantitative treatment possible, but difficult

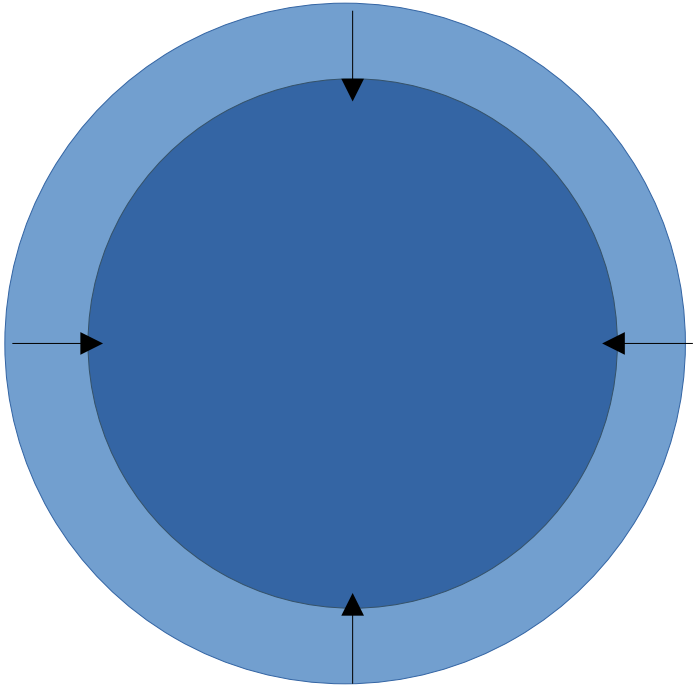
This shows that there can be a finite number of eigenspaces with negative squared eigenfrequencies

Existence depends on interplay between gravitational and elastic forces

Elastic forces always resist further deformation

Gravitational forces can encourage it

A spherically symmetric compression would lower the (always negative) gravitational binding energy, while raises the elastic potential energy





Feedback form for the course

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Mode splitting and coupling

From last lecture

Weak form of the equations of motion can be written:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle,$$

Eigenvalue problem in non-rotating case:

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

Eigenfunction expansion solution in frequency-domain:

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} |km\rangle,$$

Spherically symmetric Earth models

Angular momentum operators

Gradient in spherical polar co-ordinates:

$$\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \frac{1}{r} \hat{\boldsymbol{\theta}} \frac{\partial}{\partial \theta} + \frac{1}{r \sin \theta} \hat{\boldsymbol{\phi}} \frac{\partial}{\partial \phi},$$

Used to define the **tangential gradient operator** on the unit sphere:

$$\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \frac{1}{r} \nabla_1,$$

Associated operators related to those for angular momentum:

$$\hat{\mathbf{r}} \times \nabla_1 \propto \hat{\mathbf{L}}, \quad \nabla_1^2 \propto \hat{\mathbf{L}}^2$$

Spherical harmonics

Simultaneous eigenfunctions of commuting operators:

$$\nabla_1^2 Y_{lm} = -l(l+1)Y_{lm}, \quad \frac{1}{i} \frac{\partial Y_{lm}}{\partial \varphi} = mY_{lm},$$

The **degree**, l , is a non-negative integer

For degree, l , the **order** is an integer between $-l$ and l

Orthonormality condition:

$$\int_{S^2} Y_{lm}^* Y_{l'm'} dS = \delta_{ll'} \delta_{mm'}.$$

Complete basis for scalar functions on the unit sphere

Vector spherical harmonics

To work with vector fields, we defined:

$$\mathbf{A}_{lm} = Y_{lm} \hat{\mathbf{r}}, \quad \mathbf{B}_{lm} = \nabla_1 Y_{lm}, \quad \mathbf{C}_{lm} = -\hat{\mathbf{r}} \times \nabla_1 Y_{lm}.$$

Orthogonality relations:

$$\begin{aligned} \int_{\mathbb{S}^2} \mathbf{A}_{l'm'}^* \cdot \mathbf{A}_{lm} dS &= \delta_{ll'} \delta_{mm'}, \\ \int_{\mathbb{S}^2} \mathbf{B}_{l'm'}^* \cdot \mathbf{B}_{lm} dS &= l(l+1) \delta_{ll'} \delta_{mm'}, \\ \int_{\mathbb{S}^2} \mathbf{C}_{l'm'}^* \cdot \mathbf{C}_{lm} dS &= l(l+1) \delta_{ll'} \delta_{mm'}, \end{aligned}$$

Complete basis for vector fields:

$$\mathbf{u} = \sum_{lm} (U_{lm} \mathbf{A}_{lm} + V_{lm} \mathbf{B}_{lm} + W_{lm} \mathbf{C}_{lm}),$$

Eigenvalue problem

Eigenvalue in non-rotating case:

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

If the model is spherically symmetric we can use **separation of variables**

Angular dependence expressed in terms of vector spherical harmonics

For each degree and order, we get a **radial ODE eigenvalue problem**

Huge amount of algebra and the need to work in detail with vector and tensor spherical harmonic expansions

We will just summarise the main results

Toroidal modes

Two classes of eigenfunctions emerge from the separation of variables

Toroidal modes have eigenfunctions of the form:

$$\mathbf{s}(r, \theta, \varphi) = W(r) \mathbf{C}_{lm}(\theta, \varphi),$$

Radial part, W , depends on degree but not order – results in $(2l+1)$ -dimensional eigenspaces, this being characteristic of rotationally invariant systems

Eigenfunctions vanish in fluid regions (analogous to S-waves)

Displacements tangential to spherical surfaces, gravity not involved in their dynamics

Spheroidal modes

Spheroidal modes have the form:

$$\mathbf{s}(r, \theta, \varphi) = U(r)\mathbf{A}_{lm}(\theta, \varphi) + V(r)\mathbf{B}_{lm}(\theta, \varphi),$$

Radial functions again independent of order – **(2l+1)-dimensional eigenspaces**

Eigenfunctions can be non-zero in both solid and fluid regions

Radial and tangential components to the displacement, self-gravity contributes to their dynamics

Toroidal eigenvalue problem

Spheroidal and toroidal modes are orthogonal due to properties of VSH

For toroidal modes of degree l and order m , we can take test functions:

$$\mathbf{w}(r, \theta, \varphi) = W'(r) \mathbf{C}_{lm}(\theta, \varphi)$$

It is then easy to show that:

$$\langle \mathbf{w} | P | \mathbf{s} \rangle = \zeta^2 \int_0^b \rho W'^* W r^2 dr, \quad \zeta = \sqrt{l(l+1)}$$

Potential energy form similarly reduced to radial integral, but calculation harder

Toroidal eigenvalue problem

Weak form of the eigenvalue problem:

$$\begin{aligned} & -\omega^2 \int_0^b \rho W'^* W r^2 \, dr + \int_0^b \mu \left(r \frac{dW'^*}{dr} - W'^* \right) \left(r \frac{dW}{dr} - W \right) \, dr \\ & + (\zeta^2 - 2) \int_0^b \mu W'^* W \, dr = 0, \end{aligned}$$

Equivalent to 2nd order ODE eigenvalue problem (or Sturm Liouville type)

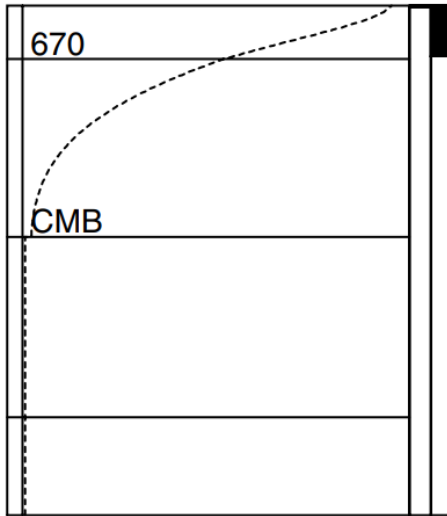
Closed-form solution in homogeneous planet using spherical Bessel functions

More generally, numerical methods needed

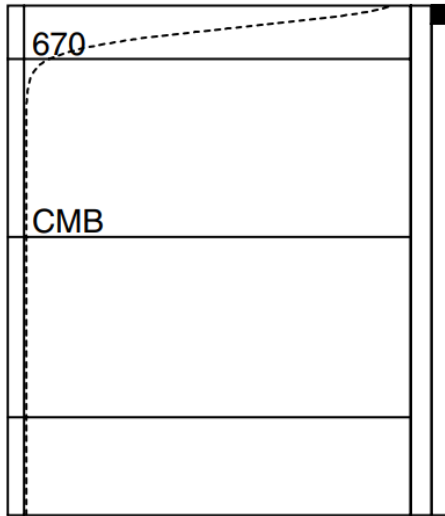
For each l , we get an infinite set of **overtones** labelled by an integer, $n = 0, 1, 2, \dots$

Toroidal eigenfunctions

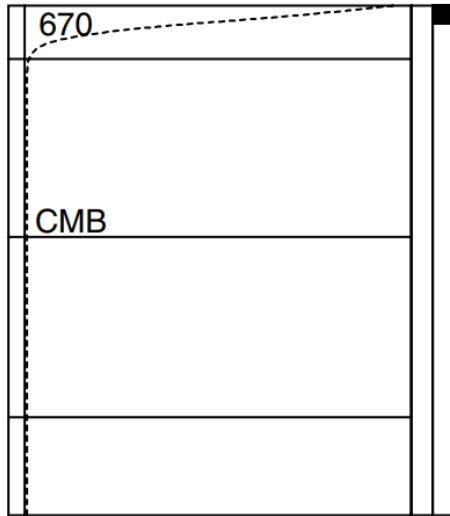
${}_0T_{10}$ Period = 619.9s $Q = 173.2$
Surface Eigenfunction



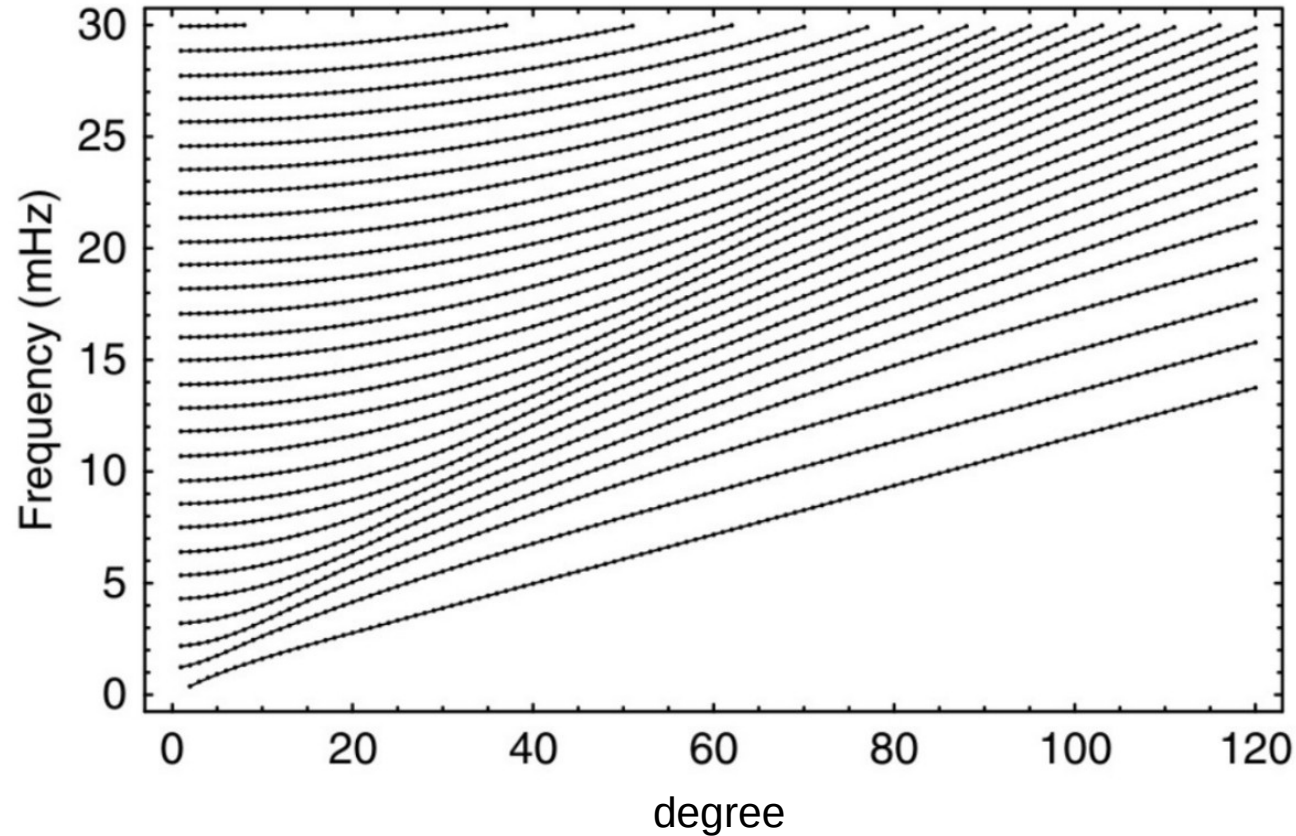
${}_0T_{50}$ Period = 164.6s $Q = 131.1$
Surface Eigenfunction



${}_0T_{100}$ Period = 86.5s $Q = 139.2$
Surface Eigenfunction



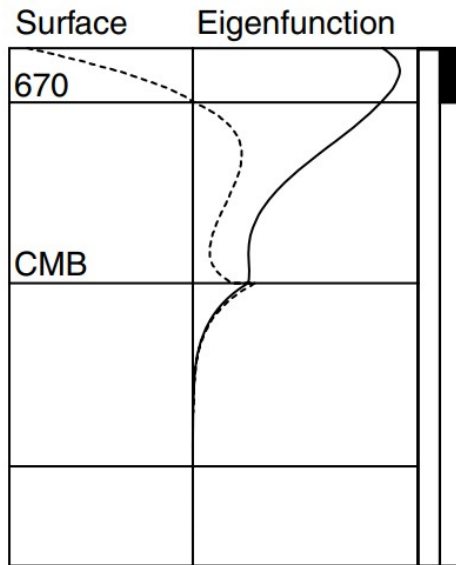
Toroidal mode eigenfrequencies



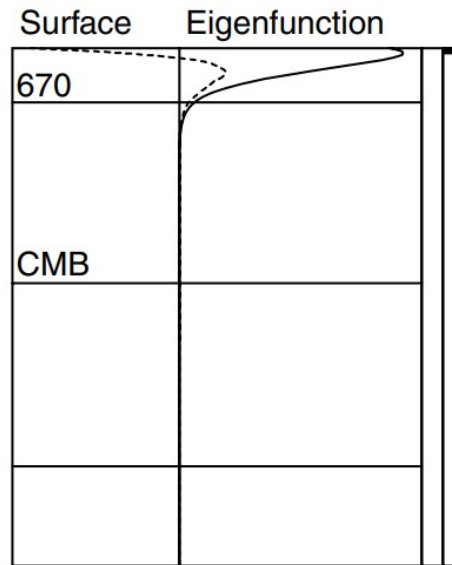
Woodhouse & Deuss (2007)

Spheroidal mode eigenfunctions

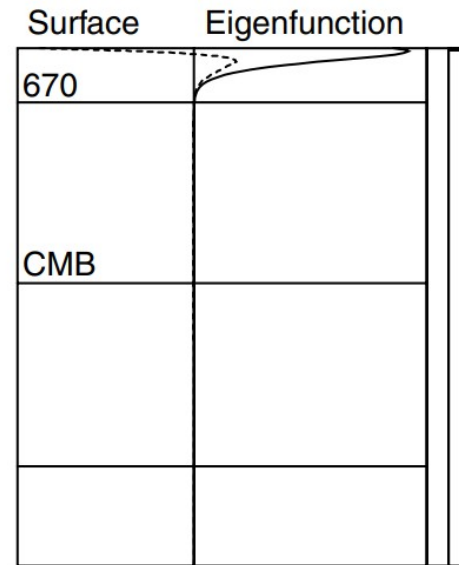
${}_0S_{10}$ Period = 579.2s Q = 327.8



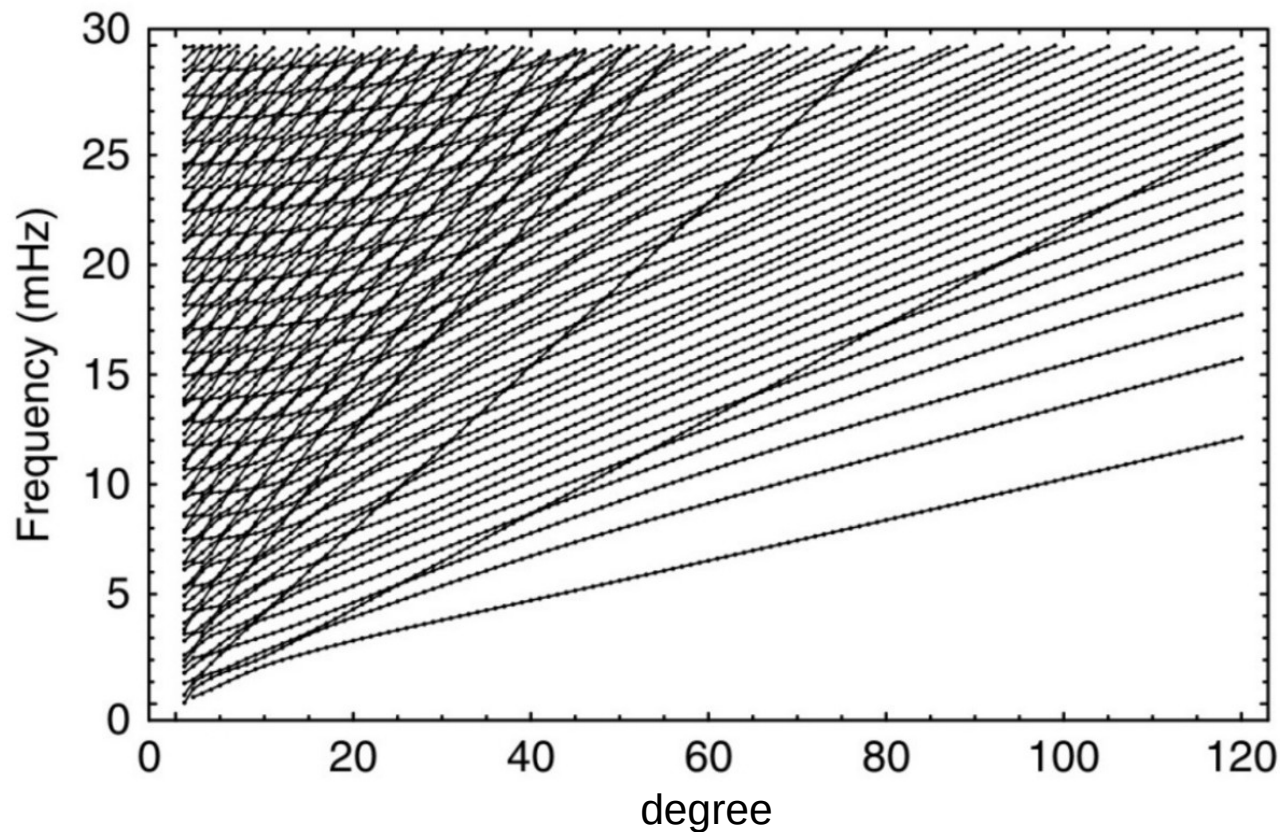
${}_0S_{50}$ Period = 178.2s Q = 143.2



${}_0S_{100}$ Period = 97.6s Q = 118.0



Spheroidal mode eigenfrequencies



Woodhouse & Deuss (2007)

Multiplets and singlets

Each eigenspace labelled by **three integers**:

- 1) **Mode type**, q , with 1 = spheroidal, 2 = toroidal
- 2) **Degree**, l , a non-negative integer
- 3) **Overtone number**, n , this being a non-negative integer

We call each such eigenspace a **multiplet**

Within each multiplet of degree l there are then $(2l+1)$ eigenfunctions labelled by their order, m , lying between $-l$ and l

Each eigenfunction is known as a **singlet**

To denote singlets, we use the notations:

$$|qlnm\rangle, \quad |km\rangle$$

First-order splitting theory

Perturbed eigenvalue problem

Suppose we have solved the spherical reference problem:

$$-\omega_k^2 \langle \mathbf{w} | P_0 | km \rangle + \langle \mathbf{w} | H_0 | km \rangle = 0,$$

We consider lateral heterogeneity as a first-order perturbation:

$$-\omega^2 [\langle \mathbf{w} | P_0 | \mathbf{s} \rangle + s \langle \mathbf{w} | P_1 | \mathbf{s} \rangle + \dots] + [\langle \mathbf{w} | H_0 | \mathbf{s} \rangle + s \langle \mathbf{w} | H_1 | \mathbf{s} \rangle + \dots] = 0.$$

Following degenerate perturbation theory, we look for solutions of the form:

$$\omega = \omega_k + s \delta \omega + \dots$$

$$\mathbf{s} = \sum_{m=-l}^l a_m |km\rangle + s \delta \mathbf{s} + \dots$$

Splitting of the degeneracy

Following the standard argument, the calculation is reduced to solution of an $(2l+1)$ -dimensional Hermitian eigenvalue problem

$$\frac{1}{2\omega_k} \sum_{m=-l}^l [-\omega_k^2 \langle km' | P_1 | km \rangle + \langle km' | H_1 | km \rangle] a_m = \delta\omega a_{m'}$$

This gives $(2l+1)$ values for the perturbed eigenfrequencies along with a zeroth-order eigenfunction for each

Generically, the degeneracy will be removed, with the multiplet **splitting** into $2l+1$ distinct eigenfrequencies

This is just like the **Zeeman effect** for the hydrogen atom

Rotational splitting

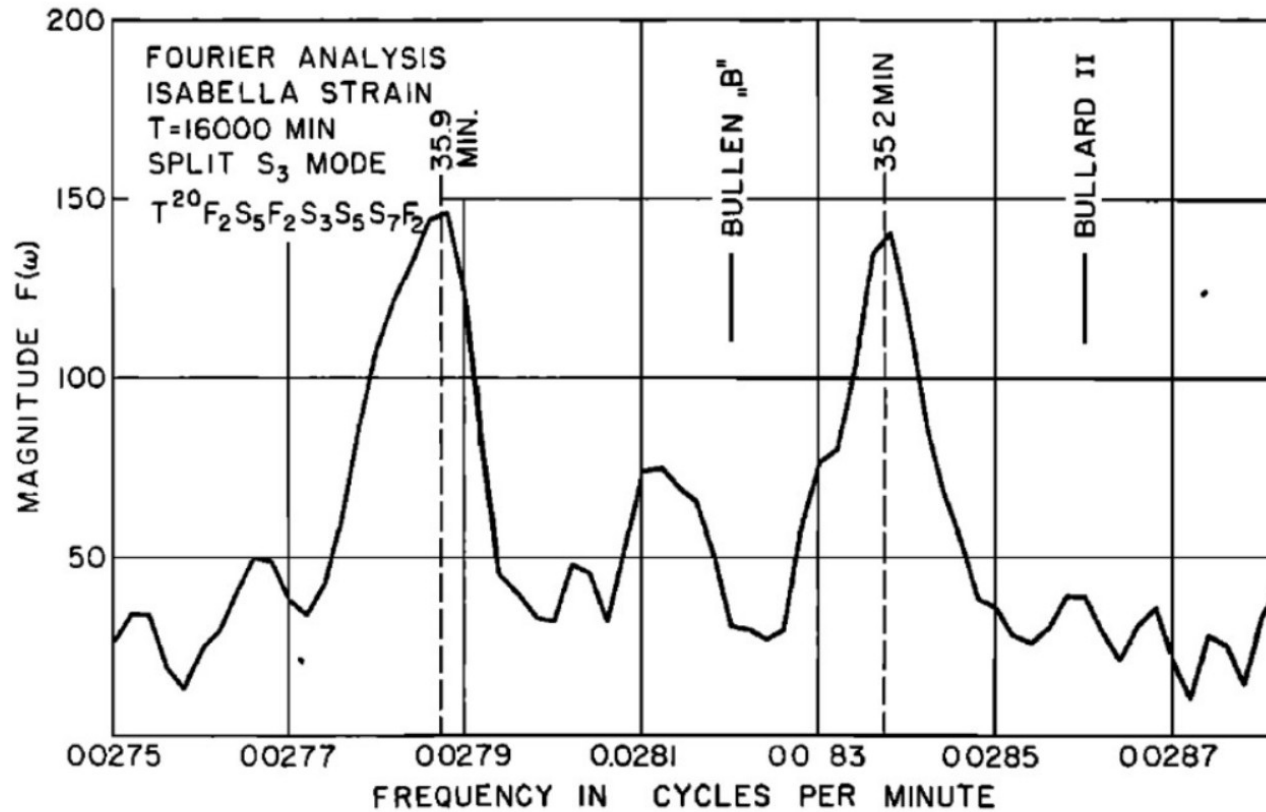
Rotational effects can be incorporated to first-order accuracy in the same way:

$$\frac{1}{2\omega_k} \sum_{m=-l}^l [-\omega_k^2 \langle km' | P_1 | km \rangle + i\omega_k \langle km' | W | km \rangle + \langle km' | H_1 | km \rangle] a_m = \delta\omega a_{m'}$$

Various **selection rules** can be derived to determine the manner in which heterogeneity and rotation cause splitting

Extensions to **quasi-degenerate perturbation theory** are possible to account for coupling between two or more nearly degenerate multiplets

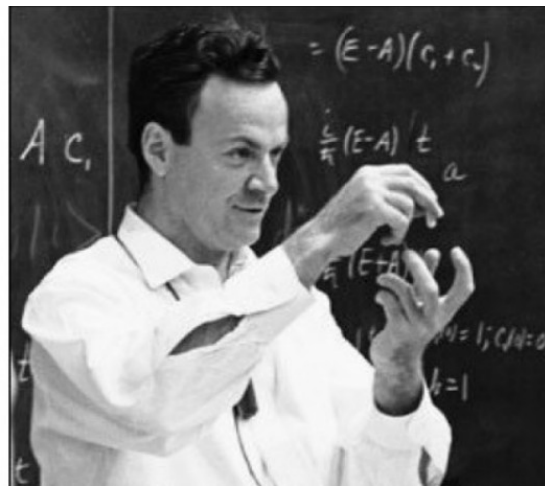
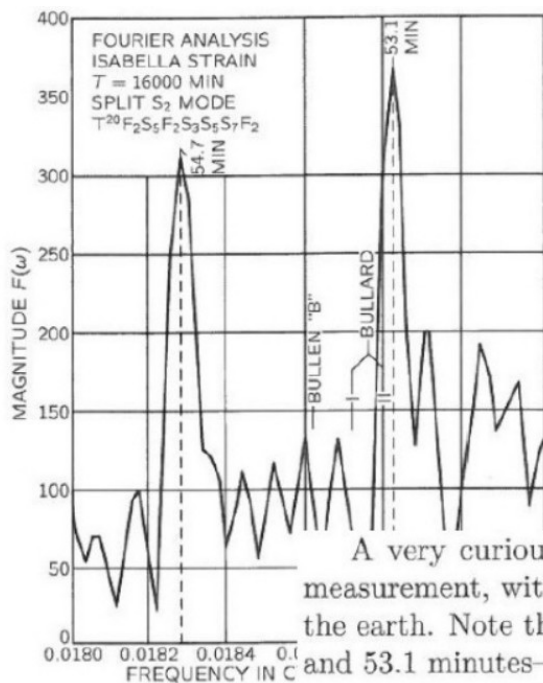
Spherically symmetric density



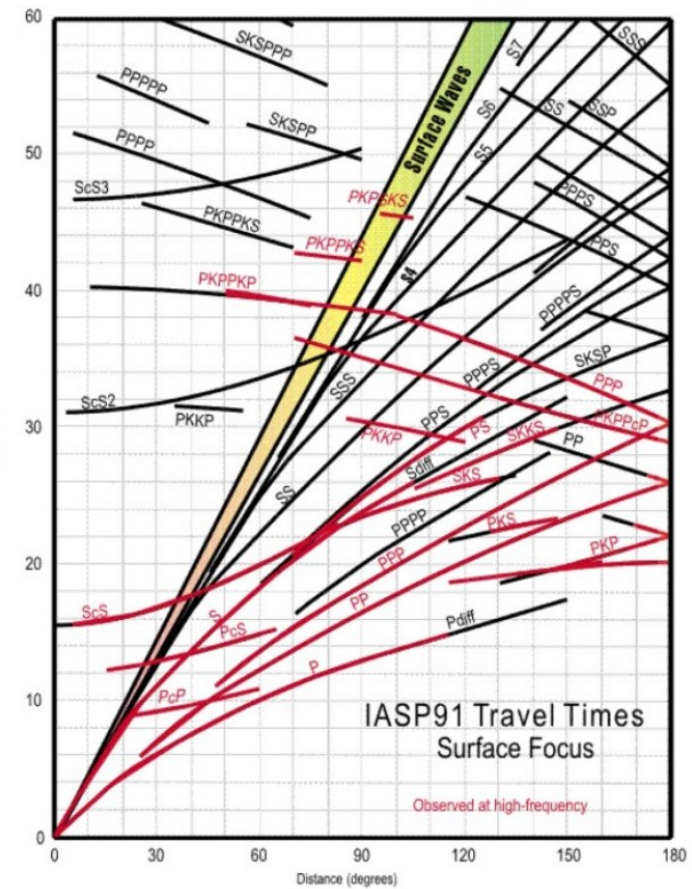
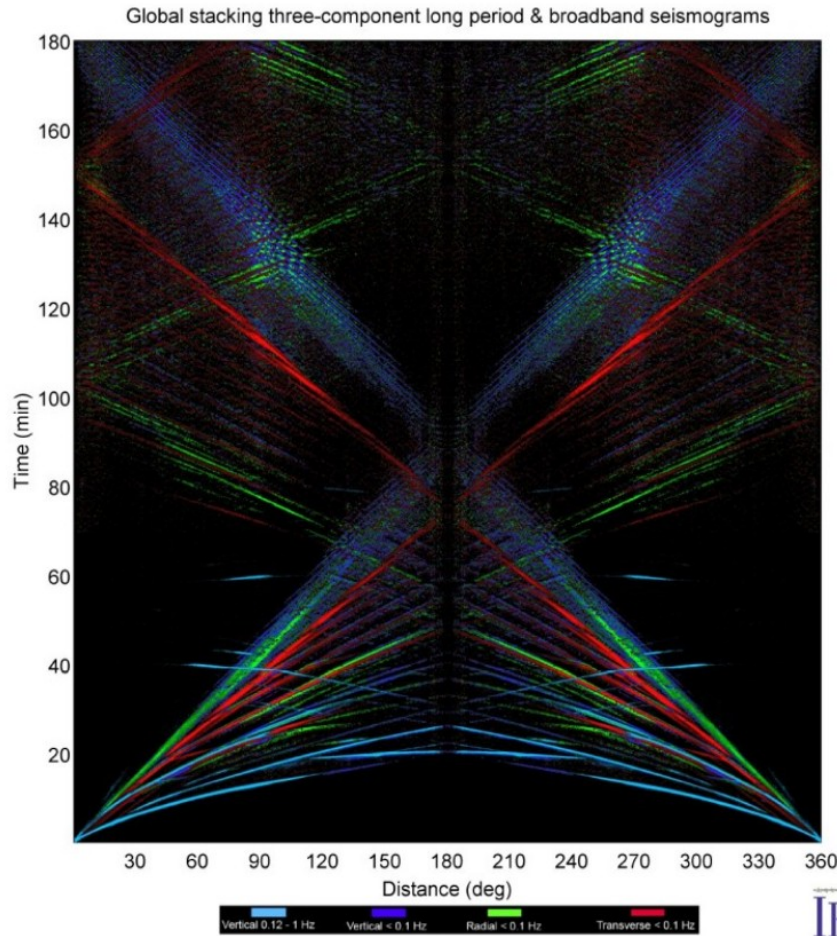
HUGO BENIOFF, FRANK PRESS, AND STEWART SMITH

FEBRUARY 1961

First observations of free oscillation spectra following the largest ever recorded earthquake



A very curious point is revealed in Fig. 51-8, which shows a very careful measurement, with better resolution of the lowest mode, the ellipsoidal mode of the earth. Note that it is not a single maximum, but a double one, 54.7 minutes and 53.1 minutes—slightly different. The reason for the two different frequencies was not known at the time that it was measured, although it may have been found in the meantime. There are at least two possible explanations: One would be that there may be asymmetry in the earth's distribution, which would result in two similar modes. Another possibility, which is even more interesting, is this: Imagine the waves going around the earth in two directions from the source. The speeds will not be equal because of effects of the rotation of the earth in the equations of motion, which have not been taken into account in making the analysis. Motion in a rotating system is modified by Coriolis forces, and these may cause the observed splitting.



Travel times for the same epicentral angle can be averaged to try to remove the effect of laterally varying structure prior to inversions

Diagonal sum rule

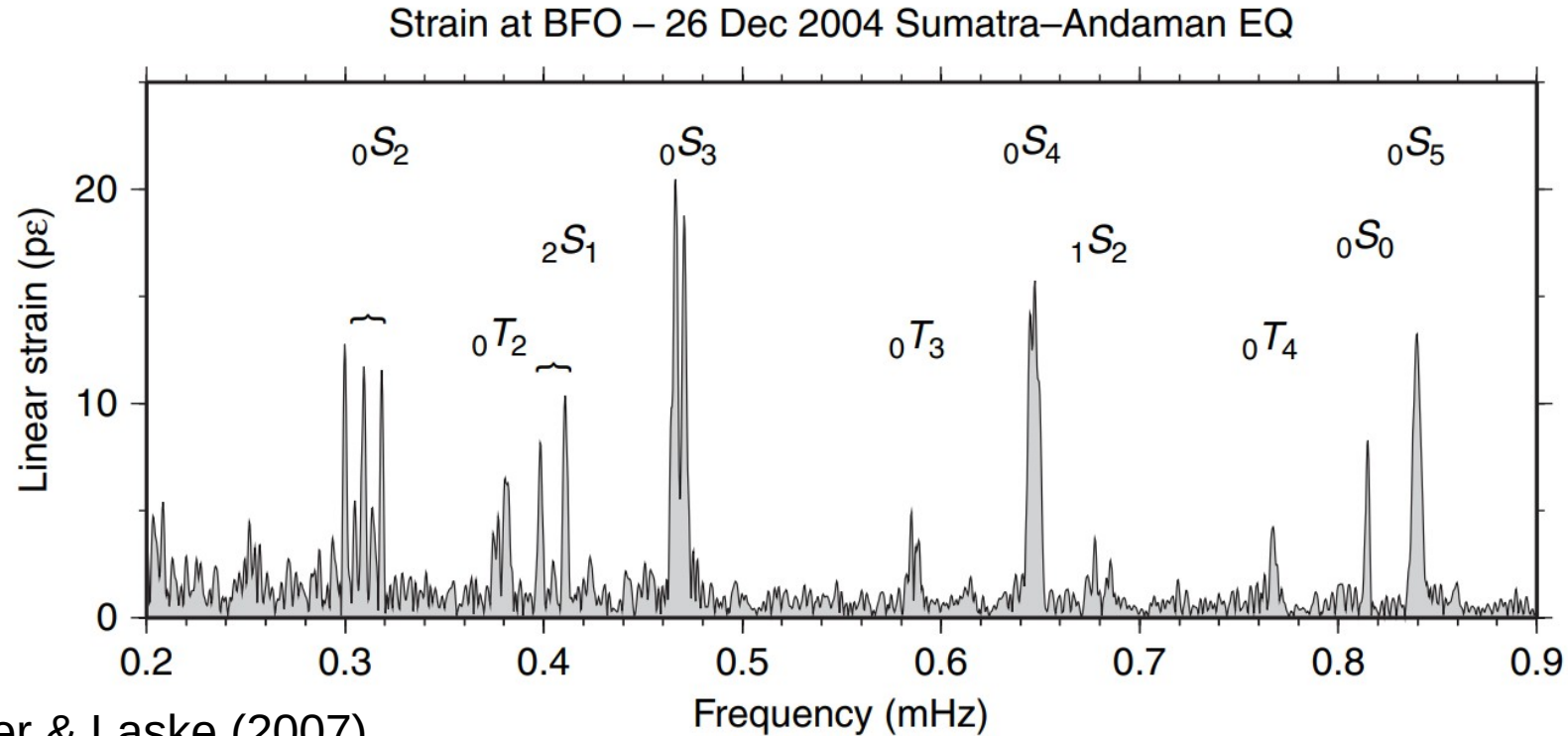
Eigenvalue problem associated with degenerate perturbation theory:

$$\frac{1}{2\omega_k} \sum_{m=-l}^l [-\omega_k^2 \langle km' | P_1 | km \rangle + i\omega_k \langle km' | W | km \rangle + \langle km' | H_1 | km \rangle] a_m = \delta\omega a_{m'}$$

Using symmetry arguments, it can be shown that the splitting matrix has zero trace – this implies that the eigenfrequency perturbations sum to zero

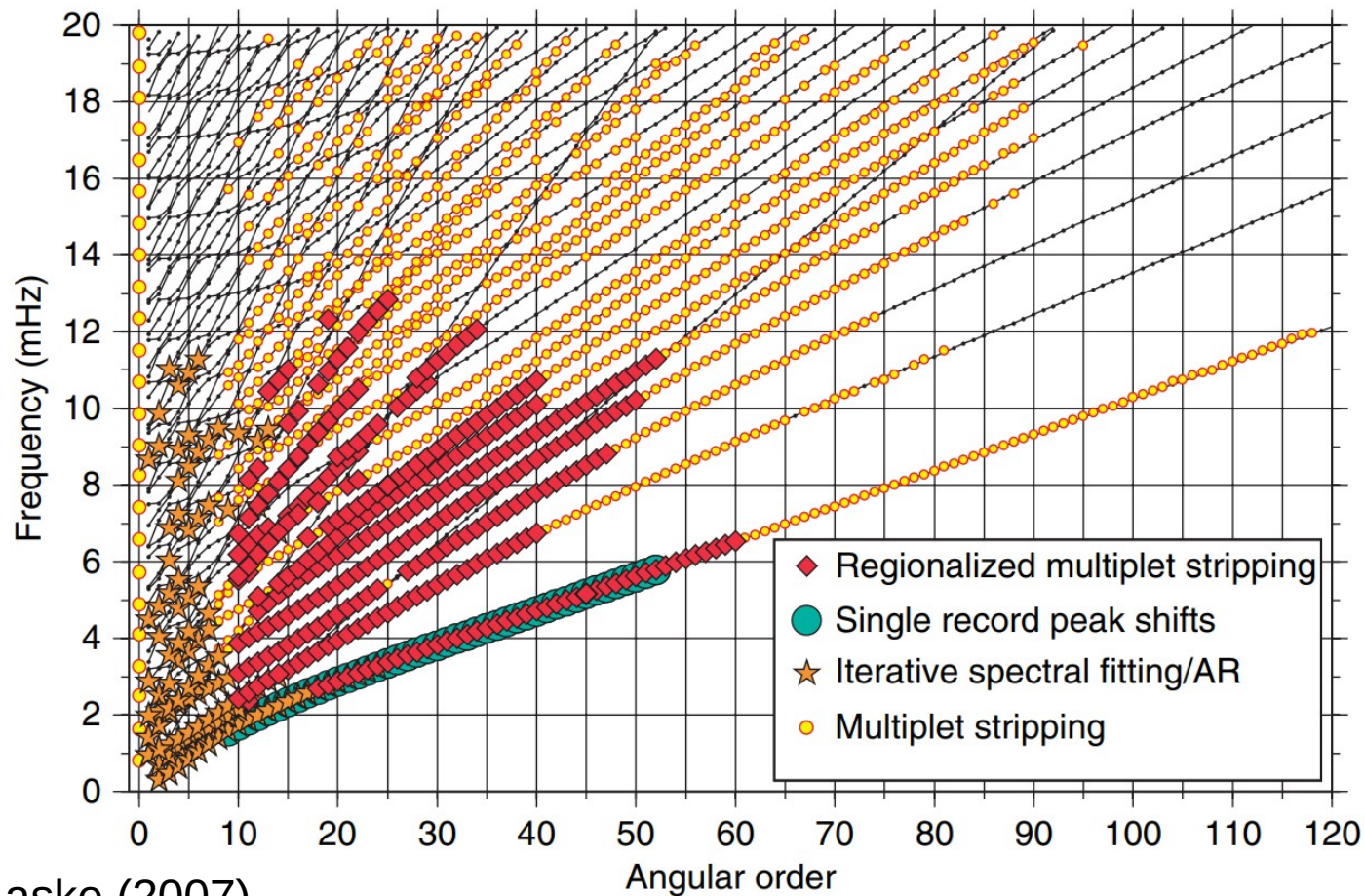
This is the **diagonal sum rule** due to Freeman Gilbert

It provides a rigorous basis for averaging observations of split eigenfrequencies within inversion for the spherically averaged structure



Widmer & Laske (2007)

It is relatively rare that all split singlets for a multiplet can be identified and measured, but the diagonal sum rule still forms the basis for more sophisticated methods for measuring eigenfrequencies

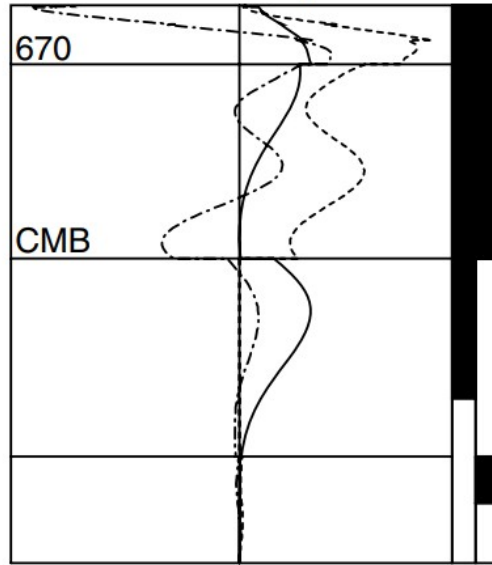


Widmer & Laske (2007)

By the late 70s, about 1000 eigenfrequencies had been measured

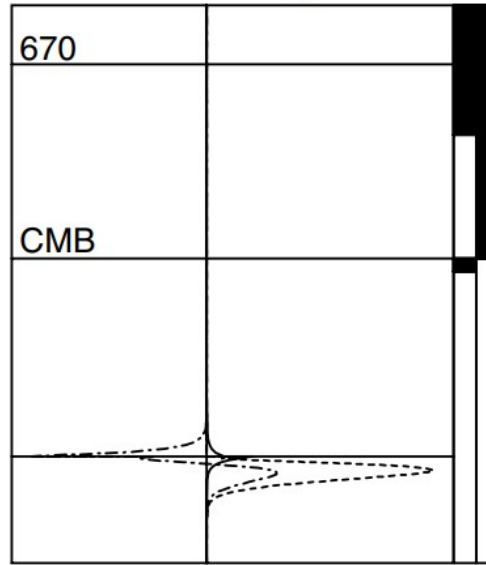
${}_5S_2$ Period = 478.2s Q = 317.9

Surface Sensitivity kernels



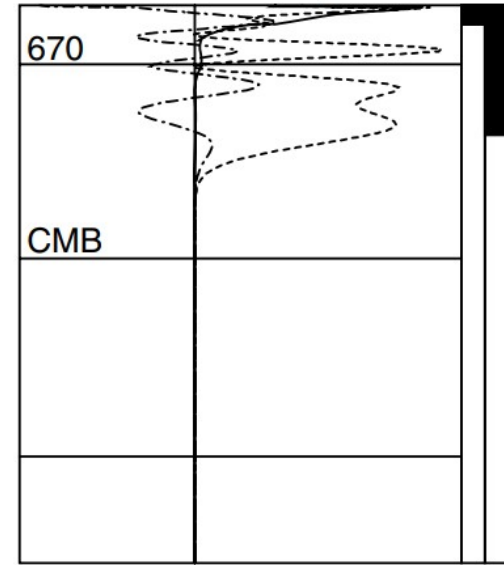
${}_5S_{10}$ Period = 240.6s Q = 89.5

Surface Sensitivity kernels



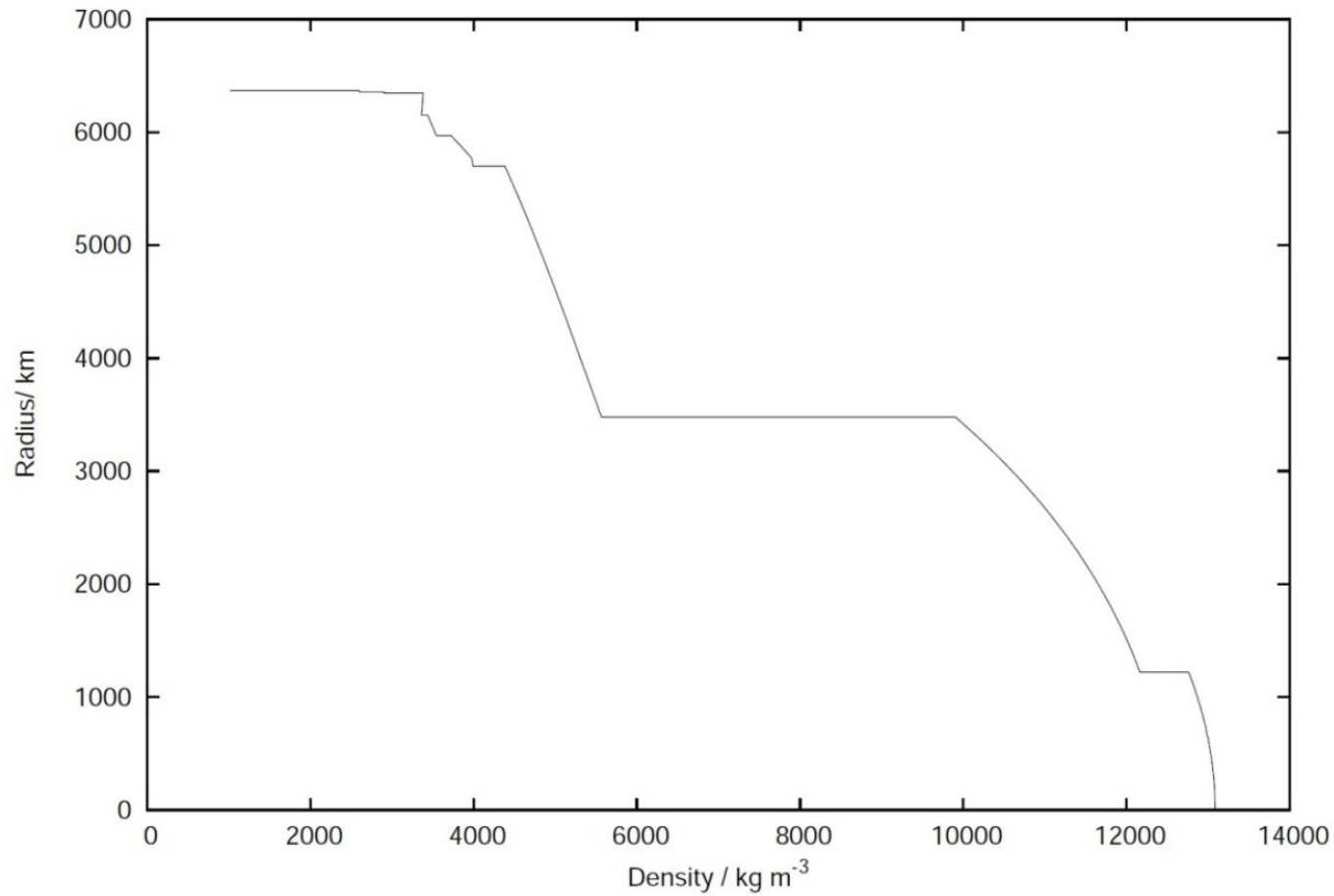
${}_5S_{50}$ Period = 91.1s Q = 241.9

Surface Sensitivity kernels



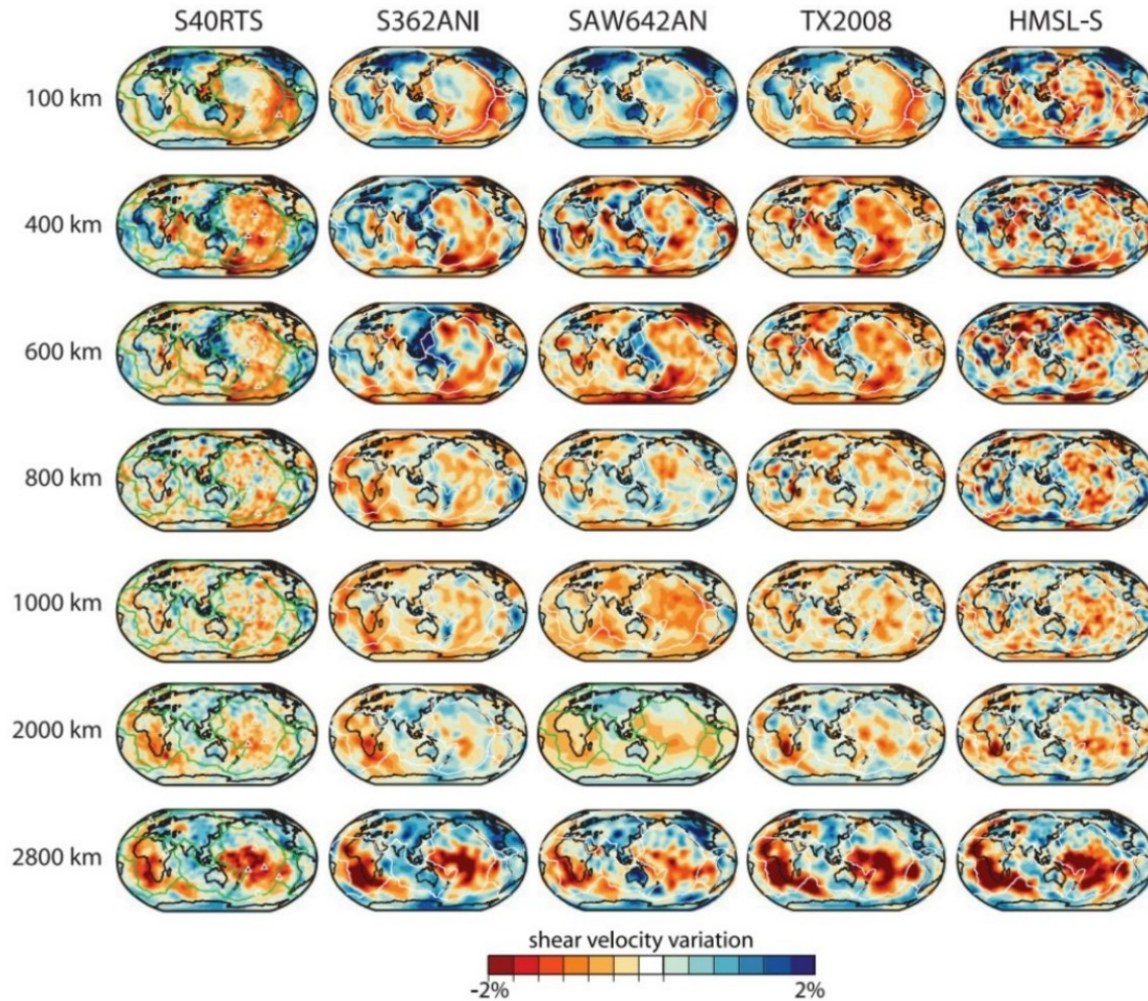
Woodhouse & Deuss (2007)

Using first-order perturbation theory, sensitivity kernels for eigenfrequencies with respect to radial velocity and density perturbations can be determined



Eigenfrequencies along with mass and moment of inertia used to constrain the spherically symmetric density variation within the Earth

Lateral density variations



What are the large low velocity regions at the base of the mantle?

Thermal or compositional?

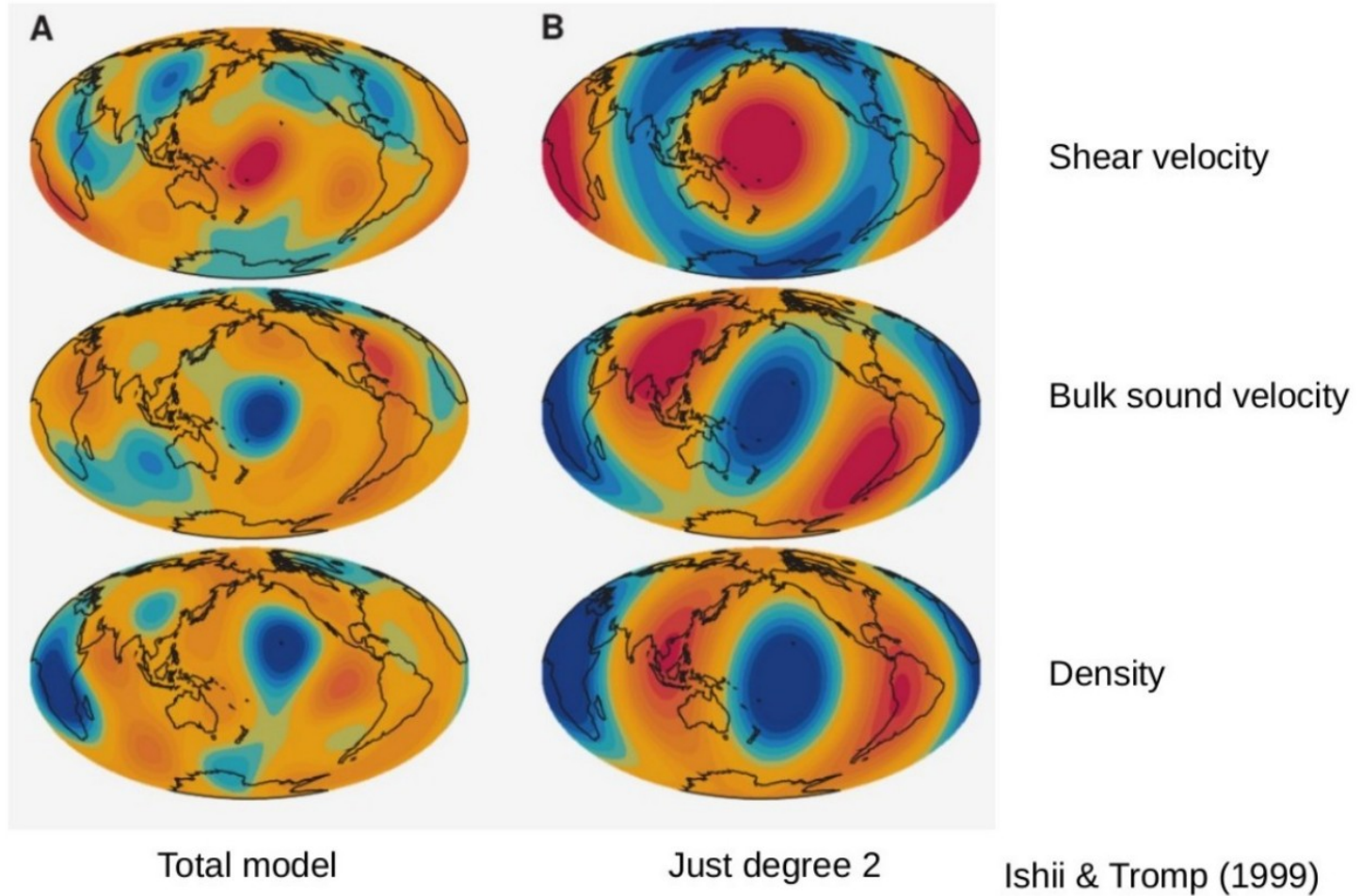
Bulk sound speed defined by:

$$\gamma = \sqrt{\frac{\kappa}{\rho}}.$$

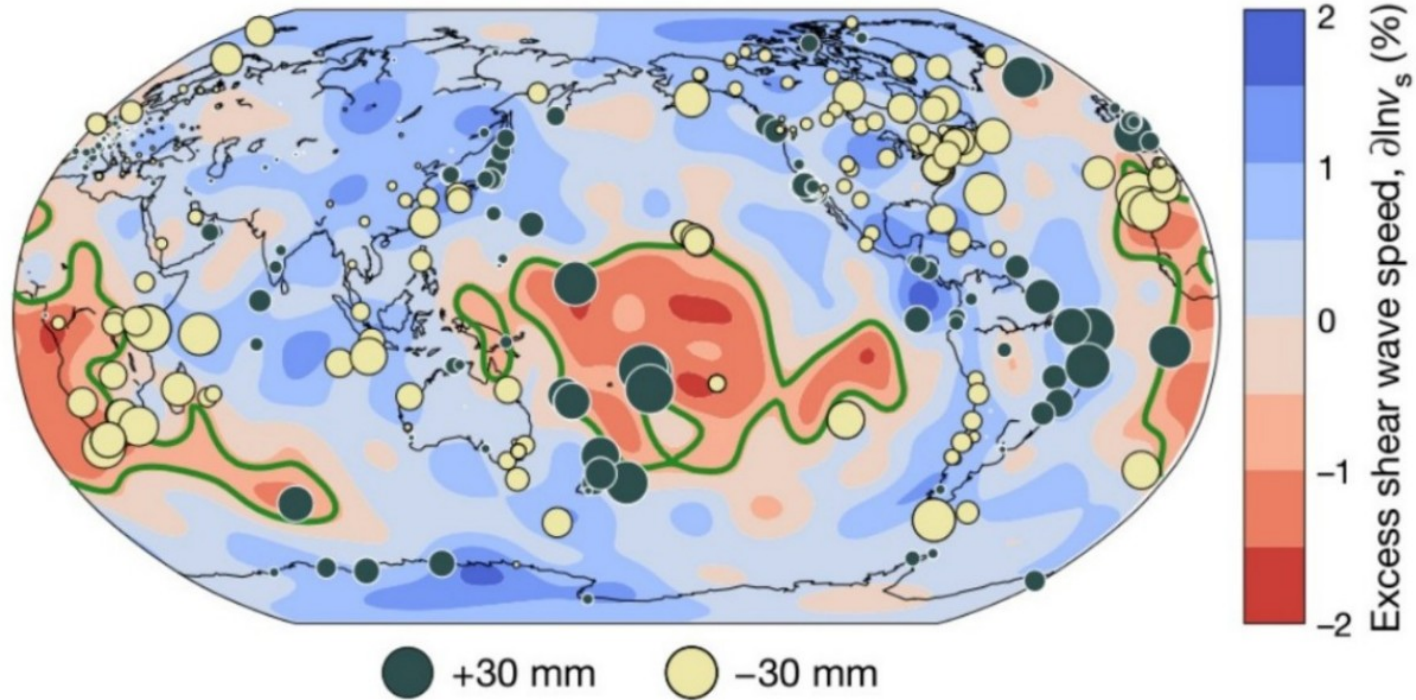
If velocity variations are due to **temperature alone**:

$$\frac{\delta\gamma}{\delta\beta} = \frac{\frac{\partial\gamma}{\partial T}}{\frac{\partial\beta}{\partial T}} > 0.$$

Joint P- and S-wave tomography suggest these velocities anti-correlate in the lower mantle – suggests a role for **compositional heterogeneity**



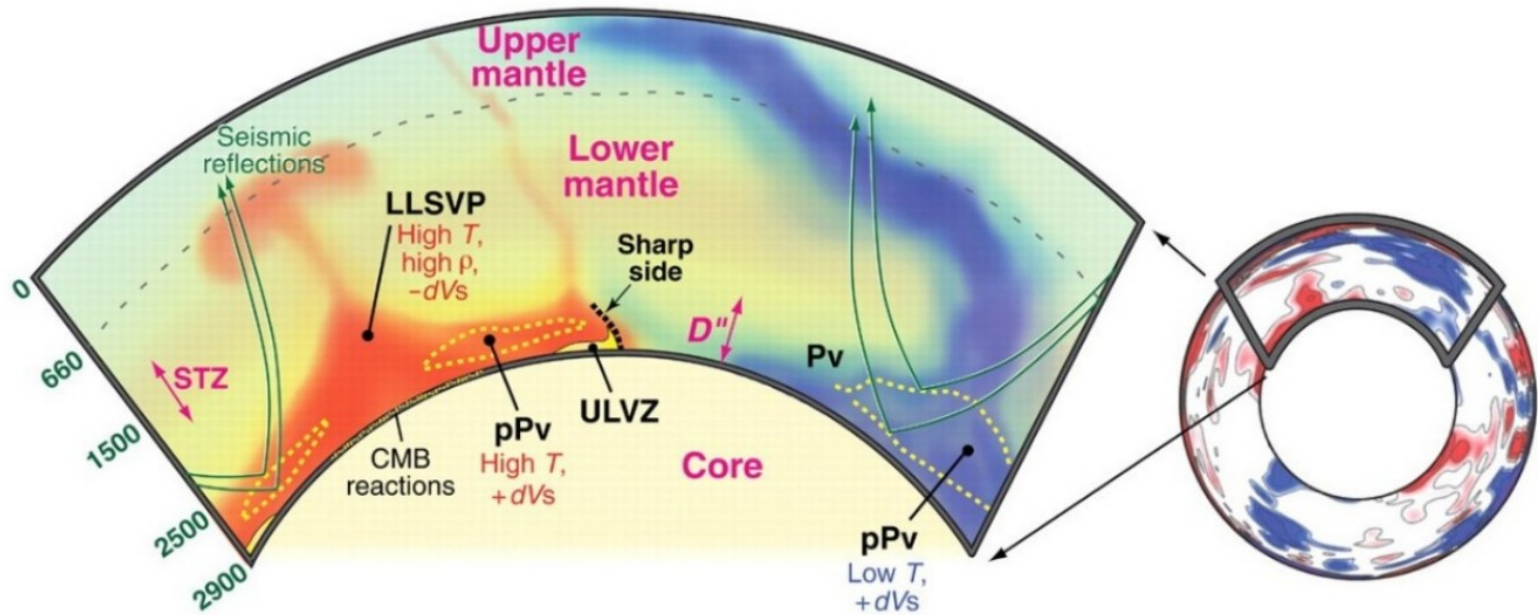
Early free oscillation tomography suggested LLSVPs are **dense features** – but conclusions have been questioned due to **approximations in forward modelling**



Lau *et al.* (2018)

Solid Earth tides also sensitive to deep mantle density. Results of a preliminary study are consistent with dense LLSVP

LLSVP origin?



Garnero & McNamara (2008)